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The Construction of the Human Being and the Human Tendencies

Our work has to be realized as a vision¹. The Montessori vision was never a cut and dried educational method, boiling down to materials, grades, and so on. Her vision was a way to help children fully develop of their own unique potential.

We do this by involving each child in a vision for life. Each child is a sense of new hope. Vision is a means of providing a path for the child toward his own realization of becoming a responsible, self-directed, alive human being. Not one with no life in the eyes or in the step. When we see someone without natural zeal, we ask what has been done to them (since enthusiasm for life is initially in most of us). We must recognize that society is systematically taking the zeal out of the children. We tie them down to compromise so they're not able to naturally blossom.

In the US and many other countries, there are more than 25 different organizations purporting to use the Montessori approach. A distinguishing aspect of AMI is the 'Training the Trainers' program – providing orthodoxy and genuineness. Some other organizations have become watered down. The ideal or vision is not consistently extant anywhere but it appears in various schools. When you consider a school, ask whether teachers are working on the same vision as your own. The Montessori name is not enough; the goal of the school should be queried. Is it the same as yours?

For the Children:

Montessori wanted to show us a logical, clear and reasonable vision for education so common sensical, its spiritual essence is elusive and can be lost. Essence can be lost beneath our own personal baggage, or from pressure from parents, administrators, regulators, etc. But who is looking out for the children?

Maria Montessori realized what the children's potential was. She spoke of us (adults) indirectly and our potential as well. The human being is the youngest form of life on planet earth. We've only just arrived, as evidenced by our prehistorical time-lines. If we remain resilient in our daily living, questioning and hoping, living in awe and wonder, engaged in the only process of our own deeper significance, then authentic realization as a species will take place. In that way, the earth will bloom as naturally as before we arrived.

The AMI teacher training course is to take students on a voyage of

discovery about the potential of humanity and their place in the cosmos. We speak of “prepared environment.” We, as species, came onto earth, where earth was our first prepared environment. Each child still arrives into a prepared environment. The first humans had to get to know this environment. What it contained, how it was to be used.

A newborn has to go through much the same process after birth. Though the environment is not the same, the process is the same as first humankind. If we can help this process in the classroom, helping the child to make himself secure and knowledgeable about his environment, then human potential is better developed. Then the human person is better oriented to his environment and able to adapt to his time, place and culture. This is the Montessori teacher's task in some way shape or form.

Montessori is *not* about materials or **curriculum**. Montessori is more about realizing the vision set before humanity when they first appeared on earth. As teachers, we will encounter parents and children who are suffering from terrible experiences which have beaten them down. We must realize the human spirit can rise above this; this is our task – to enable this. This can seem overwhelming. The first humans had to cope with elements and other calamities which they had no understanding of. They competed with animals for survival. They had to learn to overcome these things just to exist. We must realize and remember that human beings can overcome great obstacles. We are there to help children to do this. We are tempted to become empathetic with those we know who have suffered. But we need to get them working in the direction of independence, self-knowledge – only then can adjustment take place and normalization follows.

We must hold on to the vision of historical life as a series of dramatic acts and changes, actions and interplay that have gone on throughout the ages. This is how we present '**cosmic education**'¹⁵ - the education of the child by demonstrating his relationship to the universe. Also essential to Montessori teaching is that things shouldn't be too easy. Things that are worthwhile are not easily come by. We must keep this in mind for students and ourselves. The teacher's year of training is preparation for the classroom; the first year in the classroom will be even more difficult – though the work is of a different type and is with the children and the benefit of their feedback. Being prepared for the children is the great task ahead.

Montessori teachers must reflect on why one is here. For self? Or for children? At her last lecture in London, in 1951, the entire audience stood to pay her tribute. She scolded the audience saying, “You still don't understand what I am talking about. I am pointing to the child. You are pointing at me. It is the child who is the hope. Go serve the child.” She was struck by something said by a child in London one day outside a classroom: “Help me to help myself.” We paraphrase that, “Help me to think for myself.” When Maria Montessori spoke of aiding life (as opposed to using the Montessori method), she said she had not invented anything. She said anyone who observed the child could have observed the same thing. What she saw in 1907 in Italy were children with the same innate potential as the first human beings possessed.

Essential Principles:

Maria Montessori reminded us that two things are necessary: 1) the development of the individual and 2) the participation of that individual in the fabric of social life. It's the individual who must take his place as a social human being. We must always look out for those two aspects related to the children. Are we helping the individual to help him in self-construction according to his inner temperament, and secondly, helping him participate in a social life so that others in that society are served.

Once, Mario Montessori reminded us that the task is a particular service to the child – not mechanical or intellectual only – but also spiritual. Everything done in Montessori training is to help enable the teacher to serve the children in these regards: construction of self as fully as possible and social participation. Teacher training is different from the elementary classroom experience. The teachers and teacher-trainer have but one year vs. six years together. This training has been called a “benevolent dictatorship” by Ms. Stephenson⁴. Student teachers must realize Montessori training is pushing us to complete successfully a huge amount of build-up as a teacher in one year.

Maria Montessori also reminded us each person must form himself. She said: “You cannot make a man good or bad. You can help him to develop but he must make himself.” We also must remember the second plane of development can never be divorced from the other planes of development. There is a progression from the preparatory (abbreviated primary) course onwards. Trainers will continually point out connections between first and second planes. As well, they'll point to the third plane and how the previous planes relate. Actualization of the child in light of a child's personality and individuality is the goal. The readiness of child and personal interest is what the teacher looks for. There are no levels or grades – these are the wrong terms psychologically speaking. We want to avoid competition in class. We don't want students pigeon-holed by either student or teacher. The teacher should always think in terms of readiness and interest. For these reasons, we abandon labels in favor of hard observation.

Advent of Universal Education:

When we reflect on work of Maria Montessori, especially in the Casa dei Bambini, we must ask why it has not been noticed by other people. We find after searching, that some of it has been noticed. Aristotle said education must be based scientifically on human nature. The Roman rhetorician, Quintilian, who lived in first century AD, gave his opinion that children had the natural capacity to contribute to their own development, if they had decent teachers and a healthy home. These ideas have been around a long time; so what happened?

When universal education began, these ideas were lost sight of. An age was set as opposed to a measure of readiness. Various countries chose the range of 5 to 7

years old to begin formal education. Their criterion was when children could sit still and be fed information. They felt they had to be fair in beginning all children at the same time. What happened if child couldn't sit still and listen? Reward and punishment entered the system. The idea of breaking the day into time-slots was offered to avoid fatigue. What happened if the child was interested in something when the time-slot ended? The children were interrupted and had to stop. Breaking up day with recess also arose to tire out the body in order to allow the child to sit still and pay attention. Maria Montessori was a product of this educational system.

There are two forces working against each other in these settings: teacher against child and vice versa. The teacher – setting doses and – the child receiving doses and pushing back. How much is determined by teacher and not child? Maria Montessori saw it might be possible to do something different. She began to develop these ideas in 1907 in the first Casa dei Bambini. She observed that children appreciated repetition, worked in that way without fatigue and frustration, and they chose work as opposed to toys (Maria Montessori took away things unused). They were able to conform to an order of putting things away, to work independently, with different ages helping each other, until eventually, they naturally began to write. They were joyous about work even when the teacher left the room, pursuing their need for practical hand-oriented activity, so that concentration developed on its own.

Montessori's Observations:

Maria Montessori didn't begin with these ideas. She was asked to provide an environment to care for misfits. Through the environment she prepared, these results appeared. These ideas became over time the Montessori method – construction of the human being, not a curriculum. Not two forces working against each other but rather a service to a unified force. Our work has to be conducted along similar lines. This is why teachers observe carefully. They have the chance to observe and study children like Maria Montessori did. Is there something essential in human nature that enables the child to construct himself in such a way that he can become part of the environment in which he lives? Something that enables him to take his full place in the society around him? Are there factors in the human being that are universal? Something that enables him to work at forming himself? If so, that's what we need to study and understand. Then we will have something different from traditional education.

Montessori saw children explored, orientated, ordered and **communicated** among themselves. They used **repetition** in their activities until relative perfection had been reached. In doing so, they exercised and manifested their nature. The first humans on earth had to find means for sustaining their life within the first environment (land trees, rocks, insects and so on). They had to pursue food, clothing and shelter. These are the fundamental needs of human beings, that lives are sustained on. Human needs are nouns. Obtaining them is done through actions, verbs. These tendencies² are verbs, the action through which needs are met. **Exploration**, one of the tendencies, is used all the time. We use these same skills everyday. Exploration is basic to human nature. We're

inclined to, have a propensity to, explore. **Definitions** of terms must be kept in mind in discussing theory and must be well learned. This tendencies for exploration has a corollary: they have to be able to **orientate** themselves to their environment. Orientation is another human tendency.

Mario Montessori's pamphlet refers back to the earliest human beings to show the universality of human characteristics. We can imagine early man orienting themselves as we do. To find food - but also to find their way back to where they started from. On business trips we always orientate ourselves in this same manner. Closely related is a human tendency for **order**. We like to see patterns and to create organization of information perceived in our world.

Attraction to order has been conditioned through nature. The natural world has order and patterns inherently present. A corn seen always yields corn. Humankind observes that order appears firstly in nature. Humankind in turn conditions the natural world in various ways. So a tendency toward order is sympathetically stimulated within and without each day.

What makes the impact of tendencies so extraordinary? Perhaps intellect (reason) and will (love) are keys to understanding their impact. Because humans have these two gifts, they have the possibility and responsibility to handle the world well. Many use intellect but abuse or ignore the will. And vice versa. Montessori teachers have to aid children to develop these complementary gifts of such great value. Maria Montessori saw this and conveyed it through "The secret of childhood." How can the teacher assist will and intellect to act together in intentional concert? If we can't perform this service, our work becomes merely materials and curriculum, and no ultimate help to humanity.

How we work with the tendencies, helps the child in prepared environment of classroom, homes and world. The intellect uses the tendencies to build itself up. Our task is to draw out the tendencies and to strengthen them. When humans first appeared on earth, they came with a particular and specific nature. One that is a unity of these two principles – intellect and will. An organic body and an intellectual soul. That soul is formed from a spiritual and intellectual embryo.

This is the nature of humanity that comes through the ages and remains today. It is the nature that we as teachers must concern ourselves with in order to be of service to children. We help the children by serving the potential with each child – not by performing a teaching task. That's the vision part of our work that we must keep in mind. This is our primary role. We give presentations of information but that is not our ultimate focus of the work. Rather it is related to realizing potential. Though we may use the word 'teacher' versus 'director', we must remember the non-traditional goal of directing the child toward realizing potential.

In its essence, Montessori education is a study of human beings, and the process of formation that begins at conception and continues throughout life. In reality,

what we call Montessori principles really are not her principles. They are factors having to do with human beings and their work, the life they need to lead. They are reasonable and logical³. In general, Montessori principles are common sense which are not present in traditional education. Maria Montessori just happened to express these ideas after observing the need for them. As Montessori teachers, we need to get rid of the idea of curriculum and graded results. We must think of our work as setting free the human potential. This is not easy to implement.

Montessori's Formula:

Teachers are always under pressure from other staff administrators, boards of education, parents and others who don't understand our vision. They mandate standards, tests and requirements that are rubbish in relation to human beings. Montessori works if you believe it will work. It is based on a formula. If the teacher changes the formula, the same results are not achieved. Teachers must be consistent with the terms Maria Montessori set down. When the vision is not achieved, one should look for what has been changed in the Montessori formula. Teachers must hold on to faith in the children; they do not let us down if we do this. Montessori teachers who still refer to levels and such convenient labels are in error. They are making it easier for themselves; a crutch in order to keep track of everything. This is not our job. It's detrimental to children to think in terms of levels. This is part of the formula that most often is ruined. Children are being locked into a progression that is convenient for the teacher and staff.

Through accomplishments in a Montessori environment, children become different human beings. They are enabled to make an impact on the world. Recognition of the **intellect** is so important in this regard. The teacher makes the child aware they have intelligence and that they can use it in constructive manner. They become aware that they can reach their own conclusions and make their own judgments. They don't need to be led by the teacher. Children have the ability to say 'no.' But they have a right to know what they are saying 'no' to.

The **will** is a blind faculty that depends on the illumination of the intellect. The child cannot choose what he doesn't know. He must have knowledge available to make good choices. Development of the will is why the exercises of practical life are so fundamentally important. Out of all materials in the Casa dei Bambini, they include the one set of exercises that do not have control of error. There is no right or wrong way to pour water, wash hands, etc. But there are points of interest to show one how one can do something more perfectly. Table washing is an example. Initially, we want soap and water and scrubbing. But sometimes soap is spread around on the floor or there is some other similar imperfection. Until one day, the teacher can come by and demonstrate scrubbing without losing soap on the floor, offering: "Now would you like to try it that way"? This is an invitation to use that **point of interest**. If the child chooses to make use of that point of interest, he is beginning to develop his will. Development of willpower then generalizes to the playground where new choices arise, such as retaliating for arguing and fighting. Gradually, the child is empowered by experience to make better

choices. That is the development of will that we need to nurture during the time we have with the children in primary and elementary classrooms. It is developed willpower that will help Montessori teachers in training to deliver albums on time. This grows from the same development of will we employ with the children.

When we get children in elementary classrooms, hopefully the children have developed some level of willpower so we can take the development further. This helps the child to make use of cosmic education. Otherwise, we must focus our help on developing their willpower to avoid children being held back in development.

The more teachers understand humanity, its environment and tendencies, the more they can understand the children in their classrooms. Those same tendencies will operate in the children. They will need to explore. In the Casa dei Bambini, they've already begun exploring the world. At the second plane of development, they need the chance to explore the universe. This is why cosmic education (applicable to the second plane of development) is a continuing story or drama to the child. As elementary teachers, we need to practice and develop the ability to tell stories. We should fill our classrooms with stories.

A mistake many teachers make is beginning the morning with language and math. We feel the hard stuff must precede the easier stuff. Every morning, the Montessori teacher should begin by telling a story-based presentation. There are few things done with the full class; but we can begin with small groups. Stories keep cosmic education alive in the classroom. This is how to build joy and life into classrooms. If children become interested in the story, they make use of the language and math as they follow up on it, looking through books, writing stories and plays. Language and math are tools needed to explore cosmic education. But cosmic education is kept alive through stories. Later in the day, the teacher can pick up on more dry or technical lessons needed by the child such as writing, spelling and math. These are needed and useful – but first and foremost, the teacher must keep stories alive in the classroom. This keeps the classroom exciting for them.

Children shouldn't know who the teacher will be from day to day. What the teacher will wear and what trick will be pulled from their bag is a source of enthusiastic anticipation. Elementary teachers have to be actors and actresses, prepared to do just about anything except bore another group of children. This is why the elementary classroom can be so much fun.

We need to remember when first humans arrived in their prepared environment, there was no time-table set down for their learning. They used those characteristics which are part of their human nature to find out what the environment held for them to survive by. They used their senses and coordinated movement to expand their environment. With their tendencies they moved beyond basic acts to satisfy needs for food, clothing and shelter. They exercised their intellects. For example, they didn't stop at wrapping fur around the body but developed fastening to hold the wrap on. They pioneered increased functionality in their environment. They decorated their clothing. So

their interests were not just for functional purposes but also for beauty. They used the patterns they observed around them in the natural world by carefully observing and then abstracting from what they'd seen.

They entered culturally into “vanitas”; a Latin term etymologically related to vanity, which essentially means a betterment of the necessities. Vanity originally was a betterment of necessities, not necessarily a negative thing. It's why we wear jewelry and decorate; it is something with us, part of our basic nature. In this shift, humans moved from fulfilling physical needs to addressing emotional and spiritual needs. As result, we have historical developments in literature, art, music and religion.

If we implement Montessori education at the elementary level in the way it was meant to be implemented, children will have an effect on society. They think differently. They approach life differently. They have discipline. These things are carried throughout life. This service to the child contrasts greatly with the idea of merely helping the child to memorize information. The impact is much greater when we are devoted to character of the person and his long term impact on society.

To realize this vision, study of the tendencies is the teacher's first job. We need to look and observe. Is the child exploring, orientating, ordering and forming community? We can observe these things walking down the street, in church - anywhere there are children. Also observable is, whether adults are fostering or hindering the tendencies and the realization of this vision?

We as teachers need to concentrate on how to set these tendencies free. That is how we get away from the idea of curriculum. In the Montessori framework, the elementary curriculum is always around. It's everywhere. It's life and the universe. In turn, study of life in this way brings science to the child holistically. Science appears partly in biology, history and so on. It is part of the drama of humanity on earth.

Humans have a tendency to form an image (**imagination**). To create a concrete representation of something they have seen or heard. For instance, one can imagine a tool that can replicate how a beaver cuts down a tree. We had to make *something* to cut a tree at some point. This is termed abstraction.

Other tendencies to be set free are **repetition** and **control of error**. We still use these as adults. In cooking and many other endeavors we use these in daily life. This is the basis for prototyping in the business and engineering fields. These play a part in scientific evaluation and statistical analysis. The Montessori teacher embraces repetition in the student in the sense that the 'great work' is highly encouraged and often comes about through repetition. Control of error comes when the elementary child begins to evaluate himself and as a result, begins to apply himself with greater precision and care.

The tendencies are the faculties, the vehicles of the intellect that enable the human being to learn about and come to an understanding of his environment. In the first plane, the child uses these tendencies sensorially. In the second plane, the child uses

tendencies through reasoning to arrive at understanding. The environment at the second plane is the universe and all that it contains.

The only way the elementary child can get at the universe is through the **imagination**. We can't know for sure how the first humans behaved or how the earth came into existence. But we can imaginatively reconstruct these events and activities using the information available to us today. This is how we approach children at the elementary level.

The more we understand human beings in their essence, from the beginning of their time on earth until now, the more we can understand the child today – from the beginning of his time until we welcome him into our classroom. The more we can understand what human beings have made of their environment so far, the better we can understand what might still be possible in the future. Each human being needs to understand the responsibility that he has for his intellect, in how he uses (or does not use) his intellect. And he needs to understand the will that human beings possess – and his power to choose and control outcomes.

According to Dr. Montessori's world view, these gifts, intellect and will, were granted to humanity because of their special task in the universe. With reason, will, intelligence and love – human beings were given freedom of behavior. As a species, we are able to adapt to any environment and, through working on that environment, to modify it to meet our needs. We need to also remember that when humanity arrived, they arrived as independent individuals endowed with instincts, senses, intelligence and movement. These were the precautions given to preserve the individual lives and the species.

Satisfying Human Needs:

The first tasks of humanity were to satisfy their basic needs. So it seems human beings have had within them always, characteristics which belong to their nature as human beings. There appears to be a propensity towards acting in a certain way. These characteristics of a particular nature, these tendencies or inclinations, shown by the nature of the human beings to behave in a certain way, are the elements which make possible the formation of the human being. These we must understand and provide for, if we would understand what Montessori is all about. If, in fact, each child comes with all of these innate tendencies, we must help them operate in each child – rather than put obstacles in the path of their operation.

In general, our basic needs are categorized as spiritual and physical. Our needs are the nouns of the world while our tendencies are verbs. There are three fundamental physical needs: food, clothing and shelter. Additionally, we have need for transportation and for defense. Beyond physical needs, there are needs of the spirit, such as a need to progress, to feel secure and to own a sense of identity. Tendencies operate to meet both physical and spiritual needs. In The Coming of Life, key lessons, we encounter

the Fundamental Needs chart, to help us understand human needs better.

“Seeds of Change” is a book describing the impact of how our needs mold our history and that of the world. “Cod” is another book of similar use. We use these kind of books (and stories) to impress on the children how needs and actions and environment interact. For example, through meeting material needs, a culture builds up its material territory and economy. In meeting spiritual needs, our spiritual territory is built up; the aspects of a culture that identifies it uniquely by its sense of identity, its music, its lore and so on, that are absorbed by the children of that culture².

Humanity was given a spirit that some assert does not cease. They were given the freedom of walking on two legs, so that hands are free for manipulation and creation. They were given the opposable thumb so that they can make fine movements. From the first, human beings began their task of creating their own history. They chose the type of life they wanted to lead. Today, we still choose what to do with our lives. This is what we need to help children do as well.

At the beginning of the second plane of development, the children need to be introduced to the various ways human beings have come together to form societies and how, as a result of meeting material needs, they developed an economy of mutual dependency. And as a result of meeting spiritual needs, they have developed a unique culture. Tendencies are involved in and enable all of this.

In a lecture in London in 1936, Dr. Montessori asked listeners to consider man as a new energy which has come upon this earth – as an entirely new form of energy unrelated to other forms. The new energy is formed by action and work upon the earth. As Montessori teachers, we must keep the tendencies in mind, that is, why and how they operate in the children – and how to extend the tendencies – in order to make a difference in the classrooms.

Tendencies in Depth:

It is interesting to note that Mario Montessori tells us in “The Human Tendencies and Montessori Education”, page 20, that, “... tendencies do not change and are hereditary...”

What are each of these tendencies? Judging by Mario Montessori's text, the tendencies of humanity include those for exploration, orientation, order, repetition, manipulation, precision, communication, abstraction through observation, control of error and a yearning for self-refinement.

With respect to exploration, interest (volition) is fostered. We interact with the environment to explore ourselves. Exploration makes us independent. It needs to be

encouraged in all of us. It builds knowledge. It employs the will. It enables our adaptation. As humans, we also explore with our sentiments (i.e., whether we like something or not).

With respect to orientation, emotional security, which is generally important, is strengthened. A need for social and physical orientation are both present in humanity. We have to help children orientate themselves. There is stress until a student feels a part of the class and feels he has a place in society.

With respect to the tendency toward order, the child achieves an inner satisfaction during encounters with it. He sees how things fit together. He feels some level of emotional security in regularity and predictability. He discovers relationship. Sequence and balance and other natural phenomenon are absorbed and incarnated in himself in other settings. Classification (generalization) arises in him.

With respect to the tendency for repetition, it helps the child to develop precision. It is the infatuation a child feels with an activity that spurs him into repeated refinement through repetition, either by the hand or by mental effort.

With respect to the tendency for precision, the child is led to psychologically approach an ideal. This amounts to a tendency to embrace a quest for self-perfection. This brings out individual characteristics of personality. Perfection is subjective. It is not to be understood as an abstracted perfection imposed from without.

With respect to the tendency for abstraction, this was the means for early humankind's first economies, based on the natural resources and conditions present in the environment. We can understand an idea in our minds without sensorial materials present.

With respect to the tendency for communication, we feel an urge to connect with others, 'mind to mind', to coin a Zenist term, whether through an infant's smile or through grandiose written works. We learn cooperation and a variety of other ideas through communication. Our communication preserves knowledge over the ages. By communication, we establish relationship with the external world. Much study has been done on speech specifically; its development and other aspects.

Notes:

¹Mario Montessori in 1957 and 1958 gave a theory lecture. This theory presentation is based on this lecture

²Human Tendencies and the Absorbent Mind are critical aspects of Montessori theory.

³Refer to Montessori's own handbook for further details.

⁴Margaret Stephenson, around 1960, brought Montessori to the US. She helped establish The Montessori Institute of Milwaukee.

⁵Brian Swimme's "The Universe is a Green Dragon" is recommended as "Cosmic Education" in print.

Overview of the Four Planes of Development

The **tendencies and their operation help the child construct himself**. The child needs to learn about his environment, what it contains, how it works, what it's for. How he, in his turn, can use it and understand it, making it a part of himself in the process.

In order to understand how we can best help the children, teachers need to put those tendencies in relation to the stages through which the child passes in the constructive process from birth to adulthood.

Before we can be specific about the elementary level, one must review all Montessori's four planes of development. One must always relate it back to the first plane and look ahead to the third plane. We have to **understand how the planes connect to each other**.

There are guidelines given by Maria Montessori. **“To aid life, leaving it free however, to unfold itself. That is the basic task of the educator.”** If we keep tendencies of humanity in mind, we can do that. Because the tendencies are operating within each of the children. We do not do the job of educating; the child does this himself. We aid the process. That puts us in a different relationship to the child than if we're responsible for everything the child does and learns.

Dr. Montessori also said: “Trust the **spontaneous organizing power** of the human intellect.” There she's telling us that within each child, there's a power working for the process of educating, for the process of life. We have to trust that power. Especially during the second plane, we must remember not to spoon-feed the intellect. Every time one sees a worksheet being used, that is spoon-feeding the intellect. Something else she said: **“The task of the teacher is to observe, not to judge.”** If we can observe, then we can tell what each child needs and how much help each child needs. That's why our skills in observation are so important. We have to be able to **measure out just the amount of help that each individual child needs. This is how we treat each as a dignified individual**.

Again, she said: “The human **mind works by itself, moved by a power within**.” If we reflect on that, in relation to the children that we will have in our classes, then our task becomes one of making the environment so interesting that the mind cannot help but want to know more about the things that are being introduced.

If we keep these points in mind as we think of the child constructing himself from birth to adulthood, if we keep the tendencies in mind, those are the guidelines that we need to use in helping the child construct himself. What Maria Montessori was concerned about was to show us how to know what kinds of help the child needs. It was **from her continuing observations of children in different situations, in different countries, of different ages, along with her medical background and her studies in anthropology, that she was led to the conclusion that the constructive process goes**

through four separate and distinct phases or planes of development. This also means the child, during these different stages, shows different characteristics. And therefore, the child needs to be treated differently in each plane. This is why Maria Montessori concluded that the process of education cannot be one year getting a little bit harder, and the next year getting a little bit harder, as the child gets older and older. That child is asked for more results each year as he gets older.

Referring to our children as levels or grade levels, it is still this mentality that they have. Montessori from the beginning said no – that's not how we should think of the children. They are not understanding the pedagogical theory of Montessori. They hold onto the ideas that they are comfortable with. They must let those ideas go. **There are children of mixed ages that are ready for and interested in different things at different times.**

So **throughout all the planes of development, it's the child who has to be the guide to us.** It is the child who must lead the adult to see what help is needed. It cannot be the adult having a preconceived idea of a plan and fitting the child into it.

So through her years of work Maria Montessori concluded it was these four planes of development that we must focus on, to understand the characteristics that evidenced during each plane, and that then we need to see what exactly the child needs to construct herself. Maria Montessori took the **planes of development as coming between the years between the years of birth to age twenty-four.** And then she divided those 24 years into four six-year periods. From her observations, she also realized that **within each six-year period, there are 2 clearly marked sub-divisions of 3 years each.** As her work and observations continued, she also came to the conclusion that there are complementary periods within the planes of development where multiple things work together. In fact, there are **two pairs of complementary planes.**

Therefore we need to consider the **first and second planes as complementary to one another**, as belonging together. They constitute the **stage of childhood.** By end of the second plane, you have a completely developed child. The **second set of complementary planes would be Montessori's third and fourth planes.** They constitute the **stage of adulthood.**

What this means in terms of education and our work as Montessori teachers, with our work with parents, in terms of anything to do with children, is that anything a child needs during a particular plane needs to be provided in order for maximum development to take place. **Nothing can be left out. If something is left out, the child goes into next plane not fully prepared.** This doesn't mean the child cannot grow and develop into an adult; but is his full potential realized and what is that **opportunity cost to society?** Perhaps it just takes longer to develop those potentials for some. The teacher wants to make it as full an experience as possible for the child, regardless of conditions. Knowing and **understanding what the children need becomes our responsibility** because providing that is the **adult's task.**

Dr. Montessori drew our attention to another parallel. The **first and third planes parallel each other in characteristics**. Similar characteristics are exhibited and similar needs are apparent. They need to be treated in a similar way. They need that gentleness and kindness. **Often as adults, we aren't aware teens need this gentleness** because they are so sensitive but so large. We shouldn't treat them the same but similarly.

Taking the four planes of development, Maria Montessori noticed the **first and third planes are the stages of the greatest transformation**. The greatest changes take place **from the beginning of each of those planes**. Is that still true today? Consider the great difference between the newborn and the six-year old. Think of the 12 and 18 year old; they are different beings. Now, think of **second and fourth planes**. **From 6 to 12, there is lots of growth that takes place in general. But there's not the complete transformation found in the first or third planes**. Think of the 18 to 24 year olds. For some, particularly, males, there is still growth. They might get taller – but again there's not a complete change as in the first and third planes.

She also said there is a parallel between the second and fourth planes. The same kinds of things are happening. The same type of construction is going on. The same needs are evident in the second and fourth planes. What this means for us, since we're working at the second plane, is that the **second plane is the time of great strength and durability**. It is important to remember we're speaking of children who have been given the help and support required in the first plane – so they arrive at the second plane, ready to take on that plane of strength and durability. We will find kids in the classrooms that have not reached this point and we'll wonder why. How have they been mistreated? Where was proper help and treatment lacking?

Another thing Maria Montessori pointed out, is the fact that **as each plane begins, great changes take place – marking the passage from one plane to the next. It is these changes we need to watch for**. This is what really determines which plane a child is in. It is not the 6th birthday that marks the move into the second plane, nor 12th birthday that marks the move into the third plane. It's around 6 and around 12 that a child moves between these planes. This is why in the early Montessori schools, Maria **Montessori did not put four walls around the 3 to 6 year-old class, four walls around the 6 to 9 year-old class, four walls around the 9 to 12 year-old class. There were low walls separating spaces, allowing the child to see from one environment to the next and they could go back and forth between them**. This is opposed to traditional system which segregates children by age. During Montessori's teaching, there was a flow of children between classes. This should be true today as well. If we provide an environment for the flow back and forth, it is the child who decides when he is ready to move from one level to the next, as it should be. It shouldn't be the board of education, the teacher nor parents that determine this. It's the child's characteristics and needs. Financial needs of the school, as well, should come into play but only after the needs of children are met.

In public Montessori programs, there is a mandated age when children are moved between grades (or held back). And there is still something more that can be done if teachers are willing to be flexible in determining when this transition begins. But good

communication between teachers and a new attitude is required for this. It is a bit more work to track the child. But if one is running the schools for the children, this is how it would be (as opposed to running the schools for adults).

It's the rare child who at 5 ½ is ready for the elementary classroom – usually its 6 ½. This shows the wide range that only the child can negotiate for educators. Private schools have to consider their financial reality; one must realize many would go out of business if they followed this theory. This is where teachers must think through how to best service the child. Some schools keep them in primary only to find kids on the waiting list went elsewhere and meanwhile the school has lost money. There were no primary kids to take the new slots; so there is a balance to be struck in practical terms. The educational ideal is reached one step at a time in building up a school. But knowing what the ideal is, is what is needed first in order to negotiate in that direction. The message that we are trying to run schools for the children will eventually get the support of teachers and administrators and others. Teachers always stress the best interest of the child.

In the first and third planes, the greatest change and transformation takes place during the first three-year period (0-3, 12-15). The second three years of each of these planes crystalize and stabilize and perfect that growth. The second plane (6-12) shows some differences between sub-planes but it is not as radically divided. It is because of the changes shown during these 4 planes of development, if we recognize and consider them as Maria Montessori did, that a curriculum of education and the task of the teacher cannot just follow along one line of development. **The education offered has to be adapted to any particular stage of development, based on the psychological characteristics evidenced within that plane by the child.**

So if we can take those planes of development, observe the children, see if it's really true that they exhibit different psychological characteristics, perhaps we can do three things: **1) create an environment, especially prepared for the child at each plane 2) remove obstacles from the path of the child's construction of himself and 3) instead of studying a curriculum, remember we need to study the psychological characteristics and the human tendencies.** If we can take that as our plan of study, maybe we can find a way to help each child without imposing on him or her and allow each child to educate himself or herself.

First Plane of Development

The Child from Birth to Three

The significant factor to consider for the first plane is that it's during this plane that construction of the individual person takes place. This child constructs himself, constructing a person who is inseparable from the whole human being. The construction begins with the first microscopic cell at the moment of conception. And during the years from birth to three, an immense construction takes place.

This is a sub-plane during which the **child works unconsciously**; this means without consciously willing or choosing the construction that he or she is making. But the importance of this stage lies in the fact that the work that goes on during these three years, the development that takes place, form the **basis for everything else that follows**.

A mature responsible adult should be the result of the firm foundation that was laid down during the **first plane of development. Those are the years of a child's life that are most important for his self-construction**. During the years of the second world war, when Maria Montessori was in India, she spent much time focused on the child from birth to three, going back to the tiny child because she realized everything starts there and they are the most important years of a child's life. She said the university years should take care of themselves if the early years are taken care of.

At the moment of conception, the microscopic cell is endowed with a power that enables the construction of further cells and the physical form of the child – but also the form of the child as a person. That cell is endowed with the power to construct the human person. **If we consider the child at the moment of birth, he shows little sign of human intellect, as we think of it. He does not yet have a functioning human intelligence**. He doesn't have language or communication - language that the intellect needs to use. And the child has very little coordinated movement.

The **child is endowed with the sensory organs that are ready to function**. The child is also endowed with the human tendencies and within that child is what Maria Montessori called the **horme (an inner force or life force)**. And it's that hormone that urges the child on to construct himself. It's that hormone that endows the child with the power to make himself a functioning human being. The human functioning intellect is there at the moment of birth only in potential. It is made actual or built up on the experiences that the child has in the environment that surrounds him. The sensory organs are there, ready to begin taking in impressions, that will form the building blocks that the child uses in the construction of conscious intelligence.

This is why Maria Montessori talked about the **need for a specially prepared environment** – designed for this construction to take place in. It is an

environment that allows the potential to actualize. The **child has within himself the potential for coordinated movement and a particular set of movements – those that are used by the human being. The movements that the child begins to carry out are suggested to him by what he sees other human beings doing in her environment. This is why he needs to live the life of his society.** Unlike other animals, the human newborn spends a long period of time unable to carry out much movement. Because this is the case for humans, the newborn spends this long period of time taking in the environment from a more or less stationary position. The newborn does not immediately begin to grasp and pick things up. They have to build up the power to carry out these kind of movements. They need to see other humans carrying out movements to eventually carry out those same movements themselves. So all movements seen in the environment are an invitation to the child to become an active member of this family or society.

It's the **horme within the child that directs the child to the movements of other human beings and *not* to the movements of pets that might be in the environment.** A newborn won't imitate a dog or cat (until older and in jest). But hormone dictates imitation of humanity. The mind needs to be unconscious from birth to three so the young child cannot choose to act like a dog before he's learned to act like a human being. Only when they are conscious do they imitate other non-human movements.

The **tendencies are directed towards those objects and experiences in the environment which enable the child eventually to construct himself as a being of reason – a human being.** It is in carrying out activities that the child builds his own psychic pattern of the behavior of a human being. The particular pattern that he's going to build is dependent on the particular environment of the family and the particular human society in his environment. It depends on the particular circumstances that his family is living under. This is why there are cultural differences between societies. Here in the US, we tend to become more blended together. It's not as distinctive as in other place where there is a 'pure' culture that is passed on to children.

This means that some of the first tendencies to be exhibited by the child are those for exploration and classification because the child has to find out what the environment contains. He uses sensory organs for this exploration. Once the child is able to move, he'll move almost constantly. He needs to be able to classify the objects in his environment to understand what is used to sit on, to eat and so on. He has to put things together in an ordered pattern for himself. These explorations are carried out in his immediate environment, in the environment of the family and within the society of that family. Once the child is able to explore, he begins to make these distinction for himself.

Consider the simplicity of the life of a dog and the number of things required to live its life. But then think of the number of things you did just this morning – probably quite a variety of different things. We must remember the young child doesn't

know any of those when he is born, nor how much the child must take in, practice and learn to become an independent functioning human being. How can we not be in awe of the child's mission. This is why the sense of sight is so important. Before the child can move, he uses his sense of sight to study and take in impressions – concentrating on the activities around him, exploring activities necessary for his own psychic behavior.

As he becomes able to move, the child's exploration continues in a wider field, making use of movement and activity. The young child, for this reason, has a strong need for an ordered environment and ordered activities. He has to make a pattern for himself. This cannot happen if things keep changing or don't have order. Imagine if streets were renamed every day. The young child needs consistency. Once he's made that pattern, change does not have such an adverse effect. This is why tendency for order is so important and so active. The child needs order to form his intellect – to classify and organize, to orientate himself to his life – and in order to put himself in a relationship with the environment and activities going on within that environment.

With respect to exploration, it's important the child explores with his sentiments. He **wants to please those around him. He takes on the likes and dislikes of those around him.** His family is giving him guidance for constraints on his life. We see what this means with respect to prejudice or lack of prejudice. We can overcome some poor beginnings but it **later requires a conscious act of will to overcome early impressions** passed on from those in the child's environment. This is why parent education is critical. Teachers inherit the unfortunate things picked up from parents.

The teacher is greatly concerned with movement in the first plane of development. The elementary teacher must also remember that when one speaks of movement, there are two aspects that one still deals with in the elementary level: 1) balance and coordination and 2) motor control, especially the fine control of the hand. In the “Absorbent Mind”, there is a chart about movements and its development by age. We should refer to these charts to be familiar with what age certain development takes place. As soon as the child begins to walk, his tendency is toward almost continual activity; he's got to practice constantly. This is why the young child repeats these movements over and over again. Carrying a tray, for example, is a worthy practice for the young child. It is through movement that the child has a wider and wider field for exploration. This is what will enable the child to adapt to his particular society and environment. This is not magically completed within the first plane; development of motor skills continues but is less dramatic.

The tendency to work (manipulation, activity, etc.) is also present. For this young child, this tendency becomes operative when / as he becomes able to move and control his own movements. He wants to take part in that life of the family. This is when we should be including him to set the table, for example. He wants that work and activity.

This is the sub-plane of the unconscious absorbent mind. Analogies of this mind include a sponge or a camera – no thought, just absorption. This changes gradually. It's important to be aware that the young child cannot discriminate, cannot realize something is not beneficial to take in. It will be absorbed no matter what. Children can be over-stimulated and there are other various influences that can have an adverse effect. Another potential the child must make actual is that of language. There are stages the child goes through in developing language.

The first three years of life also bring independence in the ability to do things for and by oneself. That is a normal part of human development. With the young child, we need to give them opportunities for independent work, choices, activities. All of these are there in potential, and only actualized if child has the chance to practice them. These choices, which the child makes, begin in the first plane of development to empower the child later on.

Even for the young child with the unconscious absorbent mind, there must be limits for activity and behavior. He will take in the limits that are set, just like everything else. Parents don't have a problem preventing their child from touching a hot stove but they tend not to realize there are also limits to other behavior that should be set as a protection for the life of a family or society. The good of the whole family should be looked at and appropriate limits set, so all can live harmoniously. We should think of these limits as protections or safeguards for the child's social life rather than prohibitions.

If we've taken care of all this development from birth to age 3, the child in the Casa dei Bambini is ready to extend these developments. "Understanding the Human Being" by Sylvana Montenaro and "Montessori from the Start" by Paula Polk Lillard, are good books for the birth to age 3 period of development.

The Child from Three to Six

The second sub-plane crystalizes and refines the work done in the first sub-phase. By the end of the first plane, the child should have constructed himself as an individual entity. If working with children in the first plane, the teacher must take care that opportunities are offered for this individual construction. The child must be allowed to work individually – not in a group.

Maria Montessori realized the first thing the child has to do is to construct himself, know himself, and be able to evaluate himself. Then he has something of value to take into the group at the second plane of development.

This means the prepared environment of the Casa dei Bambini must be comprised of individual work, presentations and activities. It's only at end of first plane that we have a few activities with the group (e.g., walking on the line, silence game, chart memorization, word function lessons such as verbs, collective exercises in math).

The prepared environment should allow for this individual construction and also present the child with what Montessori referred to as “**keys to the world.**” She looked at the world and saw it's made up of qualities and facts and encapsulated them into the materials. That's what we find in the sensorial materials in the Casa dei Bambini; color, size, dimension, shape, form, taste, smell, sounds, touch, language, mathematics, people, manner of dress, homes, manner of travel and work, songs sung, stories told, geographical terrain, the biological furnishings. All these are encapsulated in the material of the Casa dei Bambini. It's as if we had taken the qualities of the world and isolated them for the child.

We use these materials to **prepare an environment, keeping in mind certain factors** that the child also needs (e.g., order). We must give the child the opportunity to **work with an ordered concept**. Another factor is that the **environment must be attractive**. The child must be invited and intrigued. To use and take care of materials is present throughout the first plane for the child. Another consideration is that each fact has to be presented in isolation so that the child can concentrate only on that one thing without distraction or ambiguity - so it stands out. It's **through isolation that we can show the child what is important and therefore what to focus on**. A fourth factor for materials in the environment is that each quality or fact is limited to one example of it; so **there are limitations. (There will still be limitation in the elementary environment but in some cases there is duplication.) If we did not limit ourselves to one key, we would confuse instead of clarify.**

The environment should contain keys the child can use – not impediments to his mind. So one of the important roles of the teacher is to prepare and maintain the environment, at every level. It's important in the children's house, that everything a child needs is together and ready for use. **We wouldn't ask the child to go around and assemble the pieces to do an exercise (in contrast with the elementary level, where we might do so).** The first plane child needs everything all together right then to do something. If something is incomplete or broken, he may lose interest and opportunity may be lost. This is the nature of the absorbent mind.

Another thing important regarding furnishings in the Children's house is that it is there for the children, not for the adult. Everything should be child-size. Some activities should be around things the child is familiar with from the home environment so that the Casa dei Bambini provides a chance to continue doing work that has already been introduced in the home. As well as to extend those activities and introduce new activities. Maria Montessori wanted this environment to be home-like. There should be not only little tables of different sizes and colors, but there should be plants and animals in the environment, pictures on the wall – things that serve as a continual invitation to care for the environment because it is theirs.

The environment needs to provide the opportunity for movement. Movement from birth to age 3 must be extended and strengthened – further developed. So, the children can move freely around the classroom, choose what they work with and where they will work - and who they want to sit with, if anyone. They learn how to move furniture and to walk around the rugs on the floor from walking on the line. These are activities that lead the child toward greater and greater self-control until eventually he is able to exert or inhibit all movement, as in the silence game, for example.

Within the prepared environment, we also have activities and materials grouped into certain areas. Practical life is one area. There are basic important ideas regarding practical life: care of environment, care of self and points of interest.

There are also certain grouping of materials: 'Practical Life' is the one group that does not have control of error but rather points of interest. 'Grace and Courtesy' is another group. (At elementary level we expand on that to show cultural norms.)

Activities from home should find its way into practical life work in the classroom. Thus, practical life exercises should be culturally appropriate. All activities should be developing self-control and self-discipline. Also, they should include the ability to follow a sequence. Many activities are also developing the pincer motor control of the hand as well, such as table washing. Table-washing should occur everyday as it is

one of the best activities to achieve normalization for certain children.

Another area of study in the Casa dei Bambini is the use of the sensorial materials. One of the psychological characteristics of the child is that he is a **sensorial explorer**. Exploration is carried out through senses. They give a key to the environment of the world because those materials embody that world. The world is made up of color and form and dimension. Maria Montessori took those qualities and made them concrete. This is what she meant by '**materialized abstractions**.' They are limited to one material per classroom to teach sharing and patience.

It's important to realize the sensorial materials themselves are not merely to give sensorial impressions. The child has, from birth to age 3, taken in sensorial impressions. Sensorial materials in the Casa dei Bambini build on this to classify them, allowing the child to organize them and to put them together into a pattern.

Organization and classification are the foundations of the working intellect. Sensorial materials are building blocks for developing intellect. **Sensorial materials allow for repetition** of activity so the child can work with something over and over again **until he himself is able to indirectly absorb the abstraction** – to know he can recognize color, shape or dimension.

Sensorial materials are not limited to just a particular exercise or to two exercises. They are meant to be repeated and be deeply digested by the child. Keeping this in mind, there are two other aspects of the materials in the prepared environment of the Casa dei Bambini to remember. The first is sensorial games (e.g., cards for geometry cabinet, finding a color matching a color tablet). These are important to allow the child to take impression from the material and apply them in the environment. If we don't use those games in the classroom, we're not using the sensorial materials to thoroughly introduce the world to the child; its the games that do that. Secondly, language is another aspect that provides a key to classifying. **The symbol of the word is put together with the concrete quality; then the mind can carry that idea around without the material present.** Concrete material cannot be held in the mind – only language. Games and language are critical to the use of sensorial materials and they must be done separately from the initial lessons.

Another area for exploration in the Casa dei Bambini is language. The linguist C.L. Barber refers to language as “that great machine tool which has made human culture possible.” Language enables human beings to rise above the realm of matter into the realm of ideas. From birth to age 3, the child has explored spoken language already. He's discovered that words are made up of sounds and that those sounds need to be put together in a certain order. As long as language is in the environment for the child from birth to age 3, he will already have explored the

vocabulary of his society, the sentence structure of his society. He will be able to use language himself and be able to understand what others are saying to him. Within the environment of the Casa dei Bambini, we need to help the child continue exploration of his language.

We can help the child discover four different things about language if we use the whole language program provided in the Casa dei Bambini. We must think of language as these four areas of discovery (as opposed to teaching reading and writing):

- 1) There are more words than he already knows. We do this by using words they don't know, by naming things, using nomenclature cards that enrich their vocabulary, reading stories to them, speaking to them in conversation, mixing ages in the classroom for social interaction, using dictionaries, books, songs and poems when possible, speaking about pictures displayed, by including in the classroom the presence of plants and animals, and language used with sensorial materials. (This is vocabulary enrichment for the teacher).
- 2) The child can make his language visible. We use sandpaper letters and movable alphabet to show him his own ability to form these written sounds for himself. Sandpaper letters must be traced over and over; eventually they can make their own language visible through their practice. (Writing)
- 3) The child can tell what people are saying to him, even if they speak to him through the written word. We don't ask them to read directly. We ask them to discover the meaning of writing. (Reading)
- 4) The child discovers each word we say or read or write has a function of its own to carry out, i.e., there are categories of functionality among words. He discovers sentences also have structure through reading analysis (i.e., sentence analysis) using arrows and symbols. He learns indirectly that English is not an inflected language; in English meaning is determined by word order. (Grammar)

The child also learns in the Casa dei Bambini that the language we use enables us to express creations and inventions. Montessori materials allow for a sensorial exploration and a language exploration. Materials don't teach mathematics but rather put the child in contact with those elements of the world necessary to understand quantity and amounts – to express and create invention, for economy and trade and so forth. The math materials put the child in contact with something inherent in human beings and in nature which Pascal long ago called a mathematical mind. The materials provide the possibility of exploring the pattern of numbers and geometrical form, simplicity and order.

We again, since we are beginning, isolate one central factor - the base of the decimal system. This helps the child to understand that in math there is actually very little to learn. He learns there are the three families (units, tens and hundreds) and the three

hierarchies (simple, thousands and millions). He learns that number quantities can also carry out a few limited activities. They can come together in addition and multiplication. They can separate in subtraction and division.

Like other materials, the math material is once again concrete so the child can manipulate and explore the material; eventually the child realizes he does not have to carry that material around with him any longer. We can give the child the language of the symbols so he can carry that language around with him. It's the language of the math facts that will allow the child to carry this knowledge with him and use it without concrete materials present. All that is from memorization which is just language work. An important thing for the teacher to emphasize is that it's no more difficult for the child to learn that $7+8 = 15$ than it is to learn 'truck' or 'triangle' - or any other symbolic representation. This way we don't pass on negative feelings about math.

In general, the environment is furnished with materials in such a way to enable a factual, sensorial exploration of the world. Since the child has an intellect which is operating, he cannot help but want to know more about his world. In the first plane, he wants to know facts. The facts form the foundation for creativity and the work of the imagination which comes into play in the second plane as well as later in life.

Within the prepared environment, the child also **needs to be provided with liberty, freedom, enough time to work on something as long as the child desires, to choose the place to work in, and the ability to move when he wants to. He needs freedom to talk in the environment and to share experiences with others.** The freedom that we accord the child in the first plane of development is generally limited to that which provides for the good of all of the children at the first plane. The child is free to act as himself, individually, but also has to remember that he is a member of a society. He is a member of that group in the classroom.

This means there cannot be total freedom. The **effect of liberty should be that the child develops an inner discipline – by having an opportunity to have the will develop in experience. This develops self-control.** Though we speak of the first plane as the construction of the individual, that person still has to be aware of society, at home and at school. Maria Montessori speaks about society as an organization of human beings, each feeling his responsibility in relation to the collective order. The society in the Casa dei Bambini should allow for this feeling of responsibility toward the collective order.

In the ages of 3 to 6, **the absorbent mind is consciously at work. It's the absorbent mind that allows the child to adapt to the life of his own society,** to learn about the life being lived and to understand that society. The absorbent mind works through the tendencies, taking from the environment whatever that particular individual

needs. That creation that the absorbent mind brings about is one particular human being, one unique individual who has never been before and will never be again. Helping bring that unique individual to fruition is the responsibility of the adult. That constructed individual should arrive at the end of the first plane, ready for entrance into the second plane with no potential undeveloped.

This means the **area of responsibility for adults is preparation of the environment for the conscious absorbent mind. The adult has to recognize and understand sensitive periods as those transient times when the child tends toward a certain activity** which is most easily mastered and achieved at that particular stage. The adult has to know the difference between liberty and license. It means to recognize the child's need to develop inner discipline. To realize that this is part of the child's work of constructing himself. It means the **adult needs to learn how to observe because only through observation can we know how to better serve the needs of the child. It means to realize when to be active and when to be passive in our service to the child. It means to rid ourselves of anger and pride when we are dealing with children and to cultivate humility.** It means always to remember that need for individual work in the children's house.

So, the teacher's responsibility is to help each individual to construct himself as much as possible. Our ability to think is what constitutes our humanity. It's our thinking and reflecting and planning that sets up apart from the other created beings. So our responsibility at the first plane is to assure that the child builds as strong a foundation as possible for this thinking individual. (At the second plane of development, we'll take these thinking individuals and help them extend their powers and abilities. The form of the mind and work of the child changes. The work of the adult changes. But everything we're able to do or not do rests on the first plane of development!)

The Second Plane of Development

Observation tells us the human tendencies operate differently at each plane of development according to the special psychological characteristics that belong to each plane. The first plane child is an explorer of fact. Their tendencies operate sensorially. Along with the absorbent mind, the child is helped by tendencies that function to construct his personality.

As we come to the **second plane, we find psychological differences.** We find the tendencies operate differently. We find, instead of the absorbent mind helping to make the construction, we are now working with a mind that can reason. It can think logically and make judgments. The **absorbent mind has become a mind that can also reason.** This is the mind the child will rely on more and more throughout life.

This **also means the environment has to change. The role of the adult and the particular work done by the child has to change.** The child in the elementary class does not just continue-on along the same line as that in the Casa dei Bambini. People who have worked in the primary setting must realize that things will work differently at the elementary level. What do the children need in order to construct themselves further? This is what the teacher must pay attention to.

We already spoke about development at the first plane. By the time we get the child at the second plane, **he has achieved a certain level of independence.** And now, his intellect is asserting itself and he is able to reason. It is this functioning reasoning mind at the second plane that is of interest to the teacher. The **role of the adult is to allow the mind to naturally function;** this is all. The **work of the child is to be carried out through reason.** If that factor is missing, the children might as well be in a traditional classroom.

At the second plane, we are not teaching the children how to learn so much as we are teaching them how to reason. **In order to exercise reason, we use of the natural tendencies.**

At **around the age of six, the onset of the second plane** takes place. In order to continue the construction that the child needs to undertake as a human being, he must undergo a transformation. It is as if all that preceded has to change. Dr. Montessori referred to this as almost like a new birth.

Since the first plane child has to construct himself as an individual, he has to be what Montessori called an introvert. What she meant by introvert is that activities have to be centered around oneself by necessity. The child is turned inward and focused on his own construction. Montessori was not analyzing the personality in the same manner as psychologists might do. She only meant the young child has to be concerned primarily with himself and his own activities in order to advance the construction of self. But at around six years old this changes.

In order to pass on to this next stage of construction in the **second plane, the child can no longer focus on self as exclusively**. Now he must turn outward. In the first plane, the children are happy to belong to their family. They are happy to be with their family and do things with them. On the second plane, that is not enough any longer. This can be seen in the behavior of the child if we look for it.

But in the traditional system and society, we don't look for and therefore don't process these changes. Often one or both parents are working and there is imbalance and neglect. Though the parents want to spend time with the child on the weekend, the child needs friends more in this second plane – not just parents. So parents have to be aware of this change too.

Another change is that **delightful manners and courtesies of the first plane child disappear**. In order to make that next construction, the child has to become tougher. The softness of childhood disappears. This toughness helps the child break away from the family he can gain some space in his own right. (This assumes the child has progressed sufficiently in the first plane.)

In order to continue self-construction, the child has to move outward into the wider society. The family is not enough for the second plane child. He ultimately belongs to the family of the whole world. So nature is starting to get him ready to take his place as a member of that society. He cannot do this if he remains too closely attached to his family. This is why at this age, **friends seem to be more important than family**. If previous development has been strong and secure, the child knows family is there for him and therefore, the child can venture off on his own, knowing he can find support if needs be.

Between ages 6 and 12, the **child is at his strongest physical state** yet. He's passed through the illnesses of infancy that build up immunity – though not yet the stage of puberty (which also brings new rounds of illness). The child at the second plane has durability, strength and capability. It's as if nature helps the child take on a further construction, practicing belonging to the society of families of the human race.

Before the age of six, the adaptation that takes place is to the unit of the family. It is construction toward adapting to the family. After six, the child must learn to work and learn how to enter into a wider society which is a combination of families. So he **must participate in social life** that goes beyond his immediate family.

There are certain other characteristics that mark this transition from one plane to the next. It is important to realize **not every child exhibits all these characteristics with equal strength or when we might expect**. Each is unique though characteristics described by planes are useful generalizations.

One thing is that the child does not want fussing over him. A child on the first plane will need great attention if there is a scratch. Much tears and so on. An

elementary child may have a gash in the knee with blood dripping down the leg but the elementary child will not notice or need the adult to notice. This toughness is characteristic of this plane. Adults need to explain that care is still needed but should recognize that in this plane the child does not want to be thought of as a baby. They want to be looked at as more mature and capable.

The **second plane child becomes adventurous and daring**. This is part of the push by nature to move away from the old and to find new limits. So we as adults need to be very clear about the limits we set down for them. It's not just, "go outside and explore"; they might go further than one might image. So limits needs to be explicitly made and stated.

There is a push for the child to be **assertive as part of the process of breaking away**. They become untidy and messy – even slovenly. They dislike washing. A child who normally takes his own shower, all the sudden, stops and even neglects changing clothes regularly. Adults must realize this is part of that development and **compromise must be made to allow them to explore their needs and tendencies. This is the nature of the child's change toward reason which infers that limits must be explored**. For the Montessori teacher, it means that limits have to be communicated ahead of time if possible. Washing hands, for example, has to be explicitly reasserted, to get the class in sync with good practices, since we have moved from primarily matching and imitating to organizing relationship and acting by intelligent choice. Those actions are now not automatic as before.

Before the age of six, the child liked to dress like his peers. After six, he still wants to dress like his peers but now there is a **desire for the extreme or the extraordinary**. Baggy pants, for example, represents a means to go to the extreme. And there is hardly ever *too much* encountered for them. It's at this age: they like food prepared by themselves or by others but, all of the sudden, the parent's cooking is no longer interesting. This is a great time to teach them how to begin cooking. Mom and Dad's cooking will be critically compared to everyone else's. Since they are pushing for independence, their own food, that which they cooked, will have a unique satisfaction.

It is as if they are **pushed by the hormone. There is a hardness, a streak of separation, so they can enter into the society of the world**. It is as if during these years, the child is being given a training school on the larger world and he know he must complete this course. It is the move toward society that at the second plane, causes the **children to form groups** with other children in the class and/or other children in the neighborhood. The groups are not made up of children of all the same age, but most often they are between 7 and 10. Elementary teachers have to recognize this and allow for it in the classroom. The **children have to be allowed to make their own groups**.

They become organized, having, for example, certain requirements for membership in the group. They will have their own rules for the group. It is important not to tell them their rules or behavior is silly. We must not tease or laugh at them. The children will show great loyalty to other members of this group. One sees this if one

member of the group gets in trouble outside of the group. The group will side with the child, not the adult. They are loyal to each other within the group. If a child has defied an adult, the group will admire that child for standing up in that manner to the adult. They can become antagonistic, making it difficult in the classroom if we allow it. These same children who can be antagonistic, can band together to help society when they see help is needed. Their attitude is not hardened. Often in war, children of this age have been found to hide and protect soldiers from danger.

Depending on how we handle these groups, they can cause mischief or they can bring great good into the environment. We don't want them to harden in opposition so we have to understand what is going on in these groups. These social groups can have a secret treasure; teacher must respect these types of important things. They may make up their own secret language since they like to communicate with each other and exclude others who aren't in the group. There is a strong obedience to the group. They are making their own practice-society. If they form their society, it becomes something they can value and then practice living in the structure of. If they make up the structure, it's within their level of competence and comprehension. It works for them.

This practice society is the first real society outside the family that the child enters. There is a new code of ethics and way of life. It is because of this need to develop these social structures, that teaching in the elementary class has to be, for the most part, teaching in groups. This is a contrast with the primary level. Most teaching in the elementary level is to small groups. Only occasionally will there be individual teaching.

Unless we teach to small groups, we're not meeting the needs of the children. We won't get their cooperation. It is **from this teaching to small groups that we can help the children form their own groups and their own societies. We help them organize themselves socially** by presenting a variety of group content. Our small groups should vary – not the same people each time. The **groups are based on readiness and interest only**. So the groups should be changing all the time.

For example, a group presentation to five children may only be effective for three of them. The remaining two may hear that lesson again at a later time with other children. Children's groups are formed by being with different children in lessons, getting to know others, finding out who might be a friend. We don't accommodate cliques; rather we **provide opportunities for each child to experience different personalities in the classroom in the anonymity of being in the group**.

The second plane is also the age when conscience develops, the idea of right and wrong and the recognition of rules and values. Often at the beginning of this stage, the children go through a period of what the adult would call being a tattle-tale. The teacher needs to be careful to handle this carefully so as not to foster this in the future. The child tells when a rule is broken, not to get another in trouble, but rather the child wants to hear from the teacher whether the activities are acceptable or not. The teacher should respond with 'no, this is not allowed' or rather, 'it's okay because we are doing something special.' We don't want to correct the other child at this time. We can handle

that later if appropriate. The tattler should not be involved in this. The child who tattles has the need to know what the limit is and why. To respond beyond that brings the tattler into the problem inappropriately.

He only **wants to know what is acceptable and why**. The child has to form his own code of moral values and needs to understand the basis an adult uses. If handled correctly, tattler-stage disappears around seven or eight. As a lower elementary teacher, one might have 6 or 8 of these children coming to you tattling. But the teacher has to see through the situation and realize the children's needs in order to handle it properly. It should be clear that only the teacher is to reprimand in the classroom.

Because the child needs to move into the wider society, he possesses a need to develop more independence at this plane. There are those things which should have been learned in the primary level, including the ability to think independently to some extent. The child should be able to work on his own. At the elementary level, we want to develop and extend these capabilities.

There are **two aspects of independence to consider**. There is **independence of attachment to the family we speak of above. The second is the particular independence given by the group** to which the child now belongs. It is through that group that the child is going to become free to do things by and for himself outside of the family unit. In the first plane of development, the child should have been able to find himself accepted by his family and its members. With recognition of that acceptance, the child becomes able to see his own value, both as an individual and as a member of society. Now because he has the need to enter a wider society, the child needs another sense of evaluation. The evaluation important to the child now is his own self-evaluation in relation to the group to which he belongs. The standard is going to be set by the group. This is part of the aspect of the 'new birth' that takes place of the social human being after the construction of the individual at the first plane of development.

Another aspect of this independence is that there is a **furthering of the exploration** of the environment the child needs to undertake. That environment has to include that which is away from and outside of the enclosed classroom environment. The first plane child needs the security of the family and the enclosed classroom, a safe secure and limited environment. At the elementary level, this is not enough. Their tendency is to push beyond into a wider environment. This exploration is no longer just sensorial in nature. The children now become **interested in what they cannot see with the eye, with what they cannot experience with senses alone**. If a child is to carry out this kind of exploration, he needs more independence than heretofore mustered, to do so.

Exploration at the second plane shows itself through what Maria Montessori referred to as a hunting instinct. They will catch things (insects, etc.) because they want to understand how they function and how they are able to do what they do. How the butterfly flies through the air, for example. The first plane child is interested in facts, such as the butterfly's name and the fact that it flies. Now, at the elementary level, the child wants to know the reasons and causes of what the butterfly represents. The reason for

actions and conditions.

If the **adult does not understand this characteristic, he will overreact and be appalled at the cruelty the child displays** – like pulling wings or legs off an insect. Normally, the child has not thought through the idea of harm when the legs are pulled away. The child will feel badly once this is realized. Sometimes, we bury or dispose of the situation a particular way to resolve it. And we may offer alternatives in the future to explore another way.

In the first plane of development, facts gathered provide the child an opportunity to classify and organize the qualities and content of the world. In order to get inside those facts, the elementary child needs to be given reasons for these facts. What is visible is no longer interesting; there is a need to go beyond for the elementary child. Human beings have always been driven to understand nature's mysteries, to see what was earlier hidden. This is how civilized society has advanced. Why things exist and how they function are now interesting to the child.

If a child is interested in something beyond the senses, he needs help. The **imagination becomes alive and active at the second plane to accommodate this study**. Before age six, Montessori warned us not to give the child imaginative ideas. The child should first be grounded in facts and reality. The young child cannot distinguish between fact and fantasy. Furthermore, the young child needs those facts so later his imagination can use them to create new thoughts and images – since creativity relies on a basis of fact. It is through imagination, working from fact, that second plane children need to immerse themselves. This child can only explore the invisible through the imagination. It is with the help of imagination that the child will investigate the universe and human society in its many manifestations. It's only imagination that can serve to allow the child to explore what cannot be sensorially explored.

At the same time, these children are still attracted to reality. For them, it is interesting to note whether certain stories are true or not. They like adventure stories, stories of heroic acts, adventures and great deeds.

Another **characteristic shown by second plane children is that of hero worship**. They are looking for heroes. They have a great sense of admiration of justice, of compassion, particularly directed to those younger than them or those in need, and those who have undergone hardship. These **feelings of compassion have to be taken into consideration in the lessons that we present** in the elementary classroom. We need to appeal to and make use of this characteristic. Teachers should give the child a chance to admire, to feel compassion, to find true heroes. In general, children find heroes in society on TV and so on. We can help bring more pronounced heroism to their attention through books and magazines, stories, poems and so on. In this way, we can influence their ideas of heroism and their own potential to realize such things.

Dr. Montessori said that at a certain point, **elementary children understand better with their feelings than with their intellects**. Teachers should

appeal to feelings. At the same time, there is great clarity and potential in the child's intellect. Because of this, the children have a need for work, real tasks, and big projects. They are strongest physically and most receptive to intellectual learning at this stage. This means the teacher and others have a responsibility to not let the child waste these years.

Activity is important for these children. They still must understand through manipulation. This will once again take place through repetition. But now the repetition will be made through amplification. Second plane children are not so inclined to sit and do the same thing over and over again. Repetition through amplification occurs instead. For example, squaring a binomial (two terms) vs. squaring a trinomial (three terms) may lead to an inquiry for the extreme; children on their own might continue to square using four terms or the square of the letters of the alphabet. They want to explore beyond old limits. We hold them back if we don't allow them to go off on their own volition toward those type of efforts. **If a teacher allows the children to make their own discoveries, running the class is much easier. Teachers often believe they should be 'doing' to make the classroom function.** We react from our own experience and background. We can overcome this conditioning by remembering how the child most benefits by our passive aid.

Because of the functioning intellect and the power of reason, now the child has a **mind that is receptive to the abstract.** Earlier it was receptive to impressions and facts. But that does not mean they are ready for abstraction. It is during ages 6-12 that they become receptive to abstraction. Teachers need to allow children to work with and manipulate the materials, to **make up their own examples and experiments until they arrive at their own abstracting capacity, naturally flowing from tendencies.**

It is through continuing experimentation that each individual child is able to approach abstraction. This is why we limit the material in the classroom. Teachers must give the keys only and allow the child to grasp them, work with them and come to their own comprehension of what is represented by them. **If our materialized abstractions are not limited to just one example, the child does not know what to focus on and the push toward abstraction is lost in the clutter of detailed and endless examples.**

Another reason children are trying to pick great tasks is that they want to show us what they are capable of. They want to show us another degree of independence. This is why at the elementary level we must take care not to interrupt or stop the children's work. **They need time and consideration to work until they are satisfied.** That work will become the child's own self-evaluation.

The **'three-hour work cycle'** is something Maria Montessori observed: the child comes into the classroom and false fatigue ensues soon after; but after a time, the child chooses something of more appropriate challenge, in which the great work will occur. The child needs large blocks of uninterrupted time for these large projects. They won't undertake a large challenge unless they perceive they have time to invest themselves with some abandon into the task. The human mind is distracted when a future

point is anticipated or when one expects interruption based on past experience.

At the **second plane of development, the child needs a sense of security. It has to be given differently than in the first plane. In the first plane, order, routine, a sense of acceptance and being valued is what is needed. In the second plane, the child needs to find security through being able to evaluate himself. He needs to begin to know what he's capable of**, what his possibilities are. It is self-evaluation that gives second-plane children the security they need.

Elementary teachers often find children come through primary school and have that sense of security and it seems all will be well. But then it does not happen and we are puzzled. There is an **emotional attachment with the new environment, including the other children and the teacher, that must occur**. So they begin by returning to the teacher for evaluation. 'What do you think of my picture?' We don't give praise or rewards so what does one say? In this case, they want to form attachment with the teacher. So the teacher wants to provide this attachment without overtaking the child's ownership of the work. So there is a place for encouragement which provides courage and some extra feeling for going forward. This provides adjustment to the new class and new teacher and they can move beyond looking for security through the teacher. We show them how to check their own work to help them maintain ownership of their work later.

Another **way of helping the child feel secure from day one is to let them help in maintaining the environment**. This gives them a role of their own in the environment and the group. While the first plane child participates just for the activity, the second plane child needs to connect with the function and the need for maintenance itself. This is where their interest lies more directly. **They are ready to revitalize seriousness**.

If the **characteristic needs at the second plane are to construct the child as a member of society, able to assert himself and evaluate himself, and if the tendencies of human beings have to operate on a wider field of exploration, through the imagination and reasoning power of the intellect, teachers need to ask themselves what kind of environment is needed at the second plane?** The teacher's job, as in the first plane, needs to prepare an environment that helps the child realize his full potential.

The teacher has to keep in mind that the two main factors to consider are firstly, the remarkable changes that take place, the differences between the first and second plane children, the remarkable stability that the children show after this transformation takes place.

The second factor to keep in mind with respect to the shift from first plane to second plane is the inner urge of the child to incarnate the society of the group. We must keep in mind the child needs society and needs to be prepared to enter into that society outside the classroom. This is part of the elementary teacher's work.

We also need to **keep in mind the human tendencies, especially those of exploration and orientation.** The child's exploration and orientation are going to need to take place in a wider field than that offered in the classroom – even if there is an outdoor environment involved. The classroom is no longer enough. Explorations need to take place, in part, outside and away from the school. So the environment becomes a double-environment. Whether we speak of that environment that we can prepare or the environment which needs to be explored outside our control – it's a double environment. The part we control - and the part we do not, that resides in the real world. Elementary children need to be able to work in both environments. This again, contrasts greatly with the Casa dei Bambini and the first plane child.

One of the tasks of the teacher at the elementary level is to prepare the children to work outside the classroom, to prepare the children to explore outside the classroom. This means that practical life activities at the elementary level will involve this preparation for working and interacting outside of the classroom. The children need to be prepared regarding how to set up these expeditions, how to travel outside the classroom, how to gather the information they seek in order to bring it back to the classroom and make use of it. Unless we ensure our children have a chance to do some of their work outside the classroom, they will not be developing their full potential.

These expeditions outside the classroom are not field-trips in the traditional sense. A field-trip in that context is chosen and set-up by the teacher and the children and teacher attend. This in general, is a different matter. The children need to go outside the classroom to find some information that is not available inside the classroom. It is going out to explore phenomena that can only be demonstrated inside the classroom but not really experienced; for instance, we can demonstrate and describe a river but one cannot really experience the fullness of what a river is. This requires leaving the classroom.

The **exercises of practical life therefore, need to include the information on how to get from one place to another, how to conduct oneself in society, how to be safe out in society,** how to use a museum or library or art gallery or farm and so on. It is not enough at the second plane to bring someone into the classroom in order to speak of these phenomena. This would impart information to the children but this does not help the child see the complete picture nor is it the *first-hand* experience which the child seeks. These trips need to begin, seeded within the classroom, by the child's questions - drawn from readiness, needs and interest. This means, as the adult in the classroom, the teacher needs to know what is available outside the classroom and in the community. What is available on a permanent basis and what is temporarily available. This becomes part of the teacher's work on the prepared environment.

What does the child study at the second plane? Dr. Montessori asked us to give the child the universe. “The universe is an imposing reality and an answer to all questions. We shall walk together on this path of life. For all things are part of the universe and are connected with each other to form one whole unity.”

At the second plane of development, in order to understand, the child has to have a complete vision. This is why children get involved with number sequencing, for example. **They want to find the totality of a picture.** As teachers, the adult must give this child the big picture. If the big picture is provided, then **their minds can work on looking for the parts that make up that whole.** This is what they will begin to do in the second plane.

In order to give a complete picture to the child, **teachers cannot teach one subject in isolation from the other subjects. Science, chemistry, botany, music, art, history, mathematics – are all intertwined and interrelated.** They must be encountered hand in hand. We can only answer questions in one area by relating it to another area in its whole context, if we want eventual total comprehension for the child.

Every presentation to the child should be just the next story or key along the path of exploration. Albums may be categorized but don't have to be presented as such. The teacher presents them as just the next story in the path of what the child happens to be interested in. When practicing presentations, the **teacher should be able to coordinate interest from one area into what is being presented.**

At the elementary level, the task of the teacher is to **provide keys to the universe and its functioning, its plants, animals and other content. Those keys can only be given by appealing to the imagination of the child.** The vision of the universe, all of the life on our planet, is not sensorially accessible to us. We have to help the functioning imagination so that through the imagination, the intellect can grasp the idea of the universe and all that that implies. The lessons are necessarily very different than those in the primary level. They are, for the most part, imaginative stories – illustrated by maps, charts, timelines – accompanied by experiments or demonstrations. These charts are to provide impressions – not necessarily facts. Our goal is to stir up interest, not to collect and present information. This is how we present the beginning of the universe and the coming of life.

Two of the Great Lessons are given as stories. The children can later, work on those impressions, ponder the ideas and find out for themselves the related facts. For the lessons in the second plane, Mario Montessori said Maria became a story-teller of the truth. This is what teachers should aspire to – to **become storytellers of the truth.** This opens doors to the functioning of the universe. They set the stage and introduce characters onto that stage. Therefore there is a certain seriousness to these stories. There is something extraordinary in them since they apply to us. This is how we approach generating awe in the child, so imagination and curiosity can carry them beyond.

Included in these stories are the tasks performed by substances and creatures throughout the ages, through which the rest of creation is helped. These stories are also to bring to the child's attention what would happen if these tasks ceased being performed. What if insects decided to stop pollinating flowers after doing so for thousands of years? This type of question helps the child appreciate the immensity and interconnectedness of the universe. How does it all relate to human beings? Do we have a task as well? What if

it goes unperformed? After we give stories and key lessons, the children follow up with their own activities and explorations that need to be left to their own choices and volition. They may look through a book, write a story or sit and digest it only. This is in their control and must be respected. These explorations, launched by the children themselves, **help them to become conscious of the cosmic task of all of creation. Montessori said this exploration should provide the child with a sense of gratitude toward God (or creation) and man (humankind).**

Helping the children develop this sense of gratitude means **giving the children a vision of all reality through time and space.** They can imagine the creation of earth and its various forms, achieving harmonious operation. To give the child the possibility of a sense of gratitude toward humanity is to **help them realize how the child's own fundamental needs have been fulfilled by others** and how this has gone on from time immemorial. This helps them understand how societies function. It gives a sense that it's the anonymous men and women of the past and present who have made contributions that benefit all of us.

Through creating this vision in them, we help children come to the realization that in spite of the diversification of people all over the world, humankind is fundamentally one.

At the same time we have to ask how the children will learn to read and write. Remembering that we are concerned with the construction of the individual, ready to participate in social life, we have to **keep both aspects of the child's construction in mind: the inspirational grand view as well as the specific detailed skills required to function in society.** Society asks that we, as human beings, be literate to a certain extent. We have to make sure we give the universe to the children and allow them to freely explore it but also have to ensure they will be able to fit in to society and to meet society's expectations for them.

In every elementary classroom, either posted on the wall or in booklet form on the shelf, and readily available to the children, needs to be the **public school curriculum** that the child will need to know by the time he is nine (or twelve for upper elementary). If this has not been done in the school a teacher joins, one of the first tasks for the teacher is to find out what these expectations are. They should be translated into language the child understands and then the child is helped to learn to use that curriculum.

The teacher helps the child to be aware of fulfilling the requirements on his own for himself over the long term. By helping the child use the tool of the public school curriculum, we help the child know what lessons have to be followed-up on. **They can choose how, when, with whom and so on. But they cannot choose to do nothing about learning these things. The child should understand why it matters that he apply himself toward this end. This is how freedom is afforded** in elementary classrooms. It is freedom with discipline and within limits. The public curriculum is the

minimum safety-net for the child's successful entry into society.

The teacher gives the presentations to “sow the seeds of culture”, according to Maria Montessori. According to her, we are to sow as many seeds as possible but not to pull up the roots or to inspect progress invasively. The curriculum is the universe. The aspects the child chooses to explore depend on his interest. What she *must* explore, is what the public curriculum contains. The child cannot choose to not follow-up on long multiplication, for example.

There is nothing in Montessori that says 'no' is inappropriate for the children. When it comes to work, there may be presentations with no follow-up. Or maybe the work is the same again and again as a crutch of escapism. The children cannot progress in chaos. The children generally know when they exist in a reasoned environment or not and they accept what is reasonable and consistent.

This plan called 'Cosmic Education' is the Montessori curriculum for the elementary class. It is the introduction of the cosmos and the work of exploring it. That is what forms the environment and the work for the children in the second plane of development. It is that exploration of the cosmos, the cosmic tasks of all forms found in the universe, that makes possible the construction of the child as an individual, able to participate in social life. This exposure to all of the universe, allows the child to construct himself and to know himself as an individual. It allows him to recognize himself as belonging to a society. It prepares him to be able to give himself to that society just as society gives itself to him.

One of the tasks of the adult then is to give the story of the cosmos in such a way that it **excites the children** so that they cannot help but want to go on finding out what else there is just beyond the door opened to them in the stories. Just as one speaks of limiting the materials in the Casa dei Bambini, in the elementary class we need to limit the presentations that we give and we need to limit how much information we give in each presentation. We want to give only a glimpse of what there is to find out. We want to do that in such an exciting way that the child is anxious to progress to the next inquiry. If we present too much information, there isn't enough left to find on his own. He is overwhelmed. Dr. Montessori said: **“The secret of good teaching is to regard the child's intellect as a fertile field in which seeds may be sown to grow under the heat of flaming imagination. Our aim therefore is not merely to make the child understand and still less to force him to memorize, but so to touch his imagination as to enthuse him to his inner-most core.”**

This is how the teacher's security can be drawn, as a practical matter, from adhering to the public school curriculum. **If the teacher ensures the child follows up on those items in the public curriculum, then he can sow as many seeds as possible and leave the child alone otherwise.** The teacher does not have to be worried about grading or defending what knowledge is imparted and gained.

Seeds need to be **sown generously between the years of 6 and 9** - not as

much between 6 and 12, since these children can go forward on their own, having had the foundational presentations already. Most presentations are geared for age 6 to 9 as the basis for all following work. These are all the elements the child needs to build up a foundation of knowledge about the universe and the furnishings of the universe, about human beings and their lives. They are introduced to as many aspects of these as possible between age 6 and 9.

Children between 9 and 12 are given the same presentations, keys and so on. But because the seeds have already been planted, the children have the opportunity to go into more detail and into deeper levels of comprehension. The curriculum is the same for ages 6 to 9 and 9 to 12. It is all the same universe. The difference is that in the 6 to 9 class, teachers are planting first ideas. At the 9 to 12 level, because they've heard an introduction and have done their own follow-up, they are ready to go into a deeper study of each of those same things and their inter-relationships. They are ready for **research, for example.**

The **two sub-stages of the second plane of development and the way in which the work is approached can be thought of in the same way as the first plane** is thought of. During the **first sub-stage, there is great growth** and new information. During the **second sub-stage, knowledge is extended, crystalized**, more finely developed, fleshed out.

During presentations, we should remember this two-fold approach. What is first presented is the original introduction. If a child has had a given presentation, those same lessons are used as a jumping off point in the upper elementary – to go into more facts, other aspects and details.

January 20, 2004

Third Plane of Development

From her medical, psychological, psychiatric and educational background, Dr. Montessori observed children, worked with them and began to talk about them passing through **4 planes of development**.

She noticed that it is the same human being who exhibits different characteristics during different planes of development. Those psychological characteristics are used for the construction of the individual during each particular plane.

At the end of the first period, different characteristics kick in with a different construction to be made. The tendencies also operate at each plane, making the construction of the human beings possible.

Dr. Montessori said **education should take into consideration the psychological characteristics – the human tendencies and the planes of development**. Rather than focusing on curriculum, syllabus and schedules. She also said we need to construct an environment appropriate to each plane so all of these things can function to their fullest.

If the fullest help is to be offered, each plane also needs a prepared adult who is aware of the psychological characteristics and human tendencies and who knows how to prepare that environment. That adult then serves as the link between the child and the environment.

The materials are part of that prepared environment. Therefore, they have to change from one plane to the next - because the psychological characteristics change and the human tendencies take a new direction. This is one reason why the Montessori materials are always contemporary – they've been designed to tendencies and characteristics that don't change. Throughout time and across all societies, humanity has exhibited the same psychological characteristics and tendencies.

We said before the **first and third planes parallel each other. Characteristics are similar. Their physical bodies are in similar phases** of development. Each plane is 6 years. Within each 6 year plane there are two sub-planes, each approximately 3 years in duration.

The first sub-plane is for construction and building – growing into – new material. The second sub-plane refines and consolidates what has been gained in the first sub-plane. Especially in the first and third planes, there's a striking difference between the two sub-planes.

We've also said the **first and second planes are a pair**. They form the **stage of childhood**. The first plane is the construction of the child and the second completes that construction. Childhood is over at the end of the second plane of development. So, it is the responsibility of the adult to make sure that nothing that is needed for that construction is left out. We also said the **third and fourth planes go together**. They

construct the adult. The third plane begins the construction and the fourth plane completes it.

Whatever is missed during the construction in a given period of development is never as fully and completely developed later. Montessori said it's like a dropped stitch in development. We can fill in these gaps but never will it be as easily and completely mastered.

The first plane of development is the period of the transformation of the new human being that appears at birth, to the child of 6 who is an individual, in the same way the third plane is a period of transformation. The child of 12 and the young person of 18 are quite different. **What should be constructed during this period is once again an individual but this time a social individual.** That contrasts with first plane construction where it is more the individual alone – self-construction and focus. The **adolescent is focused on him or herself – but it's now in that social context.**

In between those two planes of transformation comes the second plane of development – **a period of uniform growth and development.** There tends not to be those big peaks and valleys – no big transformation that takes place as in the first and third planes. These children have great strength and stability; they can do so much and we can help them. **If their construction has been positive and normal up to this point, we should have an individual who is eager, interested** and capable of doing immense work. He can use his **reason and imagination.** This plane is preparation for the next plane. We must make sure they are fully prepared to handle that next plane.

The **third plane of development begins the construction of the adult.** Montessori talked about the **birth of social man as happening at the beginning of the third plane of development.** At the beginning of the first plane, it's the birth of a new human being. Now at the third plane, it's like another birth, the birth of a new human beings in society. Hopefully, at the beginning of the third plane, we have a young person who has adapted himself to society. He has been able to put himself in contact with society, has been able to explore that society and has built up a security within it. This is why at the second plane the children need “going-out” - to explore that society. If the children have all the opportunities we've spoken of at the second plane, they should arrive at the third plane ready to enter that next level of development.

Montessori tells us we only have to meet the mental interests of the child – not to force them. In other words, we don't make the interest for the children, we only help to develop it. This is why she speaks of sowing the culture – as much as possible – at the second plane. The child arrives at the third plane “holding lightly in his mind the seeds of interest which will be capable of later germination, as the will becomes more directive.” **This is the time of the third plane – when the will becomes more directive.**

So we have this adolescent, this “neo-nato” (newly born). She says we may find these **psychological characteristics emerging: doubts and hesitations, violent emotions, discouragement.** The psychological characteristics manifesting depend on the child's earlier experiences and to the adult's responses to the unfolding child in the first and second planes of development. That doesn't mean just parents that can have a positive or negative influence. It's also the rest of the adult work that determines this

impact. It's important to have caught the word “may” - in these assertions. They don't have to appear and won't always. Or they may appear in varying degrees.

She says at this **third plane there is a decline in intellectual capacity**. Montessori observed this a hundred years ago; these findings have been proven clinically today. She said there needed to be a completely different type of educational approach. These young people are going through **great changes in their bodies that must be coped with**. It is **not the time to lay stress on the brain – as traditional education does**. Today, 6th graders have statistics! Montessori said we should take away all academic pressure – not the possibility for learning – but merely the pressure. She was speaking in particular of the years 12-15 – the first sub-plane of the third plane.

This is one reason why no where in the world have Montessori's ideas for the adolescent been fully implemented. The nearest approach was probably done in Holland around 1950. Her ideas are still considered revolutionary. Her ideas for the adolescent cannot be implemented unless children have come through the first and second planes and been given all the opportunities possible there; a rare circumstance.

Some of Montessori's ideas regarding the third plane. We have an individual who has constructed herself and explored society by participating of the societies of the classroom and the outside world through going out. We now need to take this constructed individual and help him make another construction of her own individuality – that of a human being serving his community. **Because of the doubts and hesitations that often emerge at this plane, he is particularly sensitive to criticism; so we must help the young person without criticizing the construction he is engaged in**. Montessori said, “It is just because this is the time when the social man is being created but has not yet reached his full development, that many defects in adjusting to social life take their origin.” It's like the psychological development of the young child in the first plane – who retreats into a fantasy world because he cannot bear the real world. Adolescents will do the same thing but their retreat is much more frightening; it involves depressions, alcohol and drugs and suicide. They can make a horrendous fantasy world for themselves.

Montessori said the vital problem in educating the adolescent is the education in societal adjustment. So this is where our energy should be focused – helping the construction of a social human being. The young person needs and wants to make his own stronger construction of himself - with the **ability to evaluate himself**. He needs to be able to recognize his strong points and his weak points and he needs to be able to take that constructed individual into society. These **young people need to be exposed to those qualities of humanity which will show him the nobility of humanity** – the heights to which human beings can attain.

Montessori has put forth her ideas on adolescents in her book “**From Childhood to Adolescence**” and this was really just an outline of her ideas. She said that others would have to implement them – as her life was too short by this point. Her ideas about the first plane were implemented in 1907. From the 1920s onwards, her ideas for elementary children have been tried out and implemented. But she didn't have time to develop and try out her ideas for the adolescents. She did some exploratory work, especially with her own grandchildren. Mario jr. - her older grandson, used to talked

about ideas she implemented with him and his companions – particularly ideas about drama.

In Holland, as early as the late 1920s, experimental Lycaea (high schools) were tried, based on Montessori's ideas. This was again, not a full implementation of her ideas. Around the world today, there are schools and programs making an effort to implement at least some of her ideas, some closer than others. The name for the program that she envisioned is **Erdkinder** (earth children).

Erdkinder is a term that should be reserved for a full implementation of her ideas – so it is not generally used. Montessori's **Erdkinder is for the stage 12-15** and it's based on the boarding-school approach. She felt that after 15, the young people could return to their communities to continue their education and prepare themselves for their personal vocation. Some would enter high school, others apprenticeships, others tutors and so on.

In “To Educate the Human Potential” she said by the age of 12, children who had gone through one of her schools should know at least as much as the average high school graduate. She felt that during the years 12-15, they could be “put out to pasture” in a sense – sent to the country – where the academic pressure could be removed. They could have a freer life at this boarding school in the country. They would be expected to work. But **much of this work would be physical work – not just working the brain.**

The **Erdkinder is a school of experience in the elements of social life.** Preparation in most schools centers on a career path – but she was more concerned with social development. She recognized that what adolescents need is work – manual as well as intellectual. So that they could develop an appreciation for both types of labor. Both are essential to civilized societies. **All forms of work have their own dignity and deserve respect.** All forms of work should be thought of as service to society, therefore all workers should be treated with respect. Because of those emerging characteristics of doubts and hesitations, Montessori recognized that young people need a **greater degree of independence and this would help build their self-confidence. Therefore they need the possibility of beginning to become financially independent.** They need opportunities for work in this regard, to earn, budget and spend their own money.

Because of the physical changes in the third plane, Montessori saw a need in the adolescents for peace and quiet. That is why she envisioned this boarding school in the country. The country provides peace and quiet and also allows for the different kinds of work she saw as important. The young people should have the opportunity to work on the land as a much deeper introduction to the world of nature and agriculture – and to the beginnings of civilization – with its origins in agriculture. Agriculture is also essential to the well-being of humanity. We depend on it for the food we eat. An Erdkinder should allow a social experience. In this boarding school setting, they would live as a society, with living quarters as a sort of land-children's hostel. They would take care of where they lived. They would **do the cooking, cleaning, laundry, shopping, budgeting. They would learn about finance and other skills to become independent adults.** There would be adult chaperones, ready to assist, sometimes actively but other times passively. They would serve the adolescent community to help it function. Montessori thought this

hostel should be away from the parents – far enough away so a parent could not just drop in, far enough away so the children from local schools also could not just drop in. But Montessori did suggest that as well as the hostel itself, there should be a guest house so that parents could come to visit by prior arrangement and be taken care of by the young community. Imagine the studies in commerce and economics for this Erdkinder community!

This place in the country would **provide opportunities for growing crops and following agricultural pursuits. The young people should also learn crafts and then the crops and the crafts could be sold in a market.** So she didn't envision the Erdkinder in the middle of no where – though remote.

Montessori suggested a plan of studies for third plane people as well. There should be **provision for self-expression in music, languages and art.** There should be provisions for developing the foundations of character and ideas of morality and values. The individual realizing his responsibility for his own acts and values. Mathematics, particularly in the sense of economics, languages – not one's own but those of other societies, the literature and spirit expressed by others were important. They should be able to become **fluent in speaking at least 1 other language.**

The **study of the earth** and living things should continue. Human progress, the building of civilizations through science and invention – she felt these would help to make the children realize the power residing in the mind's of human beings. She wanted each Erdkinder to have a museum of machinery for the young people to explore, so they could **put machines in their proper place and accord them appropriate value** – tools that serve humanity, not things that control humanity. Linked to all of this is the study of social life – of people before and after a certain invention. What was society like before the invention versus after? Has this machine benefitted the social welfare of humanity? Or, has it been a detriment? This study is linked to study of humanity as a whole. Where have we gotten to from our beginnings, with our inventions? Where could we have gotten to if we'd made different decisions. **What individuals and societies have made contributions that have benefitted all of humanity?** Connect studies to people and adolescents will be interested; this was the advice of Ms. Stephenson and it has been confirmed experientially. There should be individual study for each young person in an Erdkinder – according to that person's interest.

Some **practical considerations** are useful here. There need to be a **house-mother** and a **house-father** to look after the general well-being of these young people. They are ready to give both physical and emotional support as it is needed. Then there need to be teachers for daily life firstly, who live at the Erdkinder, to act as tutors for baking bread or making a budget as circumstances require. Then, there need to be non-resident specialist teachers who come and go. There need to be **technical instructors** such as a farmer and craft people – if the young people will learn agriculture and craft work. Then is also a shop or market which needs to be set up for sale of product. There is lots of money involved in such a project.

Montessori said, “The whole life of the adolescent should be organized in such a way that it will enable him, when the time comes, to make a triumphant entry into social life, not entering it debilitated, isolated or humiliated, but with head high, sure of

himself. Success in life depends on a self-confidence born of a true knowledge of one's own capacities combined with the many-sided power of adaptation, in fact, on what we call valorization of the personality.” The project closest to her vision is the Hershey Farm School in Ohio started by David Kahn.

Cosmic Education

Dr. Montessori conceived of the years 6-12 as the years for providing a cosmic viewpoint. She called this a 'cosmic education.' As long ago as January 6, 1936, in London, Maria Montessori gave the first of a short series of lectures, which were an extension to the 21st international course that she gave. These extension lectures were directed toward the psychological and educational needs of the older elementary age child.

Up until then her work had been with the younger children, although as early as the 1920s, there were Montessori elementary classes being conducted in Holland, particularly those conducted by Mrs. Joosten. It is in her classes, that Maria Montessori carried out much of her observations and experimentation with materials. It is also Joosten's son, Abs, who trained in India and moved to St. Paul MN, working to the establish the Montessori method there.

In the lectures of 1936, Maria Montessori began to talk about the older children and their educational needs. In the fifth lecture she said, "We have continually repeated that the child has revealed to us in a clear and human way that there exists within human nature an impulse toward work. And he has shown that upon the circumstance of this impulse, depends normality or its absence."

In 1938, Maria Montessori gave a lecture on the four planes of development at an international Montessori congress in Edinburgh Scotland. This was the first time her thoughts on the four planes of development were expressed. That talk is available in pamphlet form from Association Montessori Internationale.

In this lecture, on the first plane of development, Maria Montessori talks about psychic deviations of the child – that is, a deviation from the normal path or pattern of human development. In the first plane, she attributes these deviations to the child's deprivation of the possibility of doing his or her own personal work – a necessary and vital activity that builds up individual independence through real and personal experience.

From her observations she went so far as to say that deviations can be gotten rid of and cured in the first plane of development if the child is given an ordered environment in which he can be actively engaged and work for himself and given work he can do by himself. What is important to realize is that through Maria Montessori's observations, she realized and brought to our attention the fact that unless these deviations are addressed in the first plane of development, they tend to re-emerge in the third plane, more solidly fixed and almost impossible to remediate at the third plane. Looking at our adolescent population, we see this is still likely true. During the second plane deviations are less pronounced, less obvious, because they are masked due to the growth and emphasis in the social realm. This does not mean deviations disappear but rather they become less observable and manifest. Teachers at the second plane should be

aware of this.

One of the important things to remember working with second plane children is that if the child has had the possibility of working at independence in the first plane, one has to take care that the child does not lose that facility at the second plane. Adults can actually become an impediment to further development. Adults have to continue to give the child the possibility to do the kind of work that enables the continued construction of himself. At the first plane, we meet the child's psychological characteristics.

In this same lecture of 1938, she went on to say with respect to work, that "In traditional education, the whole system is normally based on two faculties of the human being: memory and abstraction." This is still true today. She said instead, since the mind of the child in the first plane has already been exposed to some aspects of culture through the prepared environment and the working of the absorbent mind, he is already able to absorb many facts about the world. Therefore because of the psychological characteristics at the second plane, the child tends to search out the interrelations of these facts.

At this stage, it's now the functioning reasoning mind that can grasp and understand these facts. It is work that will continue the process of normalization for the children. And that work is going to need to be carried out through reason.

In speaking about this idea, Maria Montessori explained what she meant by reason: "Reasoning keeps things in their relations to one-another, compares them, deduces from them, arrives at conclusions. And when the conclusions have been arrived at, there is a state of psychic satisfaction, and with this, a sense of calm and repose."

In another lecture given at the University of Amsterdam in 1950, she also talked about the reasoning mind of the older child and she said: "In this intellectual period, the child's questions are innumerable. He wants to know everything. His thirst for knowledge is so insatiable that generally, people are at their wits end about it. Therefore, they mostly choose the easy way and simply force the child to be silent and to learn only what we grown-ups consider useful for him. But in doing so, we also destroy his spontaneous interest.

Learning then, becomes a tedious and tiresome business. The result is all sorts of deviations in the child's personality. It should be realized that genuine interest cannot be forced. Therefore, all methods of education based on centers of interest which have been *chosen by adults* are wrong. Moreover, these centers of interest are superfluous for the child is interested in everything; a global vision of cosmic events fascinates the child. And his interest will soon remain fixed on one particular part as a starting point for more intensive studies. As all parts are related, they will all be scrutinized sooner or later. Thus, the way leads from the whole, via the parts, back to the whole. Thus, the child will develop a kind of philosophy which teaches him the unity of the universe. This is the very thing to organize his intelligence and to give him better insight into his own place

and task in the world - at the same time, presenting a chance for the development of his creative energy.”

This quote forms the basis for the teacher's work at the second plane. They will provide keys to the children and then leave them free, as long as they keep up with what they need to know by age nine (or by age twelve) according to the public system's standards; this is the one reason to intervene in their work. Teachers need to keep this idea in their minds at all times: we cannot fabricate interest for the child. We can only give the possibility of the child's interest developing within him. We have heard about, at the first plane, how the teacher is asked to present material over and over again to the children – until the child is at the point to begin repeating and experimenting by himself, based on a spark of interest and a minimum level of readiness. Even if done ten times for the same child, we keep offering essentially the same material. We never know when the right moment is for interest to develop.

At the elementary level, the same principle holds true. We have the cosmos and the furnishings of the universe. We have to present these to the children but not all children are going to be interested in all of these aspects – and certainly not all at the same time. The presentations have to be made so that the child has the chance to become interested. This is what is meant by – we have to sow the seeds.

This idea of cosmic education came about in the same way that Montessori's ideas for the young child came about. Through her observations of psychological characteristics, Maria Montessori developed ideas that responded to them. She recognized that at the second plane, the children were also interested in learning through using their reasoning minds and imaginations. So it is as if she began to ask herself, what is the link that holds the cosmos together? What is it that makes the interrelationship between all elements of the universe and the life forms that inhabit the world? What keeps the stars in their courses? Could this unknowable backdrop bring human beings to unity and peace?

As a result of this thinking, Maria Montessori began to speak of order as the base of culture. If it's the law and order of the universe, the law and order guiding life within the universe, and the very order of life, how do we begin to present this idea to the child at the second plane of development and how can we show order as the base of life and culture? What are the factors that we have to represent? How can we begin to show the interrelation of all things to do with the universe and its creation. It's furnishings, with plants and animals, with the arrival of human beings and their continuing story throughout time. This is cosmic education.

Cosmic education is not just a different scheme of education that mixes subjects together instead of separating and classifying subjects. Though teacher albums are categorized, their presentation is as one interrelated subject. We must present how things fit together and what relationships exist. One integrated subject: this matches our consciousness. Teachers must think about these connections for themselves. It is only through the teacher's ability to link things together that the children will also be able to

relate things in their lives. Doing history involves language. Writing about biology involves language. Essays on any topic should be written grammatically and spelled correctly. Topics will often have a geological and biological significance even though the primary topic may be history or language. It is important these things are presented as such, not as a biology lesson or history lesson per se. Teachers are only concerned that by age 9 or 12, everyone is up to speed on meeting the standards of the community. This is a social pact with society.

We also have to take care, because we are obviously products of our own educational backgrounds and experiences, that we don't think that our presentations are things that the children have to memorize. We cannot rely on feeling the security of asking the child to show what they have memorized. We have to help the child to understand and evaluate themselves for good reason, not just to meet the standards of another. We need to help them realize they are capable of doing things to educate themselves. If we keep the idea of responsibility in mind, the idea of grace and courtesy, respect for others, one should realize this does not mean we expect the children to brag about what they are good at, flaunting how great they are at something. It also means we don't want the child to feel badly about what they still need to work on. It means an honest appraisal of one's strengths and weaknesses – one's abilities and those one is still working on. It is this honest appraisal that will help children use their talents and gifts and ask for help with their weaknesses, with things not yet developed. This is what true responsibility means; self assessment in a supportive framework. This is what we're working toward with the children.

We elicit the child's cooperation in learning by providing him with what arouses his interest, because it corresponds to his psychological characteristics. We need to remember that at the second plane of development, this means we will not arouse the child's interest by giving him unrelated items – asking the child to memorize a list of facts, or asking him to complete a worksheet or workbook. These are unrelated items of interest. They are not part of cosmic education. Unfortunately, there are many who still don't understand this. In Montessori classrooms, people have not understood this and still use worksheets and workbooks. It is for security and ease of the teacher that this continues. But it is not cosmic education.

Instead, she recognized that what the child needs is the idea of the universe, because it is the magnitude of that idea that can develop admiration and wonder. Because the children have the characteristic of feeling admiration, of looking for heroes, we have to give them that possibility of finding heroes and let him wonder about those heroes. It is in this way that the elementary child comes to be able to realize the wonder of the universe, the wonder of life contained within it, and to wonder about the part that human beings play in the universe. The teacher's task at the second plane is not to teach a syllabus. It is there for our guidance only. Instead of a syllabus given in little sections, we have to think of this as each presentation opening up a new field for exploration, an exploration that is now carried out through the reasoning mind and imagination. The field now explored includes the universe, its furnishings, life and functions.

The child who has already gone through the Casa dei Bambini and has been exposed to the whole of the world through the prepared environment, should already have a sensorial foundation of culture. And that foundation is necessary for the child's next stage of development. The foundation given in the C. d. B. is indirect preparation for cosmic education which the child undertakes from ages 6 to 12. We have to think in terms of cosmic education as related to the development of human beings within the universe, and the preparation of the environment that has allowed for that development. And then we have cosmic education as related to the psychological characteristics of the child on the second plane. Cosmic education is not just a new approach to the different subject areas. Cosmic education has to be thought of as that form of relating the child to the universe and to humanity in such a way as it will enable him to understand the law and order underlying their existence.

Having been exposed to those ideas, the child at the second plane may realize, in himself, all the developmental potential which is his own particular birthright. And also accept personal responsibility for this. In "To Educate the Human Potential" one reads, "Not in the service of any political or social creed, should the teacher work. But in the service of the complete human being, able to exercise his freedom, a self-disciplined will and judgment, un-perverted by prejudice and undistorted by fear." This is where the child should arrive, at the end of the second plane of development. In the introduction of that book she says, "The average boy or girl of twelve years who has been educated until then at one of our schools knows at least as much as the finished high-school product of several year's seniority. And the achievement has been at no cost of pain or distortion to body or mind. Rather are our pupils equipped in their whole being for the adventure of life, accustomed to the free exercise of will and judgment, illuminated by imagination and enthusiasm. Only such pupils can exercise rightly the duties of citizens in a civilized common-wealth." Here, in her comparison of the high-school product, she does not refer to the number of facts per se but rather a seriousness of attitude about self-development and an understanding of the organic relationships of all things in the universe and having a feeling of affinity for them.

As one learns about cosmic education, one must remember that it is not a program of learning that we are after but rather using that program to help the children think and reason and understand their place in the universe. Only then can they become what Dr. Montessori alludes to as cosmic agents. It is very difficult to change adults. Maybe we can change future adults by changing the children, by helping them become something different than we create them as today.

If we think of ourselves as teachers, we must ask what is entailed for us in this plan of cosmic education. How can we be released from the idea of subject matter and curriculum? How can we make sure that we are not teaching subject matter, but rather exposing the child to a certain knowledge of all that has to do with the universe and with human nature.

Again, quoting from page 11 of "To Educate the Human Potential", Dr. Montessori said, "The secret of good teaching is to regard the child's intelligence as a

fertile field in which seeds may be sown, to grow under the heat of flaming imagination. Our aim therefore is not merely to memorize, but to touch his imagination so as to enthuse him to his inmost core. We do not want complacent pupils, but eager ones; we seek to sow life in the child rather than theories, to help him in his growth, mental and emotional as well as physical, and for that we must offer grand and lofty ideas to the human mind, which we find ever ready to receive them, demanding more and more.”

Teachers must remember to use the characteristic of admiration in the second plane child. To elicit those responses, we have to give the child something to admire, something to look up to. At the second plane of development, we also have to prepare the children to go out into society, out into the world, to begin to take his place in the society of the world. In order to do that, the child has to know how he fits in to that picture. What his place is in that society. We need to help the child realize that before human beings arrived on earth, the universe had to come. Within that universe, the earth had to be formed. And then onto the earth, had to come life. It was last of all, that humanity arrived. When they came, it was into an environment that was prepared in such a way that they could meet their fundamental needs. They could find a way to sustain their lives. That drama, that is the life of the universe, is a type of drama in which the prepared environment was waiting for that last great actor to come and take his place.

It was human beings, when they came, who manifested a special role in the drama of the universe. All life that appeared before human beings, had its continued existence bound up with the type of life it led. The continuance of its life depended on it following a certain set of rules and orderly approaches, patterns or certain cycles. Then we come to the advent of human beings. Human beings were given the possibility of making their own way, of falling down and rising up again – and of achieving formerly unimaginable feats by their own effort. The environment was there for human beings just as it had been there for all other life. But the other life forms were endowed with instincts and it was those instincts that caused them to know what to take from the environment in order to survive – but nothing more.

Human beings had been endowed with two special gifts: intellect and will. With those two special gifts, the power to know and the power to love, and from the environment which was prepared in such a way as they could use them, human beings were given the task of constructing their own pattern of life. So unlike the other animals and plants, human life had to experiment with food and find what was good to eat and what was poisonous. They had to invent and make their own clothing. They had to imagine and make their own shelter. To help them do these things, they had a special new feature of nature to use: the hand with the opposable thumb.

Something else came into existence for humanity also. When the physical needs had been met, it seems humanity was still not satisfied. They weren't able to just accept that first way of life they made for themselves. They were driven to keep extending that life in all directions seeking more and better conditions. Therefore there arose a need to learn.

At the second plane of development, the child has to be given the chance to explore how the universe came into being, and how the earth was prepared in such a way that humanity could initially survive. Because this will lead the children to find their own place in their own epoch within the universe. This is what will provide unity to the child's intellect and empathy for other people of the world. As the child begins to work and explore and research, he will begin to explore all of the paths that lead from this center of unity - the universe and the various forms of life that inhabit it.

As we present the universe to the children and the place of humanity in it, we begin to show the children the cosmic task of each of the elements that make up the universe. That task is the service that is rendered by each to the environment and to the other elements in that environment. The human being's place seems to be that of the agent of change with regards to the environment. We are the species that has, in many ways, completely changed the environment. We've transformed the environment to our liking. But along with that task and that ability comes the responsibility of not ruining it. There are certain aspects we are ruining today. We have to believe, due to our intelligence, we should be able to correct these things. It is the dimension of a sober will that brings the proper seriousness to get us back on track. Helping the child understand this dynamic is part of cosmic education.

In this plan of education at the second plane of development, memorization becomes aligned with the understanding of the underlying process of life on earth. Instead of the attempt to remember particular dates, names and events that are often unrelated within our inner organization, there will only be memorization involved in helping the child to meet standards set by society via the public system's curriculum. For example, math facts are expected to be memorized. Historical facts as well will be required. Hopefully, within the context of cosmic education, they can be presented in an integrated fashion versus an isolated fashion. There can follow various exercises for memorization – though this is not the general approach for Montessori and represents a practical compromise. We speak of order as the base of culture. It is only when things are recognized as belonging to a previously recognized order, that they can be properly understood.

What this means at the second plane is if we want the child to understand the universe and its components, we cannot give knowledge in a haphazard way and expect it to arouse interest in the children. The child needs, first of all, to observe the order of the universe. From that base, he'll be able to recognize a pattern underlying creation. Then he'll see a pattern underlying the life of human beings. It is only if we begin with a base of order that the child can come to a full appreciation of the achievement of life forms and of human beings. This is also how he can recognize and appreciate the service that is offered by creatures, one to the other, and how it is often through the gift of self, that creatures find their utmost happiness. For human beings also, it is through the gift of self, to others, that we're able to find our deepest happiness.

In her first lecture on cosmic education in 1939, Dr. Montessori said that she wanted to illustrate what she was talking about by taking just one small detail of the

earth. She took air and water and the problem or danger of them becoming polluted. She went on to talk about the task of the many tiny animals in the water and the job they have performed throughout the ages, absorbing elements that are indispensable for their lives and at the same time purifying to the water. She spoke of this balance of nature, how there is a danger of pollution, but others working to purify. She said it was like we had opposing forces and a cosmic energy acting within these forces, to maintain life. This is an example of order in the universe.

She also talked about another illustration to put before the child. One of the things we bring to his attention is the beauty of shellfish. She said all of the shellfish living in the sea, might have had the same covering, but instead they each come with a unique covering and they are each beautifully decorated either on the outside or on the inside of their shell. There is always beauty. She said, "It is as if there were something urging one to a greater design but all these creatures obey a fundamental law, a guidance which is in each of them." This is just one element of the creative universe. There is order and beauty, pattern and uniqueness, in every corner that could be presented. This created awe, wonder and admiration in children and is the inspirational aspect of cosmic education.

In her early lectures on cosmic education, Maria Montessori discussed the ways in which simple facts can bring us to touch on different aspects of human culture. For example, the consideration of rocks and geography can lead to studies in geology and chemistry. The grandeur of plant and animal life can lead to studies of biology. Currents in the air or water can lead to studies in geography and weather and climate. History is the story of human beings and their achievements. History is at the center of cosmic education because it is the story of humanity and the universe.

The understanding of our place in the universe depends on the understanding of the law and order which govern the elements that first created the universe and which then created our world, for example, the laws and order that governs the lives of plants and animals and the part they play in conserving that order today.

We also have to recognize the gifts with which human beings have been endowed: intellect and love. And we need to understand why these gifts were given. It is this great drama of humanity, not isolated from the rest of the created universe, but part of it, and with an important and significant role to play in it, that we need to emphasize for the children in our elementary classrooms. The universe and its elements and the plants and the animals are not just things and means available to us to exploit.

Through the Great Stories of the coming of life and the coming of human beings, we can help the children realize that all created things are one with us. Because they are one with us, they need to be treated with care. That is the gift to human beings which translates into being able to desire the welfare for another. When we speak of this gift of love, we mean this ability to choose goodness for another.

Through the eyes of the children, to whom we unfold cosmic education, we

adults can have our eyes opened once again, to the grandeur and majesty, to the pageantry and glory of the cosmic plan. This is a plan in which we are all called to participate. We as adults can begin to realize the interrelation of all universal elements, the interrelation of the stars and the planets, of matter, liquids and gases, of plants and animals, of men women and children. It is through these Great Stories of the truth, which relate the cosmic plan, that we introduce to the children, the idea of respect for all of us, respect for one another.

This is a message that has to go out to all of the children, to those in our schools and elsewhere. To the rich, the poor and middle-class. Unless we can get this message across to the children in the elementary classes of our Montessori schools, it really won't matter or makes much difference what other messages we do give them. To stretch their intellects, to make them classic scholars or budding scientists, to form mathematicians, to give them academic excellence, will have no true value unless they are also integrated, committed, respectful, loving, caring individuals. This is the theme and goal of cosmic education.

What we call subjects of the curriculum, enter into this theme because they have to do with the totality of the cosmos, not because they are the most important subject which have to be taught, or that the children need to learn something about. What is geography but the story of the earth and its coming into being - through the following of the laws given to the substances from which it is formed? What is biology but the coming of life on earth, to preserve the harmony of the earth and to furnish it with beauty and color and scent and form and variety of life? What is history but the story of the human beings who came to fulfill the cosmic plan, set in motion at the time of the creation of the universe? What is language but one of the stories that humanity has written themselves, the achievement of the discovery of the power residing in the minds of human beings, that those minds are made to have ideas and to communicate those ideas with others? What is mathematics but that other language created by human beings to have a way through formula, some of which are arithmetical, some of which are geometrical, some of which are algebraic, of transmitting their inventions? What is science but an invention of human beings to perfect their material territory, as part of the economy they developed as a way of satisfying their fundamental material needs. What is music but another language to set beside their language of letters and which, along with the letters, have led to song, opera and concerto, to blues and jazz and rock. What is art but still another language; a creation in paint and color, a sculpture in gold and silver, in precious stones, in gems and jewels, of the ideas of the mind of the human being. It is an expression of their spirit. The other element, the body, combines to form the whole human being.

The curriculum is the universe, its furnishings, its inhabitants and their stories. It is not mere lessons in books to be listened to in sequence, memorized, tested and graded. Teachers need to think of these traditional subjects in a new fashion as part of a whole heritage belonging to all humanity. The question underlying cosmic education is how does one present it to the children. How do we induce them to work with the ideas?

As elementary teachers, we must keep in mind that we may need to find ways to work at the primary level to help them more fully understand how the children could be best prepared for the second plane. One thing this involves is that the children should have the opportunity to work with the entire program provided for in the Casa dei Bambini. This means the teacher is introducing all aspects of it, all games and exercises and to the ultimate limit of the child's capacity in each area of study. Those lessons were all developed for the first plane child. If the child misses it, they are ill-prepared to go forward with the elementary work. It is all preparation for the study of the cosmos. Cosmic education can only be built atop what the children have already worked with and been exposed to in the children's house. Elementary teachers must carefully and kindly work with primary teachers to help them to help us all mutually serve the child to our utmost ability.

One hears how, in the Casa dei Bambini, the young child thrives on fact. Because of the absorbent mind, they take in fact, any fact, that they are exposed to. Given this, the primary teacher can be informed about this need to become familiar with the universe, to emphasize those parts of the education that provide the basis for cosmic education. In this regard, a primary teacher may prefer to spend time on the study of life that exists today as opposed to life that is extinct. There are degrees of relevance that one might consider, practically speaking.

Maria Montessori realized that the small child can be put in contact with the world and with all of humanity in the first plane of development. In "To Educate the Human Potential", she said, "What is very necessary is that the individual, from the earliest years, should be placed in relation to humanity."

The language material, the nomenclature material, needs to include pictures that will give all of the different aspects of the life of humanity. That means that these nomenclature cards should not only deal with the here and now, but with the lives of people from all around the globe and throughout the ages, as well as what is of local tradition. If we think of the geography folders in the children's house, many of those pictures have to do with how people met their needs. So in the elementary classroom, we can pick up on that work and relate them to the studies we do about human beings and their history on earth.

The children should come to our elementary classes with a foundation of factual knowledge and with language that they can use from their work in the Casa dei Bambini. With this foundation, they can begin to ask how, why and wherefore. If they don't have that foundation, often the materials in the elementary class become teaching tools rather than providing the basis for the child's reasoning mind, to begin to work toward unveiling one's own identity and value.

The plan of cosmic education gives the adult a very different role from that of a traditional teacher. Generally a traditional teacher is asked to prepare a curriculum and to help the child learn that curriculum. The adult is determining what is important to

learn and then passing the dictum on to the child. In our Montessori elementary classes, the teachers need to be careful not to prepare lessons that cover all aspects of the ideas being presented. We have to always remember that the lessons we prepare and the lessons we give are to just open a door. We give just a limited amount of information. That limited information has to give the children a glimpse of the fact that there is much more to find out. There is much more to come.

So the lessons we give are to give only a little bit - but that little bit has to be of such significance that the child is enchanted by it. The child is stirred up; his interest is inflamed. He must be put into a state of unrest so he cannot wait to find out more by his own efforts. This becomes the elementary teacher's job; to present material in this manner. At the same time, the lessons should convey the idea of interrelation and interdependence of all life – the notion that all parts go to make up the one whole truth.

We also have to remember that concept of limitation. In our elementary classroom, the lessons and their content have to be limited such that they provide keys to the imaginative factor of the human intellect. That faculty is unlimited in its operation unless obstacles are placed in its path.

Cosmic education should make it possible for the child, from time to time, to catch a glimpse of what Dr. Montessori called the cosmic mission of man on earth. It is the glimpsing of that mission which should form for the child the theme of his continuing construction, enabling him to construct himself, in such a way that he can see his possibility of service to his fellow human beings. This is especially important in view of the fact that, from the second plane of development, the children go on to the third plane of development which is, in particular, the plane for recognizing the need for service to society.

It is easy to pick up a negative perspective from the news and so on. Teachers must remember though that children are the future of humanity. Cosmic education is the preparation for individuals who have to live in this world today, with its imperfections and brutalities, its present circumstances. These people have to be adaptive to this time and place and culture. But that does not mean we necessarily have to conform to it. The child should realize he is free, within the time and place in which we live. He can only help affect change if he recognizes that change is necessary. This is what we want the elementary child to experience. He has a power to exert in the world and it is his birthright. It is important teachers realize what Montessori education is. It is about human development in the grand scheme. Remember Montessori spoke about an aid to life – not about learning a certain curriculum.

At the end of “Maria Montessori: Her Life and Work”, the author, E. M. Standing, tells us: “Even were our primary aim the acquisition of culture, we should still be obliged to put development first. For what is the difference between a man of culture and a man who is simply a mine of unrelated information, like a dictionary. The fundamental difference lies in the way in which that information has been built up. In the case of the former, it has been through a vital mental process coming from within, from

the center, set going and kept going by a living interest. That is why the man of culture possesses an organized and ever expanding system of knowledge, all parts of which are connected and ultimately united. They are united because they are presided over by an intelligence which informs and vitalizes and unifies the body. On the other hand, the content of the mind of the uncultured man is simply a compendium of unrelated facts, like a sack full of potatoes, and is so because his knowledge has been acquired mechanically just as you might drop potatoes into a sack and not by a process of integrating growth.”

Cosmic education is this integrated growth available to all people. In time, each is able to come to one's own process of being able to understand it. This is why there has to be time within the elementary class for the children to be able to savor our presentations, ponder our presentations, meditate on the presentations. They need to think about the idea of the universe and all the aspects of reality we present to them.

Dr. Montessori spoke about the importance of what she called - the rest. We should not immediately expect the child to show us what he has learned or accomplished. At the primary level, a teacher may give a great presentation that is well-timed. But the child may not show interest. The following day, interest may engage the children. There has been an interval of time that allows the child to rest and ponder the presentation. At the elementary level, the same idea applies. We should not immediately expect the child to show the concepts we have presented. We must never ask for that immediate response to what we have presented (except superficially). It is after a pause and rest that the child may become excited and curious about the information to the extent they want to go into it further. Sometimes, a teacher can see there is interest at this later time. A delay only matters when children have to conform to the public curriculum in the near term or else when there is a progressive accumulation of knowledge required for mastery. Otherwise, we are not mindful of and not responsive to the level of interest shown for a presentation – beyond our self-assurance that the presentation was well-given. The measure of a Montessori teacher is related to his ability to integrate the ideas from theory into the hands-on presentation of elementary materials. They cannot be separate in practice, though they are learned and practiced separately at least initially.

Date: September 15, 2004

Imagination

There are a few points to be aware of with respect to imagination. Absorbent mind becomes a mind that can reason at the second plane. It is this mind that needs to make use of imagination. The child now has a mind which can think, judge and reach conclusions. Teachers have to always keep this in mind in the classroom. The environment must be prepared with this in mind. Classroom activities offered to the children need to be those that will use a reasoning and logical mind.

In order to explore through his reason, the child has to make use of the power of the imagination. Teachers need to understand this power. Because it is in how well teachers understand and therefore make use of it, that we'll really be able to run our classes in the way Dr. Montessori intended.

It is very easy if we don't understand this power to use the materials as teaching tools or visual aid but this is not quite the idea. Copying and memorizing our materials is not using the reasoning mind. Montessori teachers have often not understood what it means to use this power in the child.

The purpose of the materials in the elementary class is to help the imagination of the child function more fully, so that the child can go from using the materials to seeing something in his mind's eye, that was not there before and to reach his own conclusions and valuations. To understand the formula that results from $(a+b)^3$, is to use our minds to come to a conclusion of something that was not presented to us. What will be presented, is the fact that one can take $a+b$ and carry them to the third power. If one does that, one has the formula for $(a+b)^3$.

What we want the child to understand is this formula and why it is what it is. The mind needs to be able to understand the why of that multiplication exercise. It is the imagination that can help the child arrive at that conclusion.

Before one goes into imagination, one should investigate intellect. Put simply, it is the ability to think. It is called the intellect because it has the power to get inside and to see the inner meaning. It is sometimes called 'understanding', because it can search and probe until it reaches the very core, until it grasps the first principles on which all knowledge is founded.

The first principles are the self-evident truths. The ability of the intellect is also called reason because it can go from a truth, already known, to truths not yet known to consciousness. Also, it can draw out conclusions that lie hidden in the first principles.

The intellect of humanity is the power that we have to do abstract thinking. It is the power that produces our ideas, the power that forms our judgments, that builds up our notions of reality. As well as the intellect, human beings have been given the senses and it is the senses that are able to form images. Those images are material and limited. They are attached to the object which is giving us the impression. For example, the eye can see color but the eye cannot define color as applied to all colors. The eye just sees the color that appears before it.

The intellect came. It is the intellect that can form ideas which are abstract. The function of the idea is to show the essence that goes beyond the qualities that the senses perceive. What the sense perceive tie a quality down to a particular object in time and place. The idea that the intellect can abstract is not concerned with a particular object or time or place. It is not concerned with those details. The idea the intellect can form is universal and can be applied to any time or place.

The function of the absorbent mind, that particular and specific form of intellect present during the first plane of development, is to take in the impressions of the qualities of the world, everything that is perceived through the sense. The absorbent mind then explores those qualities and forms the foundation from which the intellect can draw out knowledge of the universal qualities of the world and the facts of human existence and activities.

As we come to the second plane of development, the teacher is faced with a mind that is not so concerned with those sense qualities but rather is concerned about the reasons for them. Why does the sky appear blue? How do birds fly? How did the world come into existence? Their mind gravitates to these questions.

G. K. Chesterton, who wrote the book, *Orthodoxy*, in the chapter "The ethics of Elfland", wrote this: "When we are very young children, we do not need fairy tales. Life in itself is interesting enough. A seven year old is excited when he is told that Tommy opened the door and saw a dragon. But a three year-old is excited when he is told that Tommy opened a door. Children like romantic stories. But little children enjoy realistic stories because they find them romantic."

To help that reasoning mind operate and understand the facets of the universe, around which is it questioning, teachers have to help the imagination develop. They need to think of and understand the imagination as that ability to picture material things in their absence. In other words, the imagination can form images of things that we are not sensorially exploring at this moment. But the imagination does need previous experience of the qualities; so the previous sensorially is what enables us to do this. In order to work, the imagination doesn't have to have stimuli right now, working on the outer senses. But there does have to have been previous sensorial experience.

The imagination then, can picture any sort of qualities that it's had previous experience with. The imagination also has the ability to draw material from all of the outer senses. It doesn't have to rely on just one or just two, but uses all of them. The

imagination has three possibilities. One is to gather images. Second, is to apply these images in reality. The third is the use those images to make inventions.

The imagination also has two powers which teachers need to assist the child to use. The first is the power of reproduction – to picture things exactly as they are. For example, to picture in your mind an apple. This is one power. The other power is the power of creation which doesn't stop or need to stop at just a faithful reproduction. It follows logically then that the creative imagination has to fall under the control of the reason and will because it carries with it a burden of responsibility. This creative imagination can be for us, either a help or a hindrance, depending on what we imagine ourselves doing. There are many people imagining things that are detrimental to themselves and to others; it is not being controlled by reason and will.

Imagination also shows up in the play that young children undertake. Teachers might see them having an imaginary party, or picking up sticks and using them as guns for a mock battle. Teachers cannot control where imagination leads the children, though play must remain safe. Dr. Montessori said, “Imagery becomes a hindrance when children create an imaginary world because they find the real world too difficult for them. This imaginary world is mostly in the own minds, often spurred by severe neglect or abuse. The child creates a circumstance within the mind that he can cope with. There are degrees of this phenomenon. When a child exhibits unusual behavior, there is typically a background reason that the teacher should try to learn of and comprehend. Another things that happens using imagination is hallucinations. This is an inability to distinguish between fact and fantasy. Often this is brought on by a chemical cause. Dreams are essentially hallucinations, though they are passively experienced.

Teachers also need to distinguish between memory and imagination. Memory is a faculty for recalling past experience and of identifying it as past experience. It is tied down to a time and place and subject to a capacity for recall. Can one picture a jabberwocky? Since it is an imaginary creature, it does not exist. But having been given the name, if asked to draw one, we probably could do so – since we could make up anything that came to mind in response to the name. This is the essential difference between imagination and memory. It is imagination that we want to use with second plane children.

The importance of impressions already gained for the work of the imagination is because if we see with our minds, when we comprehend and understand, it is because we have seen and understood before. The reason we can now understand something new with our mind is because we combined together, in our understanding, images that we've detached from whatever our senses were taking in before. If one says, “tropical rain forest”, what aspects of it does one see in the environment? Not much. But if one says, can you picture a tropical rain forest in your mind, we can identify, item by item, the objects we see in that environment. We can see them, hear them and perhaps smell them according to the ability of the senses and the depth of experience with a tropical rain forest. We are by definition departing from reality though we have a basis in it if we are diligent in reproduction of real experience. This is what the elementary child

needs to be able to do in order to comprehend his universe.

While the child is in the Children's house, he is interested in the features of the world that surround him. This is why in the nomenclature material of the Casa dei Bambini, we can give the child geographical features of the world. The child, in fact, needs to have these things given if he is going to be prepared to work at the second plane of development.

Now at the second plane, the child is interested in how and why the world came to be the way that it is. Why, through the ages, has land in some places crumbled away and in other places been lifted up? How did the land come to be formed in such a way that we can live on it? These are the kind of questions that second plane children want to know answers to. The environment that we prepare has to be one that will help answer these questions. How was the earth created? How can teachers possibly show that idea to children. What we can do is make use of the images that the child has already gathered. We can put the child in contact with various phenomena that brought the earth into existence. We can show what is happening now with the forces of nature. From that point of contact, we can give the child the possibility of reconstructing for himself, using his imagination, what must have happened when the earth came into existence. Teachers, in a way, give the child a guide through what is happening in the here and now, to understand what must have happened in ages past. The presentations in the classroom will serve as keys to some of these ideas. Teachers provide a key lesson demonstrating the work of a river and will talk about how rivers carve their own path to form channels, flood banks, deposit residue in certain places. Then the children are encouraged to go out and actually see some of these things in action in the outside environment. Teachers can then help the child to reconstruct the work of the rivers in the past. If the child has a chance to understand the consequences of the work of water, with the imagination, he can also go forward into the future and see the consequences of the same happenings at a future time. Through the keys that teachers provide and the imagination of the child, the child has the possibility with his mind of traveling thousands of years into the past or the future.

One of the aspects of the mind of humanity that is important for teachers to understand is that it needs to have clear knowledge to be able to see clearly facts and consequences. Presentations must be clear in this regard. Clarity is something that is interesting for the human mind; the mind becomes restless until it has understood. As teachers approach the subjects in the second plane of development, they must remember that they don't want to think of them as subjects to be taught and learned. Rather, they should think of presentations as creating unrest in the mind – which then needs to know for itself. The knowledge put before the child has to be related to their interests in life. It has to be related to their questions, to their seeking intellect – not determined by the interest of the adult.

At the second plane of development, because the imaginative power is so great, the child becomes interested in things like allegory, myth, legend and fairy tale. He is now able to distinguish between these types of stories and factual real life stories. The

children know they are not true but what they provide is a vision of activities that is not possible to carry out just with human power. This is where the attraction to dragons, elves and so on come in. It is the realm where humanity cannot properly access or enter. The reason we are so drawn is because we receive a vision that is beyond what human beings around us can do. It brings a new sense of power to explore.

It is because of the imaginative power of this age that Dr. Montessori said we need to give these children a vision of God and a vision of humankind. Present the universe, its furnishings, the life of humanity throughout the ages, in such a way that the child is able to understand the wonders of the ways of creation and also the enormous efforts of humanity that came before us. Because of the characteristics of hero worship, compassion and sensitivity toward justice, along with the active imagination, teachers have to help the children realize how God has created and maintained order, by putting directives through the unconscious, through the urge of life, the urge to become – not only in human beings but also in rocks, air and all of the forces of nature. It is through obeying that unconscious urge to carry out specific directives that comes about the equilibrium of the universe.

In order to aid the imagination, teachers need to first of all, give the child a vision of the whole. From that whole, we can proceed to the details that can help classification and organization – things the intellect need to build itself. Teachers must always remember the characteristics of loyalty and admiration in this plane of development. The elementary teacher must appeal to the spirit of the child instead of just to the senses as the first plane teacher does. The eye of the seeking spirit is the imagination. We can no longer just present a material to the child and ask him to repeat it as we do at the first plane. The lessons have to change to accommodate the new child in the second plane.

The first presentation is often not to introduce a material to the child but rather it is to open the door for the imagination to begin to explore what is on the other side of that door. Teachers still give keys but now the keys need to serve that imaginative exploration. In the stories that are given, teachers make use of magic, allegory, personification and so on, in order to give that key, in order to help the child realization through imagination, what the sense can not take in. It is for this reason, because of this vast imaginative power, that Dr. Montessori said we can present to the child all items of culture, not as subjects, but as seeds to be sown. This is why she saw the need for a special method at the second plane of development, whereby “all factors of culture may be introduced to the six year-old, not in the syllabus to imposed on him or with exactitude of detail – but in the broadcasting of the maximum number of seeds of interest.”

By sowing seeds of interest and giving the child an enormous field for exploration, teachers also set free his capacity for imagination. By giving the child the possibility of coming in contact with what others have done, and with what others have created, the child has the possibility of creating himself under his own terms and in his turn.

In Jacob Bronowski's book, *The Ascent of Man*, he says several things about the power of the imagination. "There are many gifts that are unique in man but at the center of them all, the root from which all knowledge grows lies the ability to draw conclusions from what we see to what we do not see, to move our minds in time and space." Elsewhere in this book, he refers to imagination as "a telescope in time." He further says, "What we call cultural evolution is essentially a growing and widening of the human imagination." In another of his books, *The Visionary Eye*, he says, "What makes the biological machinery of man so powerful is that it motivates his actions through his imagination. It makes him able to symbolize, to project himself, into the consequences of his acts, to conceptualize his plans, and to weigh them one against another as a system of values. We as men are unique. We are the social solitaires. We are the creatures who have to create values in order to elucidate our own conduct so that we learn from it and can direct that refinement into the future." Dr. Montessori said, in *To Educate the Human Potential* (pp.10-11):

"It is along this path of high realities, which can be grasped by the imagination, that the child is led between the ages of six and twelve. Imaginative vision is quite different from mere perception of an object, for it has no limits. Not only can imagination travel through infinite space, but also through infinite time; we can go backwards through the epochs, and have the vision of the earth as it was, with the creatures that inhabited it. To make it clear whether or not the child has understood, we should see whether he can form a vision of it within the mind, whether he has gone beyond the level of mere understanding.

Human consciousness comes into the world as a flaming ball of imagination. Everything invented by man, physical or mental, is the fruit of someone's imagination. In the study of history and geography we are helpless without imagination, and when we propose to introduce the universe to the child, what but imagination can be of use to us? I consider it a crime to present such subjects as may be noble and creative aids to the imaginative faculty in such a manner as to deny its use, and on the other hand to require the child to memorize that which he has not been able to visualize. These subjects must be presented so as to touch the imagination of the child, and make him enthusiastic, and then add fuel to the burning fire that has been lit.

The secret of good teaching is to regard the child's intelligence as a fertile field in which seeds may be sown, to grow under the heat of flaming imagination. Our aim therefore is not merely to make the child understand, and still less to force him to memorize, but so to touch his imagination as to enthuse him to his innermost core. We do not want complacent pupils, but eager ones; we seek to sow life in the child rather than theories, to help him in his growth, mental and emotional as well as physical, and for that we must offer grand and lofty ideas to the human mind, which we find ever ready to receive them, demanding more and more."

Observation Guidelines

Be non-intrusive and non-participatory. Be sensitive to the teacher's ability to speak with you (or not) throughout the day. We have to build up the ability to watch the whole class at once, while at the same time giving a presentation or watching various individuals.

For observation notes, use some sort of notebook (no laptop computer). Make sure notes are legible enough. No re-writing is allowed. We will submit our notes plus a written summary.

In the notebook, write at the top of the page the name of the school, name of teacher (classroom) and the age-level of the class. On the front, write your name and contact information. A lost notebook means those hours must be repeated since the notebook is considered as part of the AMI diploma requirements.

You will take timed notes. Write the time at least every fifteen minutes in the right margin. Every change of activity will be noted along with the time the change occurred. Sometimes, if the activity has not changed, still record the fifteen minute interval and note activity continued.

Get into the notes as much as is possible, a complete description of everything going on. When you don't know names of things, you can write "five kids working in a group with some kind of map", for example. This is okay to begin with.

Look for characteristics of planes and the tendencies in action. Use a code in the notes to document their appearance. e.g. "C" means child communicated.

With tendencies, note whether the child was helped or impeded in the environment. Are characteristics of the plane being responded to or not. How is the child effected?

Notes should be what you see or hear, not what we think or feel about what we see or hear. It is scientific observation without commentary that we strive for. We want fact only in a non-judgmental manner. Be as objective as possible.

If there is something we don't understand, see if the teacher may be able to offer insight into the circumstance. Speak to MIM staff if needs be, to clear up questions.

Tell the child you are doing your work, if they approach you. Suggest they do their work if they attempt to hang out with you. Be elusive with eye contact if needs be.

Notes should be as complete as possible: how many children, what they are doing, what they are using (materials), what teacher was doing. Hand in whole notebook w/summary.

Date: September 15, 2004

Great Lesson: Creation of the Cosmos

This story is the whole into which all the parts fit. We can use this hand-out or rewrite it to fit into one's own world view. We need to turn in: the handout we're given with an indication of exactly when each demo was done and when each chart was shown.

For albums, we need to include illustrations of each chart. We can sketch them roughly or else take pictures. Allyn also has small versions of them to borrow and copy if desired.

The purpose of the illustrations in the albums is to make preparations for the lesson easier. Color my own charts using the full-size charts – *not* the small images of Allyn.

Never speak when displaying a chart or performing a demo. Make speaking separate. Each demo should be on its own tray.

Turn in the “God with No Hands” handout with demos and chart numbers incorporated into the handout.

On the new charts we receive, we can place a number at the top to identify it quickly and to keep them in sequential order.

This demo is not to be repeated by the children; normally they can perform the demo in their own. For their repetition, using a vinegar (added last) and baking soda version (mouth of the volcano).

Assignment: Use story handout (rewrite if we wish). Copy it into geography album, adding annotations for when charts and experiments are performed. Turn this in along with an outline of the charts and experiments that are referred to in the story write-up.

October 29, 2004

Great Lessons and Key Lessons

Note this will likely be covered in final exams

This will review some of the theory we've done and bring in additional points to crystalize our understanding.

We will remember as we talked about the second plane having reasoning minds and that their explorations need to be carried out through reason. We also talked a lot about the kinds of work we want the children to undertake at the elementary level. Therefore, the work of the teacher with regards to all of this, has to be that of opening a door, so the child's explorations and reasoning and understanding can commence.

What do we mean by opening a door – and into what are we opening a door. In relation to that, we need to remind ourselves that Dr. Montessori said, “Trust to the spontaneous organizing power of the human intellect.”

This is something we need to keep reminding ourselves of, both in preparing to work with children and when working with them; trust the organizing power of human intellect. This is when the child's intellect is at its greatest power and has its greatest clarity. We have to take care not to waste these years for the child. And we waste them when we give the children little bits of work to do. Dr. Montessori also reminds us, “The mind works by itself, moved by a power within.”

What we remind ourselves there is that if we are working with children, we need to get that movement started, provide the incentive. A car doesn't start by itself; one must turn the key and start the engine. In the same way, the child's mind has to be given something to catch onto, something to work around.

At the second plane of development, there's not as much material emphasized as the first plane of development – but rather it's an imaginative incentive that will provide the material and get the mind going. Our job is to give something around which the imagination can begin to work.

Also with regard to how we work with children, Dr. Montessori said, “To sow the seeds of culture.” She didn't ask us to keep track of which seeds were sprouting or to dig them up to see which ones were growing roots. In “To Educate the Human Potential, she says, “The task of teaching becomes easy since we do not need to choose what we shall teach, but should place all before the child for the satisfaction of his mental appetite.”

If our task is to open that door, how do we entice the children to enter in through that door, to begin a tour of exploration? It was for this purpose that Dr. Montessori gave us something very special. She became a story teller of the truth for elementary children. They're asking for reality – but one that can be explored with their imaginations and that then can bring the children to an understanding of essential truths. So it is to get this exploration started, to open those doors, that we have what are called “The Great Lessons.”

These are special lessons and yet even that word 'lessons' is in a way a misnomer. The Great Lessons are not to be learned per se. So they are not lessons in the

usual sense. We should think of them instead as prologues to a drama, as an introduction to a scenario, as curtain raisers. If we can think of Great Lessons in those ways, then we'll be able to use them to excite the children and this will set their minds on fire.

Each Great Lesson is a prologue in itself. Self-sufficient. But it opens onto a play of the drama of the universe, the furnishings of the universe and of human beings.

In its role as prologue, each Great Lesson sets a particular stage. It introduces specific actors and actresses. And it gives us some indications of the parts that these actors and actresses are going to carry out on this particular stage.

We start this search and exploration by putting the child in touch with the fact that in the beginning there was nothing. And then the universe and the world came into existence. Some of the actors introduced to the children are the minerals and the chemical elements. Some of the actors we've introduced to the children are fire and air and water.

You've seen how we introduce these to the children; we introduced them to you in the same way. Other actors are the plants. So all of the plants are actors on the stage. Some of the actors are the insects, birds, fish and mammals. And then onto that stage that we've established comes the last actor to arrive – human beings.

Just as in any stage set, any prologue, any set of actors in the theater, with all of these actors came law and order, pattern and sequence. They each came in their own time just as in the real theater. An actor or actress has to wait for the particular cue that now it's time for humanity to enter and take its place on the stage.

So we have those same patterns to introduce to the children with all of these actors of the universe.

Human beings brought their own special characteristics, features and gifts. They play out for us in the scenes of their drama - their particular actions upon the environment within which they begin to live. And then they write two plays that are particularly their own; they wrote them themselves. Plays that spring from the intellect of human beings and plays which show the great potential of the human being.

When we did early theory, we said the **first 3 great lessons are to help the children feel gratitude to God**, the guiding unconscious, to the cosmos, underlying law and order – whatever we feel is responsible for the universe and its balance. The **last two Great Lessons help the children feel gratitude to human beings of the past**.

After the creation of the universe and the setting into action of chemical substances, came life. Life had its own particular pattern that had to be followed as the world became furnished with plants and animals, each of which had its own particular actions to perform and each of which took its place in service to the rest. By doing this, each has made its own unique contribution to the overall drama and viability of life. When the time was right, humanity came into this picture.

With their coming a new drama began, the story of humanity. Human beings were a new type of creature. They were endowed with certain traits that make them different. But it's important for human beings to always remember that all of those prior agents were necessary and important. They carried out preparations so that we could survive and all other life could survive. And they continue to be important because other life forms depend on them to cooperate.

The first Great Lesson introduces the child to the vision of the whole reality – to the universe. Remember that Dr. Montessori said, "Let's give him a vision of the

whole universe. The universe is an imposing reality and an answer to all questions.” If we start out by giving the universe to the children, then we are giving something broad enough to answer the type of questions they have at this age.

Dr. Montessori also said, “We shall walk together on this path of life – the teacher and the child. For all things are part of the universe and are connected with each other to form one whole unity.”

The first Great Lesson is called “Creation of the Cosmos” (God with No Hands). This is the story that sets the stage for all the rest. Then that stage of the universe had to be furnished. And so law and order guiding the elements prepared an environment for the first living inhabitants of the world. When those elements had settled down and order had been created amongst them, then came a procession of plants and animals, each of which had been given a special task to perform.

The second Great Lesson we tell the children is “The Coming of Life” and the “Timeline of Life,” It needs to follow quite closely after the first Great Lesson. We can't wait too long in between these stories because you've got to get these various doors opened up. So perhaps there is a week in-between. And do we see how much we are giving the children to explore through these stories – how much interest is potentiated – and yet they've only been in the class for perhaps two weeks. They also see they don't have to wait too long between lessons. They see these stories keep coming. That's important too. They don't need to know math facts or how to read or write – to hear and benefit from these stories.

The Coming of Life, again sets a stage in its turn. Plants and animals, even in all of their diversity and beauty in the wonder of the work that each carries out, were not enough to carry out the totality of the grand plan that was set in motion and lies embedded within the universe. The plants and animals, a furnished environment, needed another agent to fulfill the plan. And so we come to this agent when we tell the third Great Lesson – “The Coming of Human Beings.”

The third story comes not too long after the second. If you wait months in-between, the children won't see the connections and you've let drop the theme of cosmic education. Human beings had to continue the story. They didn't come on earth just to inhabit it. They came with a task to perform. We have work to do just like the other animals and plants have their work to do. And so we have the two Great Lessons that tell of the accomplishments of human beings – of their particular activities – The Story of Communication in Signs and The Story of Numbers.

If human beings had to work on the environment, if they had to build up a tradition within the environment, they needed a language. They also needed a language for their inventions and discoveries. That is the language that we begin to introduce through The Story of Numbers.

Those two Great Lessons also need to come fairly early in the school year. Then you've opened up five doors into the totality of the universe, the totality of life-forms in the universe and the activities of human beings – their history. We have the whole curriculum there for elementary children. What a difference between this approach versus giving lessons in text books followed by activities with work books. If we open up all of these exciting doors, the child can explore where ever he wishes and in whatever fashion he wishes.

As we reflect on the Great Lessons, hopefully, first of all, you realize they

do have to be dramatically delivered. We have to engage the children through our presentation of them, enthrall them. Otherwise they will not serve as invitations to the children to enter into this drama - and begin to take his place within that drama. If you tell one of the Great Stories in a monotone – generally they will listen politely but it goes no further. We have to be enticing them beyond the story.

The Great Lessons are stories but are stories of great truths. Nothing within them should be flatly untrue. Because they are stories of great truths, they are believable to the children – who want to understand their reality with their own intellects and reasoning. They want to find the reasons for life and the reasons for the events in the world. Remember also that this age child wants to admire and wonder. They want to be grateful and feel compassion. They want the chance to use these qualities which are inherent in their characters. The stories we tell have to allow for that.

The Great Lessons are not curriculum and subject matter. Yet, every once in a while we hear from one that is experienced that they couldn't tell what the children got from the lesson and so they will give them a small quiz to find out. This is wholly inappropriate and inconsistent with the Montessori approach. The Great Lessons are told to unfold the imagination of the children about the universe and all it contains.

We need to remember at the second plane, that the children need to be given freedom within the environment they work in. Remember at the primary level, how the children are allowed the freedom to wander amongst all of the materials in the Casa dei Bambini and to choose whichever one he wants to work with. At the elementary level, they have to be given the freedom to wander at will amongst all the information in the universe that we've open up to them. To pick and choose for themselves what ideas they want to explore further.

The Great Lessons are special in the classroom. They are the only lessons that are often told to the whole class. In the case of a class that has been in existence for several years, and which has some new children who've just arrived from primary, the lessons is announced to the whole class. Everyone is invited and new children are expected to attend.

Children hearing Great Lessons for the second or third time often hear things they missed the first time. They begin to explore in different directions than before, different from the last time they heard the story.

The first Great Lessons if possible, should be told on the first day of school. If you do that, then you really let the children who've just come into the class know that this is really something new and different. We're not just going to continue on from where we left off in primary.

If it's not possible to tell Creation of the Cosmos on the first day, as new teachers who've taken over an established class that wants to get to know the children a bit first – tell the story as soon as possible – at least by the end of the first week. But remember the earlier the better. New teacher's are often moved by the reaction of the children who have had this early telling of the story; it was so exciting for them. This sets them up – excited to come to school.

For those children who aren't reading and writing or who don't know their math facts, these stories can help get them interested in developing those skills. And before the year is over we will talk about how you fill in those basic skills that the children need. We do that at the end of the course in recognition that not all children

should need that help.

It's important to remember that none of the Great Lessons are to be followed up with work assigned by us that the children have to carry out. The stories are to set a stage and get the children intrigued. They should be followed by "the rest", so the children can imagine meditate and reflect by and for himself – before going on. It is through the Great Lessons that education at the second plane of development, serves the psychological characteristics of the children.

The exciting reality is that cosmic education which is unfolded through the Great Lessons, provides the children with a never-ending search and exploration. And it's only this sort of vision which will satisfy children at the second plane of development. It's through this foundation given by cosmic education that the child is prepared for what Dr. Montessori calls "the later germination when the will becomes more directive. That is the task of the child at the third plane."

Just as the sensorial work in the Casa dei Bambini lays the foundation for cosmic education, cosmic education provides the foundation for the child's task at the third plane.

Without the Great Lessons, an elementary class can very easily become just a matter of material and curriculum, subjects being taught, information beings given. The Great Lessons open up the fields for exploration and the impressionistic charts and timelines and demonstrations that go with the stories, touch the children's imaginations. So that they can form their own pictures of the past. So that if they choose to, they can decide to write their own account of what they feel may have happened, using the stories they've been given as their models; but only if they choose to. This is the difference between telling our stories and showing a video with ready-made pictures. It is why we must take care not to give the children too many books – which also has already-made pictures that reduce the need for the imagination to work. You must pick books very carefully. Limit the number of books you put into the environment in the same way you limit the other material you put into the environment.

Allow the children to find their own books. They might have to go to the public library or bring one from home for that. They might find five different creation stories, and that's great if it's coming from them – and not us or another authority. The facts that we give need to be limited so that they do serve to just open that door. It is built on law and order and then just let the children start to explore. Be careful that you don't give too many answers to the children. If you are too willing to be helpful and to provide too many answers, you can become a crutch for the children. They don't have to know what you know and what you don't know. You want *them* to go and look for their own answers.

What about the upper elementary 9 to 12 years olds. They also need to be included in the Great Lessons, whether it's the 9 to 12 teacher telling the stories to just those children or whether they are invited into the 6 to 9 years old class, to hear the stories again. Either way is done. The stories for these older children open up detail for much more sophisticated explorations. Each of the chemicals has a formula, for example. That's the kind of work that the upper elementary children should get into. They should explore some of the different creation stories that come from around the world.

The foundation for this more specialized study is built up through their work in the 6 to 9 years old class. So, think of the broad picture as being given in the lower

elementary class and then the details of that picture being filled in in the upper elementary classroom.

The Great Lessons are what free us from a step by step curriculum of studies. It is the interest of the children that should guide their studies and explorations – not a curriculum set down by the teacher. If we can trust the children enough to allow them this kind of freedom, then we have really understood the needs of second plane children.

In a lecture given in 1936, Dr. Montessori gave very specific directives about our teaching when she said this: “There is no need for subjects to be marshaled collaterally. Each learner should be free to go ahead in each subject when he is ripe for it. Exception being made for those subjects which are linked together by their very nature. We tied the child to the materials in his sensory exercises at the first plane but we left him free to explore his environment. This must be the method for all the later steps in advancing. He must be given the instrument and the strength to use it and then be left free to find things out for himself.”

In 1937, she gave another directive. “In many schools, the question of respect for the child has been interpreted by the teacher to mean that she must stand aside entirely and just give the apparatus. This is not so. The teacher should be the animating spirit of the class and should awaken interest with her teaching and give the direction and then comes freedom and repetition of the exercise.”

In 1942, she gave another clue about our task. She said, “At the second plane, we cannot say, as we said for the first plane, that it is best to know how to withdraw. Here, it is necessary that the teacher know how to inspire.”

We've got those Great Lessons to get us going but we have to carry on from there.

Key Lessons

As well as Great Lessons, we have what are called Key Lessons. If the Great Lessons are to open the doors, the Key Lessons are to help the child in his explorations after that door is opened. Until we get those doors open, the child doesn't know what there is to explore. But now we've got the doors open – and we still need to give some guides, some indicators of his choice of exploratory directions. At the same time, we recall we have to limit how many of these we give, so that we don't take away from the child, the possibility of following his own interest.

The Key Lessons follow on after the Great Lessons. They are to help the child set off on his own exploration. They are to help him know what details to investigate as he is exploring. Like us, the children will be surprised about the immense diversity in botany. The Key Lessons are those lessons that are absolutely necessary for a child if he is going to understand a particular situation or case that is relevant to his broad exploration.

Key Lessons are not meant to give all of the information; there wouldn't be anything left for the child to discover. It's also important to limit the Key Lessons to what

is important and necessary because that helps the child focus on the major ideas instead of being distracted by minutia. Key lessons need to be presented by the teacher who must remember that the child's mind can work by itself. By the second plane, that child's mind is capable of abstraction; so it is not necessary for the teacher to give every single case and step involved in a progression.

For example, if you've presented cubing a 2-digit number and the child has worked with it and understood it, do we then need to present every single step in the same for cubing a 3-digit number? Naturally not. We present only what is different; otherwise we hold the child back rather than helping them move forward.

Mario Montessori used to always caution elementary teachers to not underestimate the child's intellect. He also used to remind teachers to always remember the human tendencies. The tendency within this age child to exert maximum effort, for example. This means the child himself wants to stretch just beyond what was heretofore easily reached. If we give the child only what they can readily understand all the time, they will get bored since they are not stretching and growing. They soon lose interest. We need to give them the keys so that they can go on working of their own volition. They can then stretch their intellectual powers more keenly than we can. Unless we are challenging the children in this way, we don't enable them to make the full construction of themselves. It is when they are asked to undertake a challenge, to exert a special effort, when they have to struggle a bit and succeed, that they begin to realize what they are capable of – what their full potential powers are.

It is from this kind of effort that second plane children develop self-assurance and self-appreciation. This is what keeps them on the path of normalized behavior.

Key Lessons are for the absolutely essential steps in a progression. Or the absolutely essential items without which the child won't be able to arrive at an understanding of the processes around them. They have to be limited and appropriate to each child. Key Lessons have to be distinguished as those lessons which have to be given to a child versus those things which *might* be given to the child. Those things are variations, extensions or games – not major points. These are not Key Lessons.

It's also important to remember that not every child needs every Key Lesson. This is, once again, where observation is so vital. If a child, on his own, has moved ahead two steps, are we going to say “well hold on, let me give the presentations that show this.” There are teachers who become impediments rather than a help when they do that. There are some children who will need each Key Lesson. If you accept a cross-section of society, this is to be expected. Others will not need them all; they move ahead and the teacher moves with them. So we give necessary lessons but they arise from necessity. Let the child do what he is capable of doing.

We've already heard the 5 Great Lessons and a variety of Key Lessons that can be picked up afterward. The trick is to remember that it is only a Key Lesson if a particular child needs that key.

Dr. Montessori also said, “Every useless aid arrests development.” If we are picking up on too many details or give details to children who don't need those particular

details, we are arresting development.

November 22, 2004

Theory of Going Out

The term going out is a Montessori term that has to do with the children, their work and activities in the elementary class. It has nothing to do with field trips – which are organized *for* the primary or elementary children. Going out is part of the work of the classroom. It has to be included in the activities there as much as biology and math is included. Going out has to be one of the developmental factors in place in the classroom.

The going out we talk about has to do with the **psychological characteristics** and the **human tendencies of the elementary children**. This is the age when explorations have to be conducted to some extent outside the classroom, out in society. One of the **needs of this age child is to be prepared eventually to take his place in his society**. He has to be prepared to live in that society. So children between 6 and 12 need to explore that society and what it does; what kind of life is being lived, what kind of work is being done and what kind of environment is that society living in?

In the first plane of development, we have to remember that Dr. Montessori talked about the child's need to adapt to the environment of the family. That means to adapt to the life and work of that family. Through doing this the child builds up security. As that first plane child get to be 2-3, he is ready to move a little away from the family and he is ready to come to the Casa dei Bambini, extending the group beyond the family but still maintaining a close tight-knit group. Both the family and the Casa dei Bambini consist of limited close friends or family members. So it's not beyond what the child is capable of handling at that age.

Adaptating to society means being able to take on all that pertains to the life of one's own culture, the likes and dislikes, the particular way of dressing, the habits, the particular ideas and ideals, customs of the group and the particular language spoken.

If the child is given freedom to act within that environment, to act by himself and for himself, he is able to adapt to that society. He is able to live securely because he has built up knowledge of what the environment is like and what life within that environment is like. So he feels able to cope with his own life.

This closed, limited sort of society is not enough for second plane children. They have to eventually take their place “out there.” And it's as if nature is pushing them to start to practice to stand on their own two feet. They've got to be prepared for whatever that particular way of life is going to be. If we are **to help this construction, it means that we have to make sure that there are opportunities for the child to get out into that society** and to experience what is going on out there, what that life is like.

The freedom that has to be given to the children in the elementary class is a freedom which will allow them to prepare for life - and the whole of that life will not be lived in the classroom. They have to have the freedom to explore the knowledge that is outside the classroom. That body of knowledge that has been build up by all groups of

people and all societies throughout time.

Responsibility

Along with that freedom that we need to give them to explore outside the classroom, we have to help the children build up their own individual sense of **responsibility – towards that environment and towards the people** in that environment. In order to go out into society and successfully live amongst it, one has to recognize one's responsibilities toward that society.

One of the ways in which we help the children do this is through making them aware of the requirements of the public school curriculum. The **public school curriculum is an outside factor that has to be dealt with. And that builds the child's responsibility.**

Independence

Another aspect that is important to the idea of going out is the idea of **independence**. It was during the first plane of development that the child started his own independent life. From the moment of birth, when he frees himself from the mother's womb, through learning to feed herself, to move about for herself, to do things for and by herself. Gradually, she is able to take her place as an independent being within the family.

With the development of language, **the thoughts that the child is developing also should gradually becoming independent ideas**. He has the chance to express himself through language, to share his own ideas, to communicate with others, to explore the thoughts of others. And then the Casa dei Bambini gives the child the opportunity for a further development of independence within a larger group of people. The materials in the Casa dei Bambini allow the development of the power of thought, allowing the child to come in contact with information about the world and its qualities, its language and activities.

At the **second plane of development**, the child now needs to start becoming independent of his own family, to leave them behind to a certain extent, because he has to go out **away from the family in order to explore that union of families which makes up the outside society**. That's why when we talked about the psychological characteristics of the children, we said that **this age child becomes tough and has fortitude and great strength**. The surer the foundation the child has made within his own family unit, the more he is able to move out and away from that family to explore the outside society. He knows that his family will still be there for him; this is the sure foundation.

Material and Spiritual Territories

What work is that society doing that has brought about the **building up of the material and spiritual territories**? It is this aspect of the life of the working society

outside **the child now has to have the opportunity to explore.** The child needs to be helped to understand this. In other words, the child needs to be able to explore the geographical, biological and historical environment that surrounds his society, and that has been used by that society to build up these two territories. The child can't explore this unless he gets out into society because we cannot bring all elements of society into the classroom. We can't bring all aspects of societal work and environment into the classroom, all the cultural activities, for example. That's a difference between the first and second plane of development. In the first plane, we can bring the qualities of the world into the classroom for them. We can no longer make a sensorial environment that provides enough of the world for this elementary age child. **Now, we are dealing with a reasoning explorer who needs to use his imagination.** It's therefore impossible to prepare a classroom environment that provides enough stimulus for these high-end psychic realizations.

Keys

What we do have to do is prepare an environment that provides the keys, the lessons in our albums, but then those keys have to be used by the children and there has to be something onto which those doors open. For the second plane child, that's the universe and the societies that have grown up in the world in which we live.

We have to be careful at the second plane of development not to think that we can compensate for a lack of going out by bringing aspects of society into our classrooms. We can certainly try to do this; bring in a geologist to lecture, for example, in an area where our expertise is insufficient. But that does not make up for going out. We provide the key in the classroom but must also show the application in society by going out.

It's not just information of an academic nature that these children need, but information that **can only be gotten through experience** with people in their places of work or in places where they carry-on their activities. In order for children to develop all their potential at the second plane of development, we have to make it possible for them to do some of their work outside the classroom.

Cosmic Education: Foundation for the Third Plane

We have to remember that the first plane of development is the plane for the construction of childhood. The second plane completes the stage of childhood. Unless the child is given all the opportunity he needs for this construction, he'll arrive at the third plane with developments incomplete, potential unrealized. He'll be hampered in his next development.

So one thing we must be very clear about in our own minds, with the schools we work for and with parents, is that the children of this age have a psychological need to get outside the classroom for part of their work. They have to be able to explore that outer world of the society of humanity. If the children haven't had this possibility

during the second plane of development, they arrive at the third plane which begins the construction of the adult, without the independence necessary for this construction to be undertaken.

When we think of cosmic education, we have to remember that it's the plan of work for the second plane for elementary age children. We have to remember that there's a difference between education by syllabus and curriculum, versus cosmic education. **It's cosmic education in all of its fullness that allows the child to construct a person who has the independence sufficiently developed to understand the societies of the world and how they've come about – and thus to be able to understand the world at this present moment, how it is that we've arrive at this state – and how it might best go forward.**

It's through the freedom to go out into the world outside and to explore it, finding out what it's made of, how it functions, its interdependencies, that will help the child to understand the cosmic plan which underlies the interrelatedness of all societies and of the world and universe.

In order for the child to really make use of going out, we have to provide the child with the keys for that outside exploration. We're not just going for a walk in the park. The keys are given by our presentations of the materials, of the lessons, the stories, through out firing of their imaginations – so that their intellects begin to question: “What else is there beyond this key that's been given”?

As the child develops through the second plane of development, his questioning mind should begin to realize that all the answers cannot be found in this classroom. In fact, we don't want all the answers to be found in the classroom. That's why the books we place there need to be limited. It's not really a good idea to have a school library. We want to make going out necessary for the children. Going out should be a factor that is needed to satisfy the imagination and the curiosity of a second plane child. We start off with the Great Lessons, the key materials, the illustrations, the impressionistic charts, the timelines, the materials that ask the child to think and reason. The math and geometry material, the language materials – all of these form a foundation to allow the child to wonder about what else there is.

Whetting the Intellectual Appetite

That's why, as we tell our stories, as we give out presentations of materials, as we present the experiments, we have to be very careful to limit how much we present, how much we say, how much we give. Because if we give everything, there's going to be no going out. One of the great temptations as an elementary teacher is in any area where you have an expertise, you'll be inclined to give too much. This is natural and must be guarded against. Their eyes will be glazed over and there won't be follow-up. As elementary teachers, we need to remember the function of the key – one key – to open a door. Our presentation should open that door for the child, but in such a way that the child is invited to go inside and find out for himself what has not been given. That's why our presentations have to be carefully prepared for, pared and pruned. Our words need to be restricted, details taken out – that we might otherwise be inclined to include. So what we

do give will strike the imagination. We need to think of our lessons and our stories as providing the skeletons, the bare bones; but those skeletons have to be fascinating. It's up to the children to then clothe those skeletons, to find whatever information can be found to clothe those bones. This becomes the great responsibility of an elementary teacher and it's going to be on your ability to limit that's going to depend the work of your class. If you give too much and do everything, there's no reason for the children to do anything.

The limitation in the lessons is just as important in the elementary as the limitations of words was in the Casa dei Bambini. There our focus was on showing but with very few words.

This is generally something people get better at with practice. You want to keep asking yourself - "how little can I give but how exciting can I make it" - so that the children can't wait to get started on their own explorations. That's part of the art of teaching at the elementary level. If we can train ourselves in this way, then the children will go out to find out what else there is to find around this topic, around this little bit that was glimpsed.

By the same token, you do have to give enough information to get the children started - enough illustrations, enough detail and an indication to the children that you're only telling them a little bit and that there is more to find out.

Relating Work Inside and Outside the Classroom

In our classrooms we can demonstrate the work of air or water. We've seen these. But in our classrooms, we can't really see the work that these two agents do on the earth. You can really only experience the profundity of those natural effects outdoors.

We introduce the children to the development of societies, of how they have come together and how the organization of work has come about. But we can't really see if that's true except by going out into society.

In biology, children can see that seeds sprout and roots grow but does that only happen in the classroom? How will the children know unless they go out to check this out for themselves. It's appropriate to have a garden where part of it can be explored, plants pulled up and so on. They have to see all of these things outside in the world.

All of these belong to the life being lived by that society. They don't belong to materials in the classroom. The materials in the classroom are only going to have meaning to the children if they're connected to the world outside.

We've talked about helping the children realize that it's often unknown human beings who've given us so much of what we take for granted. So the children are helped to realize that before something is constructed and made, the ideas had to be in somebody's mind. The children need to go out and see that this is still happening today. Who comes up with the changing styles of cars? There are people who do this as their job. The children can see that this process from history goes on and on.

The children, exploring the world outside and the work going on there, begin to have a different kind of respect for that world. Suppose the people who collect

our garbage weren't going to do that any longer. In today's society, is it possible for each of us to dispose of our own garbage? We are dependent on other people for doing some of these things that we can't do for ourselves any longer. Glimpsing this idea helps the children realize the organization of society and the sharing of work. We can explain the process to the children in the classroom but unless they have the opportunity to go out and see it in action, it's only an explanation and not necessarily meaningful to the children.

From the key lessons in the classroom and illustrated presentations, should come the child's need to go out. We should be stirring up their minds, raising questions in their minds, and they should be unable to find all of the answers to those questions in the classroom – if they've been sufficiently stirred up. It's from the explorations that the children themselves do – some in the classroom and some outside – that comes the construction of their own body of knowledge, gathered and synthesized from within themselves, from their explorations, not lessons of the teacher or dictates of a society.

Dr. Montessori told us to trust the spontaneous organizing power of the human intellect. That intellect cannot organize if it has nothing to organize around. It cannot construct if it has nothing around which to construct – though this is hardly a concern except to a logical mind which by its limited nature cannot comprehend the ultimate reality of time and experience and the intuitional knowledge that springs from them. Nevertheless, the point remains that no one can form another person's conscious mind – the content of that consciousness. No one can build up a body of knowledge for someone else. Each person must do that for himself. That's why at the second plane of development, the children have to be given the freedom for those explorations. Otherwise they don't really learn but memorize a few facts. They become regurgitators and imitators capable of taking from society but not of helping it progress.

Double-Environment

When we talk about the environment at the second plane of development, we're talking about a double environment – the classroom and the outside world. Presentations of materials, the giving of the keys, illustrations of the stories take place in the classroom for the most part.

The work, the exercising and the activities are done through the explorations both inside and outside the classroom. This is a crucial point the parents need to understand before they put their children in an elementary Montessori class. Ideally, it should be included in an explanation of the two planes of development that should be given to the parents before the child enters the Casa dei Bambini. The children's house provides a little bit of the construction; to complete that development, he needs to stay through the second plane. It should be explained to the parent why going out is necessary. Otherwise you will have some unhappy parents wondering why their child is spending so much time away from school. They look at these as pleasure trips. We must also remember the parents are probably going to be familiar with field trips and this is not what going out is. Field trips are organized by the teacher and generally, everyone goes. For traditionally educated people, going out is quite a foreign idea. Since the parents

don't have a similar background, they have trouble understanding what going out is.

Freedom to Pursue Exploration

What often causes frustration to children in traditional classes is that they are tied down to the teacher and the classroom and the lessons given there. They are often not allowed to go beyond what is given in that curriculum – or at least not given support in doing so. **In a Montessori elementary class, we have to provide the children with the means to explore as far as each individual wants to go.** You can't put the means for that kind of exploration inside the classroom, no matter how well furnished any classroom is, it will provide a limited field for exploration.

Going out means giving the children enough freedom to decide for themselves what they want to find out, not what we think they should find out or what we want them to find out. The work of the children at this level is not a work of projects determined by the teacher. The work has to come from the children's own interest and from their questioning minds.

Sometimes teachers find it difficult to give the children this amount of freedom – to work and to explore outside – because they wonder how on earth they can possibly keep track of what the children have done. If you've given the keys properly, which is your responsibility, and if you've prepared the children to explore outside, which is also your responsibility, they cannot help but learn something when they go out. But **it's not your business to keep track of what each child learns.** That's why we post the public school curriculum or have it in a book on the shelf – because that's what we do have to make sure the children cover and learn.

Often the children are given a test once a year. Even in private schools, often they are tested once per year. This is the norm at the elementary level. That can provide a safety net for the teacher. It provides a check and balance. Not only must we present those things that are required, **but we must know whether the children have understood them.** Maybe we look at test results and see the children were low on a specific aspect. We need to do something about this either through presentation or in helping them follow-up. As long as those mandated aspects are taken care of, whatever the children work on and explore is their own business. We only have to worry about those things they are tested on or what they are required to know at the next level when they leave us.

Very often teachers feel insecure about this. One sign of that insecurity will be the teacher adding in more 'stuff' in the classroom – extra command cards, extra experiments and so on. We ask, “where is this in the album”? **We aren't meant to be material makers - that is not our mandate. Once the teacher-made materials are done the first time, this is enough.** We are also not meant to be walking encyclopedias. The curriculum is in the outside world. Our job is to help the children access it, to teach them how to use it – and then true learning, learning by themselves, will take place. You cannot put cosmic education into the classroom. You can implement a limited curriculum in the classroom but at the elementary level we're hopefully and ideally not dealing with a limited curriculum.

Just as the qualities of the work are encapsulated in the materials of the children's house, so **cosmic education is encapsulated in the Great Lessons and Key Lessons** given in the classroom. But the universe holds the rest of cosmic education. **What we have to defend is the construction of human normality. That human normality is constrained within the terms of independence and responsibility** – which are opposite sides of the same coin. And it is developed and conserved through work just as it was during the first plane of development.

While we have to **ensure the independence of the intellect in the child, we also have to help him be responsible for his independence**. We prepare him for his place in society by making sure that he knows what he is expected to know by 9 or by 12 years of age. We make sure we have taught that and make sure he's followed-up on those things. The rest of it is there in cosmic education; the interrelatedness of all that has to do with human beings and their life on earth.

We have to always be mindful of that sentence that Dr. Montessori heard a child say: “Help me to help myself.” Not just “help me” which makes a dependent. Don't abandon me and leave me without help – because that leaves the child feeling insecure. At the second plane we help the child by giving him the keys to cosmic education. But then we have to allow him to find his own way to the cosmos. Through that searching for his own place, he puts himself in relationship with society and the world and the universe, in such a way that he can feel secure in that environment. This is the kind of security that second plane children need to develop – just as in the first plane, where they have to develop security within the family.

So as we give our lessons, we need to make sure that we're giving the child a sense of wonder, a sense of mystery of there always being something more – beyond what they know today. We have to always be stretching them, encouraging them to always go just a little bit further. This can't be done in the four walls of a classroom.

Practical Applications

Remember when we talk about going out, we're talking about the child needing to get out into society to explore. First of all, we need to give the children the chance to explore the society that is already within the classroom. They're living within a society that has a certain structure to it. They need to explore the structure of the society in the classroom. This structure comes about because we have a group of children of various ages gathered together into one class. This means **it's important that the ages be the correct ages for the children at the elementary level. It means 6 to 9s in one class – not 5 ½** - because there are so many applications to the primary program and the school wants to take in more children, moving 5 ½ year olds up. Remember the psychological characteristics need to change to be a second plane child and rarely does that happen at 5 ½ years old. In the first place, **these children can't cope very well in an elementary class**. They are like fish out of water.

It also means an elementary class should not be 5 to 7 year olds. One of the things to look for when looking for a job is what age and criteria are being used to determine when children move from one classroom to the next. It also means **you should**

not have a class of 6 to 10 year olds since the psychological characteristics of the upper elementary is different from the 6 to 9 year olds. They are beginning to have a sense of individuality, like first plane children who are more focused on themselves. It is difficult to accommodate upper and lower elementary classes together; there is a different structure to the function of the class.

Within this society of the classroom there are certain things we need to take care of with the children. They should come up into the elementary with a grasp of the idea of respect for the environment. The children's house was their place, their environment and through the practical life exercises and grace and courtesy exercises, they should have developed an idea of how to take care of that environment. Now the very constrained and exacting order of the first plane child is lost to a certain extent. A first plane child will set up her work and want all her pencils lined up just so. Elementary children are happy if the pencils are merely on the table, since they are building up an internal order. But that doesn't mean anything goes in the elementary class. The children should never be allowed to be disrespectful. Grace and courtesy starts in primary but continues throughout elementary; we establish it if needs be. We model it in the classroom. They are not so concerned with external order so we can't be persecuting them about the layout of their materials or how neat their work is – or they won't work out of frustration. But by the same token, there is so much paper, pencils and material, that the room can turn into a disaster if you don't have some expectations in place. This is why, when the child finishes working in an area, it needs to be straightened up with materials put away. It means at the end of the day, the entire environment needs to be tidied and cleaned. If anything will be left out, it needs to be straightened up.

For teachers, we have to carefully consider the setup of the classroom. Carefully plan how things will be arranged before beginning to set things up. Then we have to help the children to put things back into their places. So we need to be available when they are done working with something to show them where to put it back. Otherwise they'll stick things into any empty spot on the shelf. This is building up responsibility in the classroom that prepares them for going out. If they are not responsible in the classroom, they will not be so outside either.

We also have to help the children take care of the environment just as we do in the children's house. But in the children's house, the children are happy to clean just because they love the activity; elementary children aren't like that any longer. In the elementary class, we have to train the children in a different way to take care of the environment. We help them do this out of a sense of responsibility toward the environment. This begins with the classroom and the outdoor environment connect to the classroom but then extends outward to society at large. We have to help the children see for themselves what needs to be done in taking care of the environment. One way of doing this is first of all to **post in the classroom a list of the jobs** or the activities involved in taking care of the environment. This is the **first stage**. Week by week, let the children choose which of the tasks they're going to be responsible for. Then the **teacher has the job of making sure they've done their jobs**. This implies you give them enough time at the end of the day to do their jobs.

Depending on the degree of the care of the environment the children learn in

the primary class, will determine how quickly you can move on to the next stage. Instead of you drawing up the next jobs, take your list away and have **the children determine what the jobs for caring for the environment will be – guiding them if necessary to make sure everything is covered. This is the second stage.** Week by week they will choose which job they will be responsible for.

How quickly you move on to the **third stage** depends on earlier work. This **involves taking away the list completely and telling the children that now they are expected to take care of everything that needs to be taken care,** for the life of the class whenever they see something needs to be done. It's that degree of responsibility they need to have – to see for themselves what needs to be taken care of and then to do so without anyone asking or telling them to do so. If we depend on a list forever, ecology will never work. **Care of the environment of the world means being able to see what needs to be picked up or thrown away or cleaned up and acting upon it ourselves.**

So in the classroom, how quickly these three stages are met vary greatly. Some years you can get to the third stage early on. Other times, it takes to the end of the year to reach this point. Through these stages we teach respect to the children.

Another way in which **we teach respect to the children is through the rules we establish** in the classroom. These **rules should be stated in a positive way.** “We walk” versus “We don't run.” If you state things negatively, you're setting up a challenge for some children. If you state it positively, **it helps them accept the responsibility.** If they can abide by the rules in the classroom, they can transfer this ability to the rules they encounter when they go out. If they go to the library, there are rules to follow – such as being quiet. If they go to an exhibit, they will need to keep their hands away from things on display. Their behavior will have to conform to whatever rules they encounter in society. We try to prepare them for this while they are in the classroom. This is the responsibility of the teacher - to set the expectations.

We are giving the children greater and greater independence. It is independence that calls for self-control on the part of the children. They have to be able to recognize what is acceptable and what is not – and to act accordingly. Occasionally, they encounter a new rule, maybe something they disagree with or don't understand. This leads to discussion among the children and with teachers. We hopefully can explain reasoning. **Sometimes it comes to abiding with something we ourselves don't agree with.** What is important to remember is that we can reason with the children. Explain to them as best you can and reason with them. **If we are reasonable with them at this age, they will tend to be reasonable in society.** If we set up a dictatorship, the children are rebellious and unreasonable. **We must have the final say, but we need to reason with them.**

Another aspect to the children's acceptance of rules and regulations is that it helps to **develop the will of the child – their ability to choose to conform to the life of society.** This recognition of reasonable commitments is another way of strengthening the will of the child and helping him to choose to behave in the right way. We have talked about how **the will begins to develop in the primary level by way of points of interest where we show the child one way to do something – but they are free to make their choice** whether to accept that point of interest or not (“soap suds on the floor will not

happen if you...”).

Ultimately, **this ability to choose well should translate into good work choices in the classroom.** Are you using your time wisely? Are you doing the work that you need to be doing? There is free choice after our presentations. What the child chooses should depend on their interest – not our assignment – but also on what they need to do to go forward into new things they've been led to.

The preparation for the child's going out forms many of the exercises of grace and courtesy in the elementary class. We need to make sure the children know how to behave when they go out. This includes how to ride public transportation, how to greet people, how to get someone's attention, how to ask questions, how to thank people. So, before we can send the children out, we have to know they know how to act and behave; this is the teacher's task.

After we've done this preparation (made sure the children know how to carry themselves and care for their environment, inside and outside), then we have to make sure the going out is going to be part of the work they've started in the classroom. We can take them to the beach once in a while if that's appropriate – but this is not going out.

Going out always stems from work that begins in the classroom.

Another common mistake is that teacher's use the idea of going out as a reward at the end of a project or some work. This is rewards and punishment. That's not how going out is intended to work. We **give the presentation**; it starts there. We get the **children's interest stirred up**, perhaps throwing out suggestions for projects, ideas, follow-up – and **the children get started on whatever interests them**. But they **cannot find everything they are looking for, so they have to go out** – to find the different leaves, for example. We have to help them think through what exactly they are looking for. Did we look for it in the classroom first? What further aspect will be added to the project I'm working on? Is it something I have to find out for myself or is it a group of us that can find out? Where do I need to go to find that information and how will I get there. With whom can I go and what other preparation is needed? It's this kind of thinking through process that should go on in the mind of the child. So it has to be related to an interest of the child. And teachers need to be aware of what is available in the outside environment, available regularly as well as from time to time – seasonally and so on. The children will help you find out about things as well, as their interest is peaked. It also means we need to be aware of all the different areas: music, math, sports and so on. **The teacher has to be the primary source for ideas.**

What kinds of trips enter into going out? We can take each of the subject areas and brainstorm for different trips. Starting with geography, what are some ideas you have for going out related to geography. Kettle Moraine National Park is a good start. We have a great natural history museum in Milwaukee. Going to a river is interesting. Visiting different land forms.

For biology, there are other outings of interest. We could visit the zoo. A botanical garden or forest is interesting. Medical schools are good when related to human biology. A botanist at a university provides a good resource for outings. Fish hatcheries and ponds or certain types of sanctuaries are appropriate.

History has plenty of examples. Visiting old palaces or war battle sites is good. Historical tours, walking or driving, are useful. Architectural landmarks are appropriate. Interviewing elderly people.

Language is another area for going out. A radio station or newspaper operation is interesting. The Concordia language camps in Minnesota is good. The library is always available. Book or poetry readings are striking outings.

For mathematics, the opportunities include the grocery store or any place where currency is involved. Pointing out geometric shapes in buildings or in other forms is stimulating. Cooking or watching cooking, which uses recipes in certain quantities is a possibility. Construction sites might be enlightening. Bridges, art and composers are sources of outings related to math. Surveying and surveyors are a source of mathematical outings.

For music, visiting rehearsals, a conservatory or recording studio may be interesting to the children. Folk music and festivals may be interesting. Instrument makers may provide new insight. For art, there are free exhibits and art museums that may tie in to children's interest.

Different preparation is necessary for different kinds of trips. We need to help the children think through things like what will the weather be like, if outdoors. Will they need boots? You may have to deny their attendance if they overlook or forget certain preparations. The **children should, at times, suffer the logical consequences of their planning or lack of planning.** Make sure the children themselves are doing the thinking through and planning of the outing. You help to prompt and track their planning but this is different from doing it for them.

As we said, going out must be connected to their work. If they will find information and bring it back to the class, how will they do that. Do they need to take notes? Do they know how to take notes? If they study bridges, for example, will they bring a drawing back? Have you taught them how to make a sketch? Are they prepared to bring information back in whatever form that is possible?

Who goes out and when? There are three possibilities for who goes out. The **first is when the whole class goes out**, though this is rare and happens typically when the attraction has limited availability and it is important enough to include everyone. This is more like a field trip but you can still have the children plan the outing. If the destination is several states away, certainly the teacher is more involved in the planning than usual. The **second possibility** is for a small (up to 5) group going out because their group needs more information for a project they are engaged in. It's a point of arrival that they are now ready to go out. They develop the necessary responsibility and the necessary skills to use going out effectively in their work. The **third possibility**, again rare, is with the 9-12 class, where one child needs to go out for something he is working on individually.

The characteristics of **the 9 to 12 year olds; they start to become more individually focused.** Toward the end of this period, some of the children will work more independently. So there may be a need for one child to go out. There are issues to be worked out in today's society; it is rare that children can go anywhere unaccompanied.

There are typically companions. The important thing is training these companions. They have got to understand that they are there just to protect the safety of the children. **The children make all of the other decisions.** Even if they get on a wrong bus – the companion gets on the bus with them and let's the children work their way out of the problem with minimal assistance. Not everyone is cut out to be a companion. Some parents are not good choices for this. Assistants are actually the perfect person for being the companion for going out. We can train this person and they know the children. Make sure that when you look for a job, ask “does your school do going out”? If not, “what are the obstacles so that we can make it possible?”

There is another area that belongs to going out. It comes under the heading of **community service**. Remember when we talked about the characteristics of the second plane of development and we said the **children feel a sense of compassion**. This is the age to pick up on that. Through going out in community service, they recognize that there are other people less well off than we are. This is especially appropriate at the 9 to 12 level. There may be people around the school who are shut-ins. Or they have a broken leg and need help shoveling snow. They might serve in a soup kitchen or read to one that is blind. We help them to appreciate giving back to their society. Community service needs to be a part of what goes on in the 9 to 12 class.

Your understanding of going out and what it means as well as how to implement it, is crucial to your success as a Montessori elementary teacher. Going out is what makes a class a Montessori elementary class. It's not just a classroom using Montessori materials. It's not enough to tell the Great Lessons, to give the Key Lessons and to inspire follow-up work in the children. Unless we have them going out, we're not offering cosmic education. This is the framework that prepares the children for the next sub-plane – the 12-15 year olds, where they are ready to go out into the world of work. Just as the materials of the Casa dei Bambini prepared the children for cosmic education, cosmic education, which includes going out, prepares the children to enter that sub-plane where they begin to work in society.

October 22, 2004

The Practical Aspects of Going Out

A Montessori classroom is not a Montessori classroom without going outside into the environment. So, it's the responsibility of the teacher to find a way to get the children outside. In order to get going-out to work, the teacher need to be fully committed to it. If not, you won't be able to do it. You will find yourself in a school where they say it is impossible; you must become an activist and an agitator. When you interview, ask administrators about their policies on going outside. You will have other battles to fight besides this one so ensure this is no problem up front.

You need to educate administration about this aspect of your classroom. Why is it important? This needs to be conveyed. They are responsible for practical considerations – one being insurance. Each insurance company is different from one another, but they do insure schools that allow going out. They ask certain things. They ask for example, that every year, every parent signs a form that gives schools permission to take the children off-campus. This is usually all they ask. The requirements may be more stringent in public schools than in private ones.

Once you have convinced the administrator how important it is, talk to parents about it. You will need their support in several ways. You could give an initial parents night about 'going out.' Here, you describe to them exactly what it is. From this face to face contact with parents; you can infect them with enthusiasm about it's benefits.

You will want to go into the philosophy about going out, mentioning that the children need opportunities to explore and that their opportunities are limited if they have to stay in the classroom. Mention that our goal with these children, is to help to construct a member of society. To do this effectively, the children have to get out into society, in order to see how it works. It helps to refine the child's sense of responsibility. It allow the child to develop a sense of independence if going out is done properly.

Another aspect is development of the child's power of observation. Because the child is interested in so many things and they like to investigate and research, they have to be able to go out to libraries, to museums and to universities, to name several examples. So, teachers have to educate the parents about the theory of going out. Then, you enlist their cooperation in helping you to get the children outside the classroom. They may be called upon to take the children out. But before you ever let a parent take your children out, you have to let them know exactly what that entails.

For starters, let them know that they should have high expectations of behavior when the children go out with them. They shouldn't tolerate poor behavior in the car, on the bus, or at the site where they're visiting. This is sometimes hard for certain parents. They like to be liked. Or they are slightly permissive with their own children. These parents need to know what your expectations are for the children. You can let them know that it's okay to ask the children to be quiet in the car. It's okay to ask the children

not to run around the store and touch everything. When the children are going out, they should have a focus. That focus could be something as simple as a dowel to make a flag. So they should locate and take the dowel and nothing else.

And then the parent needs to understand that he or she should not make decisions for the children. They should not offer advice. They should not remind them of anything. Their task is to transport them to the site and to make sure they are safe. Not all parents are capable of not interfering.

When you do have parents help you with going out, it cements your relationship with them. They get to know you and they get to know how your class works. You need a good relationship with your children's parents. Also, when parents help in going out or in any other capacity, it gives you insight into them and indirectly into their children. It could explain a few things you have observed.

You may pick and choose the parents you want to go out with the children. It is okay to say “no thank you.” In fact, you must say this to some parents. You can do it politely and you should explain your reasons for it. This is difficult but necessary. It's harder to have a parent go along, who undermines what we are trying to accomplish.

Ways of Going Out

There are different ways of going out. It could be your assistant that takes them out – this is the main purpose of your assistant. It could be a parent as discussed above. Sometimes in schools, the head of the school wants to take them out. Or, you can take them out. This is encouraged when possible. So, who stays back to be with the class? The assistant can or a head of school or perhaps an implementer. Another way to get the children out, depending on where you teach, is that they can go out by themselves. This is usually only possible in a smaller rural area.

Kinds of Going Out

There are basically two kinds of going out. Longer distanced ones and local goings out.

Local outings are usually connected with the responsibility and the work the children are doing in the classroom. If they are responsible for a guinea pig, they need to go to the pet store to get its food. If they are making semaphore flags for the Morse Code, they need to be the ones to go to the hardware store to get the dowels they need.

For local going out, you need to prepare two environments: the inside environment and the outside environment. The first thing to know regarding local going out, is what is available in your immediate vicinity. Where is the pet store and the hardware store. Where are the parks and woods. Once you've identified the places you expect your children to go to regularly, especially the small businesses and the library, you should visit these places and introduce yourself. Let the owners know your children will be visiting regularly and that they are Montessori children and finally, that they will

have come with a purpose. Leave your card with personal information in case they want to contact you at a later date for some reason, such as a problem that arises. Dr. Montessori said about going out, that the children are the ambassadors of Montessori education. So, we want the public to know who we are. It is common when you take children out, to have people come up to you and remark on their good behavior. Their good behavior stands out in contrast to the norm.

In preparing the outside environment, we examine the route for getting to each destination. If the site is in walking distance and children are walking, we want to make sure they take a fixed route. We want to know exactly where they are going in case we must reach them. As you examine the routes to different places, know where side walks are and are not. Know where crosswalks are and where there are lights to cross. Know about the major streets and avoid all hazards if possible. Identify construction zones on the children's path; they are natural attractions for children. If they see a pile of dirt – it calls out to them. Identify any and all potential dangers.

If children are not walking but using public transportation, you must find out what kind is available. If they are taking the bus, you need to know where the bus stops are and when they get on and off (sometimes these are different stops). You need to know what the schedules are for transportation and be alert to the fact that schedules change. You need to know the cost of course as well; the children will carry their own money and know what they will need.

For inside trips, have an area designated for going out material. This area includes among other things, a notice that is filled out and taken to the office of the school every time the children leave the class. On that note, there is the date, time, who is going and where, and who is accompanying them if applicable. Also, there is a list of parents who can be called for taking the children out. There is a wallet for them to put their money into and they can put a receipt into it once money is spent. The wallet also has your personal ID card in case it is needed.

So we have prepared our double environment. Now we have to prepare the children – because you can't let just any child go out. Hold a group meeting with them – especially with new children and let them know that going out is part of the elementary class. It is different than the primary class. There will be times for going to the pet store to get food for the guinea pig or grocery store and so on. Explain the practical reasons for their having to go out. Also, let them know that before they can go out, their parents have to sign a consent form. - which means that their parents must believe their child is responsible enough to go out. Also tell them that not only do parents have to think they are responsible enough but you as a teacher need to believe the same thing. Let the children know exactly how you are going to determine that.

“I will be able to tell when you are ready to go out, because you will make good choices in your work. I will see you work cooperatively with other people. I will see that you can handle some of the responsibilities in the classroom. I will see you show respect for the environment and for other people. There is always harmony in your social relationships, because you show grace and courtesy.” This will set a standard of expectations for them. For some children they are going to have to aspire to that benchmark you set. Some will already be capable from the start, but not all.

When a child wants to go out and you don't believe he is ready, you need to tell him why. You must tell him why - in fairness to the child. It is only in recognizing where they went wrong that they can choose to correct the deficiency. You are laying out clear guidelines for them. You can also lay out ground rules for when they actually get to go out. The way to do this is to encourage them to think of some ground rules of safety. When they come up with those rules themselves, they have actually thought about it and are aware of what's important. If you just list the rules, then they are your rules to be broken.

Guidelines

Here are the ground rules to establish. If they don't come up with them all, say, "Well, do we need a rule for such and such"? Let them refine the rules.

- There are no fewer than two children and no more than four
- They must follow the designated route established. Walk with the class, on the route, beforehand.
- They must cross at a stoplight or crosswalk.
- They must stay on the sidewalks when possible.
- They must stay together in all cases.
- They must go straight there, do what is intended and come straight back.
- They cannot dawdle along or physically handle things unrelated to their task.
- If they are using public transit, they must pick a seat and commit to it.
- They must be responsible for the money needed and for the receipts.
- If money is lost, they must replace the funds as a natural consequence. One is in charge.
- They must know how to appropriately get the attention of store workers. Role-play this to ensure they are respected.
- They must let you and the school office know immediately upon return to the school.

When you set up going out, you are ultimately responsible. You must think through all of the details. Part of the success is to allow only those children who are prepared go out; not all should be entitled to this. Check in with the parent to see how it went if they are involved; if there was a difficulty, talk to the child to ensure improvement. (This is very serious since only one bad experience will destroy the entire going out privilege.) These are extremely important lessons in how to accept responsibility and to be accountable.

After actuating the going out program, go back to merchants and destination to see how they felt the children were doing – vigorously at first.

Long distance going out is another thing – not a field trip. Field trips are mandatory trips for all children. These are recommended to occur less frequently – they are for very special purposes, one time opportunities. The going-outs are designed to find out more things in which they are individually interested and therein lies the contrast. It is rare for the entire class to be interested in the same thing at the same time.

There is again a double environment to prepare. The outside environment must be investigated for what is available. One thing to keep in the classroom for this purpose is a telephone book, as well as a telephone. Your children should know how to use a telephone and telephone etiquette. You keep a list of acceptable parents to help with going-out. You might have a card file that has listed the different places that the children can go. This file could be grouped according to subject. So when you find about a good botanical resource, write the information on the card under 'botany.' Have a list for science, music, art, history, geology, zoology – all categories. Keep extra cards so when the children discover a resource, they can also add information to the list.

Another sort of file in the classroom that has resources for going out, is a file of brochures and advertisements. We never throw them away but hold them in case they become timely. Also, keep maps needed in your going-out classroom resources. Keep schedules and maps for public transportation, for example. Also keep notices for children about pending trips; they can set up a trip and forget about it. Notices serve to remind them about up-coming trips. They also need notices for the parents who will be going on the trips. On these notices, they should put where they are going, the date and time, the cost per person, payment arrangements and finally, any miscellaneous information such as special attire, lunch arrangements, tolls and so on. Have a class calendar that the children can use to record going-out once it has been scheduled. Whenever a child schedules a going-out, not only is it added to the class calendar but put it on your own personal calendar.

How to Set Up a Long Distance Going Out for a Small Group

There should be a general going out folder. This folder does not sit in the going out area – it stays within easy reach. Its contents can be used in planning a going-out. This begins with a child that expresses interest of some kind. He can sort through your files for more information. He must determine the place to be visited and then, who will go. Whoever has been working with a particular subject are appropriate candidates. Next, how will the children get to their destination. They may have to look at schedules for transportation as well as costs. They will consider which parents will accompany them. If the option is to use parents, the children will have to determine how many parents will be needed; that is determined by the kind of car they drive, number of seat belts and whether they can sit in the front seat with an air bag. This is all information the child setting up the trip must determine.

The child will have to determine the overall expense and the cost to parent

as well. This includes parking expenses, toll fees, food, entrance fees and so on. Once they determine what the expenses are, the children will determine how the trip will be paid for. The children typically divide the parent's expenses among themselves. If there is a child who cannot afford the entrance fee or pay part of the expenses of the drivers, it is great if you can use some of your class budget to pay for this child.

Now, they have some of the practical matters solved. They must next arrange the date. First thing to do is to check with you for available dates so picture day or something like it is not chosen. The next thing is for the child to call the place to arrange the date. Because they have knowledge of the range of options, they can have an intelligent conversation with the person on the other line. It is always good, before they actually call the place, to do some role-playing so they have all of the questions in their mind that they need to ask. This might include the time the place is open, how much it costs, maximum or minimum people that can attend, any chaperone to child ratio enforced, or any other restrictions that must be observed. The children must be reminded about grace and courtesy on the telephone – 'thank you', for example, before hanging up.

Once a date is established, that is the point at which the child places the outing on the calendar. This is also a point when the children will take a blank piece of paper, on which they write the person's name who is in charge. They also set up here, where they are going, the list of people who want to go, the number of drivers they will need, the cost and anything else pertinent to the trip. This piece of paper is the one that is put into the going-out folder that sits on the teacher's desk. It remains there because having a successful going out is a process that sometimes must be orchestrated. Sometimes the number of parents required may be lacking and other issues arise. They may bring the wrong car – the Miata - instead of the van. If it's set up for Wednesday, and the outing is Monday, the children may need to be reminded to send notice out.

Once it's all on paper on your desk, look at your own calendar, to help yourself to give them gentle reminders along the way. In the 6 to 9 years old class, you need to give them many more reminders. In the 9 to 12 class, you must expect them to handle most of it.

About money, when you have an outing, each child should handle his own money if at all possible. It is only in this way that they can learn to be accountable for money. Sometimes, when they set it up, you must pay for it in advance and it may required a single check. But it is better if this can be avoided and each child is responsible. When you go with them, take extra money – just in case.

The day before the trip, every time, sit down with the children and remind them of your expectation. When you get word that they have not met these expectations, they will not have the excuse "I forgot." You need to make it clear that a bad choice was made and that in life, there are consequences connected to bad choices.

Once the children go out, when they go out, generally there are thank you notes to write, as appropriate, and they must write them. They can write them to anyone who has helped them in executing the trip. Because you want to ensure the safety and success of your trip, when you send the children out, send them out with a travel first-aid kit. You should have emergency information, for each child who will go, with the driver

and others involved. All that information is contained on a 5x7 card normally and it can be referenced as needed. It should include doctor's names, insurance information and so on.

January 31, 2004

Parent Education

You (the teacher) need to be involved in parent education. Don't let administrators tell you they are handling parent education. It should not be the educational director. This is your opportunity to be involved, to build a rapport with parents.

As elementary teachers, your main responsibility will be to help the parents understand what Montessori elementary is all about. This sort of parent education should start long before you get the parents – before the children are enrolled. Ideally, before the children start in school at all, there should be parent education about both the first and second planes of development. We have an obligation to help them understand the developmental stages their children go through.

In particular, in terms of Montessori, in the Children's House, the children develop a lot of tools – independence, movement, decision-making and so forth. We could list the first plane tools picked up. It's a **Montessori elementary class that then allows the primary child to use those tools**. We need to build up an expectation that parents are enrolling their child in this Montessori school for at least the entire stage of childhood. First and second planes go together and **parents need to understand that elementary is the experience that solidifies everything gained in the first plane**.

If we can get to the parents early enough, we can work with them so they understand what Montessori is all about. Sensitive periods are already passed if we wait until the child is already in the elementary level.

The **entire school should have the attitude of working towards gaining the parent's cooperation** – through their education. It has to be the parents, the child and the teacher – all working together for a single effect. This is the Montessori triangle or trinity so critical. If we can work with the parents soon enough, we can be friendly with them – not necessarily friends – but comfortable and friendly, so that it is easier to talk with them. They get to know you, the teacher, and to feel comfortable with you. They need to know you in order to recognize you as a competent person. Their feeling comfortable with you is also important.

We need to explain to the parents what Montessori is and what it is not. It is **not just a place to put their child to prepare for the 'gifted program' in the future**. Some parents think Montessori will give their child a head-start. Then as soon as they can move them into the 'gifted' program, the child is moved.

As we get to know the parents, we also have an **opportunity to assess where you might be able to use them as a guest lecturer**. Sometimes, that's through their profession and other times it is cooking, making costumes, making crafts or stain glass windows. Informally, we can gather this information for future needs. Sometimes, the school will have some sort of social – picnic or the like – where there is a chance to meet parents.

What **topics about Montessori do we want to share with parents?** Human tendencies is an important area to convey their significance in the Montessori system; to orientate, to order, to explore – are urges given by nature according to the human constitution. All human beings have them and they apply to one's entire life. In talking with parents, **put tendencies in the context of planes of development.**

With respect to planes of development, describe the non-linear development of children. Each plane has its own needs that must be met and are met in part by Montessori materials. Make sure parents understand these are developmental stages of the child. Let them know Montessori studied children – and this led to the formation of planes and tendencies.

In comparing the first and second planes, let the parents know of the following changes:

- absorbing (what) mind versus reasoning mind (why)
- individual activity versus group activity
- self-construction versus social construction
- need for reality versus uses of imagination
- practical life versus cosmic education
- things versus relationships
- localized environment versus a double-environment

Now that parents understand planes better, you can refine the parent's understanding of the tendencies in the context of a given plane. First plane gives an opportunity to explore in a different manner than in the second plane. It is **because of the planes of development, characteristics of those planes and the tendencies, that we prepare specific environments and how we work with the children.** This should help them understand why, when coming from primary class, they will find the elementary class quite different. We want the parents to understand ahead of time to dispel a perception of chaos when they first visit.

We need to help parents understand what the children's work is like, and how they **may want to keep working on something** for quite a while. Just because the school-day ends does not mean they are done – they may go home and pester parents to go to the museum. We need to help parents take them if at all possible. In connection with this idea, we can bring up homework.

We want the parent to understand why we don't want the child to bring home traditional amounts of homework. In general, what Montessori views as homework is the child continuing to pursue his interests – in whatever form that takes. They will make their own homework and the parent needs to accommodate this. Their homework also involves life, cooking, cleaning or whatever. They should be reading everyday – this is a reasonable expectation – as is being read to everyday.

The name “cosmic education” is dangerous if we cannot describe it

properly. It is the plan of education that Montessori developed for the years from 6-12. It is the **aid to life for this plane** of development. It is therefore more than academics. We are educating the whole child – not just academically. We give a **big picture** and refine from there. They **learn about relationships** between things in elementary after focusing more on things themselves within those relationships in the primary class.

Make sure to bring in the **public school curriculum**; otherwise a parent wonders about the chances for success outside the Montessori world. Over the course of 3 years, public system curriculum is covered and this must be emphasized. On the other hand, we must say Montessori is concerned for the child beyond testing and performance grades. In terms of the idea of competition, it is true the child is growing in a competitive society. We find our children don't suffer when in a competitive environment though we don't ask them to compete with anyone but oneself when it comes to academics. They can compete with themselves. The parent needs to know if a child is reading at grade-level or not, for example. So let them know how we know how the child is doing; use work journals, portfolio, etc. - keeping the parents from abusing the tools in use, for their purposes.

Going out also needs to be introduced to the parent. This is part of the double-environment related to the planes of development. Let them know their participation is appreciated and hoped for.

Besides theoretical topics, the parent wants to know **what the child will be doing and learning**. We need to show them some of the materials. Invite the parents into the classroom and **show how materials are used** – or let the children show the parents how they are used. Especially if you have a parent you want to show up, let their child present materials at an educational session. It is often the parent you need to see that will not readily show up. It's not uncommon to do a session on geography or something that is easily generalized, but also language, biology and math – specialized areas – must be explained. We could show the parent something we are comfortable with. Once we get started, it will get easier. **Parents may also want to observe the classroom**. Many schools don't allow observation until after a certain date. We need to prepare them for what they'll see; children together at different levels of work and working in groups. Encourage the parent to **watch the entire class – not just their own child**. This is especially true since a child may act differently when the parent is in the classroom. Once we are experienced and comfortable, we can even allow a child to present materials to their own parents.

Some of the larger schools arrange the teachers at different stations in a gymnasium, for example, where different lessons are given at different stations. This gives the parent a variety of lessons and a variety of impressions that inform them about Montessori.

We must be **available to the parents for questions they have**. If they visit the classroom, give them a vehicle to ask questions later on. One nice way to do this is to have a place for observers to sit (have guidelines set ahead of time like not conversing or getting up to walk around) with a **pad of paper right there to jot down the question** and the time for the teacher to call to answer and discuss the question as needed. **Parent teacher conferences** are common also. We educate them there also. Doing back-to-back

conferences for a class of 30 is exhausting. One way to alleviate that is to set up office hours – one or two days a week - in which parents can visit after setting up an appointment. **Post the times available on the door** or through a handout to go home. In this way, meetings can be more timely. We can also contact parents and ask them for a meeting. **Keep track of who you've met and those you've not met with.** Maybe a phone conference is all that is needed.

There can be **occasions where a Montessori school is not the best place for a child.** Through communications with the parent, you can counsel them toward a better environment for their child. If parents aren't happy with how we're running Montessori classes, the child will likely be affected as well. Private schools can ask a family to move on if needs be. If a behavior problem shows up with a child, hopefully we've developed the ability to speak with the parent about it – and do so right away before it grows into something more. Are they noticing the same things at home? Sometimes, you'll find out there's a messy divorce in progress or something else that explains things; we then know how to help the child in the classroom.

When we have conferences about the child's work, we must **consider what is shared with parents and how.** They have a right to know what is happening, what areas need improvement and what the child is mastering. They need to know what they can do at home to help the child as well. But typically, we may not want to cover specifics about how long a project is worked on, for example – to prevent the parent from chiding the child to do more or getting involved inappropriately. **Talk to the parents about freedom and responsibility and the balance required.** Work diaries are between teacher and student – not for parents.

The **most important thing to convey is that we care about the child** and we are there to help that child. This means we need cooperation of child and parent; but they first need to know that we care.

It is **often difficult to get parents to conference** about their children. Think about providing child care for the parent, to allow them to come to a conference – and maybe even entertainment for the children. Get them going home telling parents they need to come! Private schools can serve wine and cheese. It is a public relations challenge.

February 1, 2004

Freedom and Responsibility

Unless we understand **why freedom and responsibility are necessary to each other and what is meant by social development**, this can be an aspect of a Montessori class that is difficult to explain to people – especially parents. People have a hard time understanding how the child develops socially. **If he is working by himself, choosing his own work and working on his self-construction – how does he become socially adjusted?**

We need to understand the implications of the Montessori principals of freedom, responsibility and social development. We need to understand what they mean and how these developments are possible in the children. Perhaps today as much as ever, these ideas are so important to society because **people will be more concerned with how they feel about something than what they think about it**. Both are important but it should be reason we live by and not emotion.

In “To Educate the Human Potential”, Dr. Montessori gives us a program that offers safe-guards at both the primary level and at the elementary level. She says, **“Not in the service of any political or social creed, should the teacher work**, but in the service of the complete human being, able to exercise in freedom a self-disciplined will and judgment, un-perverted by prejudice and undistorted by fear.” One of the things Dr. Montessori, herself, was extremely careful to avoid was having her work connected exclusively to any particular group or country. Mussolini, in her time, wanted to make every school in Italy a Montessori school – and she refused. Montessori, in Italy, has never recovered from that time. Mario and Maria had to move to Spain, until the Spanish civil war – then to Holland, then on to India during the second world war. This commitment by her determined the history for the rest of her life.

We have to be careful to get freedom and responsibility as well as social development into their right place in the classroom – they each are critical to each unique human individual. Each individual needs to make up his or her own mind on aspects effecting the aspects of humanity.

In a book called “The Meaning of Man”, the French philosopher, Jean Mouroux, confirms Montessori's thoughts on self-development when he writes, **“The first problem that arises is that of the meaning and purpose of our freedom. It is not given us for it's own sake or for any end we please, but quite definitely that we may realize ourselves**. There, we have a starting point to which every reflecting man will assent. To develop all our capacities and attitudes, our physical and spiritual powers to fulfill our human task and play our part in the world, and thus to construct and create ourselves, to bring ourselves to birth as we would be, that is assuredly man's true greatness and the essence of spiritual liberty.” This is a restatement of Montessori's philosophy. The construction of the human beings, in liberty, to his own human potential, so that he can find his own place in the world. That is an **aspect of cosmic education**.

Dr. Montessori's plan for the human being was to **train the reasoning**

power, so that the intellect can make judgments. It was her **recognition of the constructive powers of the child for the formation of the adult human beings that made her so aware of the significance of liberty**. We also need to remember that in the early years of her work, the traditional schools were authoritarian, strict, often segregated ages, sitting at desks much of the day (like butterflies pinned to a board – per Montessori), based on linear intellectual growth, and so on. This is still often times the norm.

Montessori began to talk about **liberty within a prepared environment**. It's important to remember that it is that complete phrase – not just liberty – but within a prepared environment that allows a child to construct himself as a human beings. So those **two concepts – freedom and prepared environment** – have to go together. The environment allows the child to find his way, by taking responsibility.

From about the mid-1930s, until her death in 1952, Dr. Montessori was **not only concerned with the child, but also concerned with peace** - peace for the growing child but also peace for all humanity. She realized that peace is of its essence, related to freedom and responsibility. Those factors have to be recognized and used if there will be true peace.

The task of the parents and that of educators becomes how to help the child in her work of constructing himself, as a fully developed human being, including an awareness of what it means to have freedom and what it means to be responsible for keeping that freedom. It was from her continuing observations of children that Montessori gave us guidelines for this. In “Education and Peace” she tells us, **“The human personality must be given a chance to realize every one of its capabilities. There is a need for whole men. Every side of the human being's personality must function.”** To function fully as an integrated whole – as a human being realizing all of his potential – the **person has to be able to recognize the reality of the freedom that is given to him and the necessary limits on its exercise. He also must recognize the responsibility that he bears** for that freedom. He has to become capable of accepting that responsibility.

Dr. Michael Gross, in his doctoral dissertation entitled “Montessori's Concept of Personality” wrote, **“The reconstruction of a humane society is contingent upon the strength and unification of the human personality, the full elaboration of which depends upon a method of education rooted in the laws of development.”** This is what Montessori is all about – rooted in the laws of human development.

Dr. Montessori's principles and practices depend on a recognition of how the child constructs himself through the four planes of development, through a recognition of the human tendencies, through recognizing that at each plane of development, the different psychological characteristics make use of those tendencies. In this way the child can explore her environment and master it – in her time, place and culture. In “Education and Peace” Montessori also wrote, “Two things are necessary. The development of individuality and the participation of the individual in a truly social life.” Her principles and practices allowed for this.

In the Children's House, the materials are there for the work of the

individual, a repeated activity with a single piece of material that allows the child to gain some bit of knowledge or understanding which the material is designed to impart. Individual work is needed – until we get towards the end of the **first plane**; at this time, we have some materials and **lessons designed for small groups such as The Store Game, Division with Bows, Function of Words – all these use small groups**. At the elementary level, we take that individual whose been given a chance to construct himself, and we work with him to help him recognize his place in society. We help him find the place where this unique individual can fit in, where he can begin to practice that recognition of society.

Again, in “Education and Peace”, Montessori tells us, “The fundamental freedom, the **freedom of the individual, is necessary for the evolution of the species for two reasons. One, it gives individuals infinite possibilities for growth and improvement. And it constitutes the starting point of man's complete development. Secondly, it makes the formation of a society possible.** For freedom is the basis of human society.” Elsewhere in this book, she talks about how society today does not take the personality into account. Though this was written 50 years ago, this is still true, perhaps more so. She refers to the way people are regarded as a colony without individuals. She said her method was to give aid to life and help affect the total development of the human personality. This means, for each individual with whom we come into contact – must be regarded as individuals. Always in the work of teaching, we are dealing with individuals. But **in the second plane, we have to ensure that those individuals are prepared to see themselves as members of society** and that they are able to take their place in society.

As we think on these ideas of freedom, responsibility and social development, we have to start with the first plane of development – since that's where this construction begins. The child of the first plane is born with the potential for achieving the construction of a human being, with all that's necessary for the formation of that human beings – with the possibility of developing a reasoning personality. **He is born with the potential for a developed intellect, a conscious functioning intellect.** He is born with the potential for coordinated movement, for functioning independently for and by himself and for constructing human language. The task of the child at the first plane is to make all that potential actual. The **child is able to do this by working in the environment.** Through this actualization takes place. For this to happen, the child needs life being lived around him – activities going on around him. He needs the life of his family and a prepared environment. Through his own activity, he gradually figures out how he should be moving and what he should be doing for himself. As he becomes physically able to, he sits up, crawls, stands, walks and he is constantly drawn to those activities – that he sees other perform. He wants to get involved.

Unless the child has the freedom to get involved in these activities, he is not able to develop all of his potential. This activity helps him learn about his family, to gain knowledge about this life is lived. As the child becomes more and **more able and independent, we also have to help him become responsible for his actions.** He needs to be responsible to the extent that he is capable of appreciating how he is living within that life of the family. This means there must be limits set for the child. This is the only way he can begin to become responsible. By the time he is 2 ½ or 3 years old, the normal

child has developed coordinated movement, formed his language, has taken on a certain degree of independent functioning. We should therefore have helped him within the family unit to arrive at this point – to become a child of three doing his own work, starting to dress and undress himself, carrying out activities within the life of the family. He must **become a participating family member and again, this means there has to be limits to the child's activities – as there must be in any social context.**

We have to remember that **children can only act reasonably if they know what our limits are for reasonable behavior.** It is through the **horme – the force of life** – that the child by 3 should have become a functioning human being at the three-year-old level.

As the child moves into the Children's House, he comes into a greater unit of society. Now he is with a group of his peers. There are more objects for activities to be carried out. He comes into this environment with movement already constructed, with language and a certain independence. **That degree of independence is going to depend upon the freedom given by the family.** Literally, some children are still dressed at 6 years old. We need to continue to give the child independence for further activities as they grow. We need to **help the child become responsible for his activities in the environment as well as for the care and respect of that environment. We also need to help him become able to recognize the rights of others.**

One group of exercises in the first plane is the practical life materials and exercises. These are materials that give the child a chance to develop a greater degree of responsibility for his own actions. **They are allowed to make certain small decisions and gain experience with this process – through points of interest, we help them develop their will. Practical life lessons are the only experiences that have points of interest – there are no control of error involved.**

The choice of work provides another freedom. But **along with that freedom to choose his own work comes the responsibility of actually working.** It is not a freedom to do nothing or to act inappropriately to the environment. We must remember though, in the first plane their work can be simply watching. Our **powers of observation need to be trained to see the difference.** They have the **freedom to choose** from among all activities that have been presented as well as the opportunity to work on that activity for as long as they choose. They also have the **freedom to meet with others** in their society; they can talk together. They begin to develop this social contact with each other. They have the freedom to move around the environment. Dr. Montessori talks about **society as “an organization of human beings, each feeling his responsibility in relation to the collective order.”** This is what we strive for in the Children's House; the child is working toward his own construction but is carrying out that work amongst others who are doing the same thing. Since they are **working together, each has a responsibility to the whole group and to the environment.**

In order for the child to make his full development here, its **important that he stay in the primary classroom long enough.** We've spoken about why 5 year olds should not be moved into the elementary class. But **responsibility is another reason.** Each child should have the possibility to come into the environment as the youngest – to be helped, cared for, guided and so on. Then, they go through a middle stage where that

help is not quite as needed – until they reach the stage of being one of the oldest, one of the leaders, one of the responsible members who is now giving the help to the youngest members of the class. If we think about it, each child should have this opportunity 3 times as they go through primary, lower elementary and upper elementary class. If they move out of any of them too soon, they don't make that solid construction toward responsibility.

The child **at the end of the first plane** should have achieved that freedom of a being adapted to the life and society of the family and close friends to which he is engaged. He **should have arrived at a certain degree of responsibility – in relationship to the environment, family and to the other members of his social unit.** He should be independent enough to behave rationally and courteously with anyone he meets. The child during the first plane needs liberty to achieve this – liberty with a prepared environment. He **needs a prepared adult to work in that prepared environment – one who recognizes an endpoint of arrival and help the child realize his own responsibility for that freedom. The child also needs a social unit that's not too large for the individual to encompass.** These are the factors that must be in-place to enable the first plane child to arrive at what Montessori called normalization – the incarnation of the personality – which is the hallmark of the whole human being. **This child should be able to take independence and responsibility, the manifestations of a unique human potential, on into the elementary class and into the second plane of development.**

Their tendencies are still operative and they have characteristics specialized for the stage of development. Now its different from the first plane. The child **still needs a prepared adult to form the link** between the child and the environment – but that link is formed in a different way than it is in the first plane. The **child still needs freedom to continue his construction and still needs a social unit** in which he can further develop his sense of responsibility. At the second plane, that **social unit is two-fold; it is the classroom society and society outside the classroom** in the community. To understand these factors, means to understand classroom management at the second plane. These are the factors we need to understand and to take into consideration and prepare for. It is **recognition of the characteristics that are different from the first plane characteristics – that guides us as to how** we present lessons and materials to the children. This means also we have to recognize the **need for interacting with that outside society.** We have to recognize the great power of the intellect at this stage of development – meaning we have to **challenge that intellect** – not feed it baby food. We have to **make use of the imagination** and the reasoning power of their intellect. Now the child's explorations are to find out how things work and why things are as they are.

We have to make possible an orientation to how society and the world function. We need to take into account the child's need for a **different kind of order – not external but the order of the mind.** This is the order that reason needs. This is the **stage for moral development.** This is a great concern for this age child because of those **strong feelings of compassion and need for justice.** This age child is **attracted to hero worship.** He needs **allegory, personification, imagery and a sense of awe.**

Within this environment then, the child must have the freedom to work at

whatever he wants for as long as he wants, but that **freedom resides within the limits of ensuring the child has met the standard established by society** by the time he leaves us at 9 or 12 years old. This is why the **public school curriculum** has to be obtained and translated into terminology the child can understand. This includes specific Montessori lessons or materials that introducing this required information. It must be posted on the wall, in a booklet on the shelf – **available to the child as a tool we can use to help them recognize what they are responsible for knowing**. If we don't use this tool, we take on all the responsibility of knowing what they have to learn and ensuring they've learned it. But the tool of making it available helps them learn to take on responsibility. So the children still have the freedom to make their own schedule – but within the limitation of ensuring they work on what is required by their society.

Then we give the children another responsibility; to **keep his own record of work** day by day, week by week. This is something that should begin as soon as the child comes into the elementary class. We need an attractive book for this, that are lined to guide their writing. It would be **used from the first day of school, with the child writing the day and date at the top of the page; on the left is the starting time for his first activity followed by a description of the activity** – regardless of whether it is a lesson or independent work. On the right comes the ending time for working on that activity. So, we might “8:10 Checkerboard 10.” We have to **give new children lessons on how to keep work journals and this begins on the first day of school**. This work journal then becomes a record of how he has spent his time at school. The **teacher has the responsibility of keeping track of lessons given and to whom it was given**.

Work journals help build responsibility for the future. It is learning how to plan to get a job done – a valuable skill in any context. It is not a planning tool per se, but does orient the child toward **developing an ability to use one's time wisely**. Making a list of to-do items is useful as a tool but the goal of journals is not so limited; it is in a broader sense that it helps us abstract how time is used generally speaking.

So far, we have the **public curriculum and work journals**. There is a **third tool. We have a weekly meeting with each individual child**. As one gets to know the children and they become more responsible, sometimes meetings fall every two weeks. In a 9-12 class, we may only have to meet once per month with some children; by the same token, there are children that need meetings more than once a week. These are individual meetings between teacher and child. The child brings his work journal listing completed work and work in-progress. This is **not the time to edit the child's work but rather the time when glancing at the work – seeing the follow-up for various lessons – and suggesting what needs to happen next**. Here we can find out if they need a lesson repeated and make them aware of where follow-up should be focused. As time goes on, the child should become more specific and targeted toward their own development. Maybe the next week the child still has not done anything about the needed follow-up; here **we can up the pressure a bit and have the child commit to when they will do the needed work**. We record these commitments for teacher and for student. Then we need to **refer to that calendar and make sure that child is doing what was discussed**. If they happen to be busy at that time, even if concentrated on something useful, **the commitment has to be brought to bear on the child's awareness of expectations of others**.

These weekly meetings should be no longer than 5-10 minutes each. This means the adult has to be organized and the children must be as well. It is not uncommon in an elementary class to see an afternoon per week set aside for these weekly meetings. The other children in the class have to recognize that we are not available to answer questions during this time. Otherwise, the meeting suffers and the message is soft. Another thing that helps is to **make a list of who should be met with and post it. Then, tell the children to keep an eye on their name on that list – to be ready when we meet with the child before his turn arrives.** This avoids the 'getting ready' loss of productivity. Besides checking on their work and lessons, ask them here if there are lessons they felt they were ready for or really wanted. And this was **lesson planning for the following week** based on their input.

Using **these tools with the children ensures the class is running properly** and that they children are becoming responsible for their work and personal development. We need to help them recognize their responsibility towards care of the environment – indoors and out-doors. The on-going exercises of grace and courtesy also help the child begin to more consciously appreciate his responsibility regarding how he treats other people. The **going-out is another factor that gives the possibility of freedom along the recognition of responsibility** for work and activities in the society. Public (community) service helps the child develop a responsible attitude towards those less well-off than we are. **The way the elementary class works, the work of the children within that class, and the independence of the intellect are our aids to the child's development.** These should enable the child by the time he comes to the end of the second plane of development, to have a **functioning intellect that makes responsible choices, actuated by reason and not emotion.** Then the child will take into the third plane, a personality that has a sense of responsibility for its own ideas and actions. Then, the young person is ready to start the stage of development of adulthood with no potential left undeveloped.

February 2, 2004

Record Keeping

The topic of record-keeping has several aspects. On a practical level, one needs to plan lessons for **each week** and then from a weekly plan, one plans lessons for **each day**. In a broader sense, a teacher needs to have a **3 year plan** for a Montessori classroom. (There is no way to give all these lessons every year – only some are given every year.)

As a teacher, one must have a **system for keeping track of which lessons each child has had**. It is most important that each teacher has the ability to choose a system he is comfortable with. It needs to have enough detail to be useful but not so cumbersome that it is burdensome to maintain. Each teacher needs something workable for him – so that he uses it. Depending on the school, there may already be a system in-place for keeping track of this. Is their record-keeping required and is there a system already which one is required to use? **Record-keeping should be required!**

Generally, some sort of **report goes home** to parents; this can happen multiple times during the year. **Records on the children will support creation** of these reports. **Conferences** need to be held with the parents – and again, records are needed for this. Other meetings and phone calls that are ad hoc often require records; even **attendance** is record-keeping – and this must be done.

Primary teachers need to keep records that can be passed on to the 6-9 year old class teacher. There should be some indication of the presentations the child has had as well as what he has worked on independently. This won't tell one what the child remembers but at least it gives an indication of how far the child has gone into materials. From this one can determine a starting point. **Be sure to ask about policies, laws and so on that may also require records. Be sure to ask.**

Some records are for the teacher's personal use. Some will be input to periodic reports and other records will be maintained as part of a permanent file. Conferences with parents is an example of when additional records are needed. It is a common practice for teachers to have parent conferences – often twice per year. It is a good idea to have written notes for personal use – beyond what is officially recorded. It can be difficult to be in back-to-back conferences for 24 sets of parents; so written notes on prioritized subjects is helpful. Begin by establishing a **positive rapport with the parents**. Remember the **parents need to know you care about their child – this is the bottom line**. Then they will be able to hear what you have to say more easily. Make sure to note relevant questions the parents ask.

If there are **problems or issues to be dealt with**, perhaps record how parents react when the subject is brought up; are they shocked or do they relate to what is said. If there is agreement on a plan of action with the parents, be sure to record that too. If it is really important, type it, date it, sign it and ask the parent to sign in. Clearly document any situation where a parent is accusative in any manner. Get others involved to **objectively record the behavior or whatever issue has arisen** – and to promote

objectivity and fairness. That person(s) might also be present at the conference. Remember we are not necessarily diagnosing every problem – some issues should be handled by other specialists.

Phone calls and other **communication with parents – some of them at least – should be documented**. A phone call or email can be for something positive as well as for something negative – keep it balanced as much as possible for the parents so they don't begin to avoid you. Emails can be printed out and filed – so this is an advantage. In general, it can be crucial to document what is agreed upon. Read it back to the parent and be mutually clear about expectations. It is also useful to note who is initiating contact – teacher or parent.

Meetings with parents are another place where records are needed. Keep track of what parent education has been done, including what was taught, the date and time and how many attended. This information can be used to plan future meetings taking the best attended or best in another vein as the set of best practices in the future. You may want to root out misconceptions they may have; in this respect, it may be useful to record questions of parents. Be sure to **record any special concerns** they may have. Many of these kinds of record-keeping are for the teacher's personal use. Sometimes, it is, in turn, incorporated into other reports.

In the classroom, keep **observation notes**. It is a good idea to take about 5 minutes each day – when not occupied with anything else – to just watch the students. We want to see how children interact, how they pick work, whether they are productive or just killing time, the **level of concentration** they enjoy, any individuals that distract the class. Look at the proportion of individual work to group work. Who works with whom? **What is gathering dust on the shelf?** If no one is working with that material – that knowledge is not in the environment. If there is a classroom assistant, this is an important way he or she can help us. Assistants should not present materials – but there are lots of ways they can help us. They serve as another pair of eyes in the classroom. They can make observation notes for us.

Depending upon the kind of **report the parents receive**, previous notes may be required to **fill in details**. It is common to **comment on social development**.

Then there is **lesson-planning**. As stated above, there must be a convenient, comfortable system for the teacher to use. The system used at the institute was developed by Phyllis – which makes a sheet divided into rectangles with a child's name in each and perhaps their date of birth – used for a weekly lesson plan. These forms are made out at the beginning of the year. Once the grid is made, make copies for each week and record lessons to be given and lessons requested during child conferencing. It is kept on a clipboard with the previous week's page underneath. **From lessons on the sheet, the teacher can determine who will attend a given lesson**. Once the lesson is given, in each child's box, highlight the lesson name to show its completion. Pages are numbered – 13 would be 13th week of school. This meant the exact date was not needed. Then there needs to be a **record of lessons by child**. So each child has two sheets – divided into columns (4) by subject. This becomes an accumulated list of presentations.

These **forms are very visual**, telling the teacher at a glance if any student is

not getting lessons on one hand (by a blank rectangle), or how many lessons a student is getting (by the list in a column). Normally, lesson plans should be with the teacher at school. The list of lessons by child can be maintained at home.

In the **beginning of the school year, present and present and present** – and this must be fed by being organized and prepared. After a few weeks, there should have been some interest developed in the class and perhaps the teacher can let up a bit. But we must make sure to **guard against getting caught up in managing behavior as opposed to stimulating interest**.

In preparing weekly and daily lesson plans, remember we are observing and noting what we think the children are **ready for**, where there **interest** seems to be going and what lessons need to be given to help them keep up with the **public curriculum**. It is better to over-plan than under-plan. If something is planned but is not done, that lesson moves to the next week's plan. Other things **can be dropped** for the moment; this can happen if one interest is eclipsed by another. **Stay flexible**. The children determine what is presented by their **readiness and interest** – though it is our responsibility to keep a **balance that gets in all the lessons over a 3 year period**, including the public curriculum. The **only lessons that must be given to each and every child** are those that support the public curriculum. Don't consciously bring groups of friends together for lessons; **mix up the social groupings**. Downplay the idea of peer leaders, but leverage those personalities if this benefits the class.

Children are keeping their work journals and we are keeping lesson plans. This helps us know how lessons are being used by the children, when children are giving each other lessons and how fruitful their follow-up is. A **presentation given by a child** will not be listed in the teacher's weekly plan but will make the list of lessons received.

Generally, **at the end of the year**, there is some sort of report prepared for the parents. There are modified traditional report cards and other variations. The **ideal format is an essay type of report once per year**. If this can be done, we have the **opportunity to comment on social, academic, physical** and whatever other type of development is pertinent.

In terms of social development, we might want to comment on the child's level of independence. Does the child choose his own work? Does the child tend to lead or follow, work alone or in groups. How does this child resolve conflict in the classroom.

In terms of academic development, we ask other questions. Is the child reading at grade level? What **special interests** does the child demonstrate? Not every lesson is detailed but perhaps one might comment on some aspect of a discipline the child showed enthusiasm for. **Be specific** for the parents. One might comment on the quality of the work – is it concentrated or in fits and starts – is it done with or without care – is the child taking pride in his work. Is the child self-motivated? Is he **keeping up with the school curriculum**?

In terms of physical development, what **fine motor control** does the child have? Does that child enjoy physical activities or does he need encouragement? In what manner is the child uncoordinated? If there are **special physical difficulties**, what extra

help is suggested? Are there any particular aspects of physical development where the **parents need to be involved** at home in order to address it? Does the child need more opportunity to play with friends? If so, it's difficult to get the child working at school – where the social outlet is irresistible. Are there areas of testing that should take place – assuming the matter has already been discussed in parent conferences?

A copy of this year-end report should be **kept at school and a copy sent home** to the parents. This type of report gives maximum latitude to give a true and complete picture of the child than a traditional report does. **Always be specific.**

February 2, 2004

Nine to Twelve Class

Adapted from notes of Carson Wright

Cosmic Education Provides a Vast Field

There are a few guidelines that we should keep in mind if we are working with the second sub-stage of the second developmental plane. The first thing to remember is that **cosmic education is the theme for the aid to life**, that Dr. Montessori envisioned for the entire plane of the years 6 -12 year old. It is a way of looking at life that allows a child to explore and follow her own interests. The first plane of development prepares the child so that she is ready for this possibility at the next (2nd) plane of development.

Cosmic Education means to give an understanding of the interrelationship of all that has to do with the coming into being of the universe. It has to do with the place of our earth in that universe, the furnishing of that earth. It did have to become a prepared environment for all living creatures and finally the preparations were made for human beings to come; that's all part of cosmic education.

It also means an awareness of law and order as a necessary factor in the creation of the universe in the lives of plants and animals and in the lives of human beings. It is in keeping those laws and the realization of that order, that cooperation has come to regulate the chemical elements, plants and animal life, and has given to human beings the chance to accept that regulation for their own lives. Then of course there is a difference; chemical elements, plants and animals obey the rules unconsciously. Animals bring about cooperation and service to each other by living their lives naturally.

Human beings are different because we have to choose how to live our lives, what decisions to make, whether to be of service or not, to make lives of those who follow us better as others have done for us; we therefore carry a debt of responsibility. So the essence of cosmic education is to help the children understand their unique role in the universe and that each of us carries a responsibility. We have to show them the value of cooperation and service. And it is in this way that human beings can show their greatness and importance. And it is in this way that we take our rightful place in the universe.

In the 6 -9, class we introduce this universal theme to the children. We do this by giving the children our great lessons. Those stories introduce the drama that is the universe, its furnishings and nature. After the great lessons, we have **key lessons** which give the children the

opportunity to explore further and to better understand how the elements of the world work, the lives of plants and animals, and what human beings have achieved. What Maria Montessori referred to as the “seeds”, are sown in the great lessons; some of those seeds are further nurtured and developed depending on the interest of the child and what key lessons he has been given afterwards. That goes on through out the entire 2nd plane of development.

Other seeds that are planted lie fallow throughout the entire plane of development. **The will becomes more directed in the third plane. We might not see any results until adolescence.** When Dr. Montessori talked about sowing the seeds of interest, she said not everything that we discuss could be gone into great detail in the 6 – 9 class. That expression, “sowing the seeds”, means giving a general impression or opening a door and introducing an idea for deeper studies. That’s why **all of the component factors of the creation of the universe needs to be introduced to the children between the ages of 6 – 9.** They are introduced to all of those factors and **they can pick and choose what they are interested in** and what they want to explore further (guided by public school curriculum). It is **this approach of cosmic education that allows us freedom.**

From Seeds Come Deep Roots

In the 9 -12 class, the understanding of cosmic education is deepened because the universe is so vast and all encompassing and there are going to be many areas of exploration left - that we might not have even thought of in the 6 – 9 class. For those areas that have been investigated, **this age child will go further in depth** with a topic. This is why **in the beginning of the year they need to hear the great lessons again**, while being told more information than they previously had heard. The children having previously heard them are going to be listening to them in a different way. They will bring to each sitting, further interest, or maybe they will recall their past experiences. They may be reminded of things they have already heard and investigated. Their understanding is deepened, they hear these stories in a different way, **they understand in a new way - new explorations can occur.**

In the 6 – 9 class the children cannot possibly learn everything that is possible to know. The 6 – 9 year olds have been given an impression of all of the general factors that make up the universe, but they haven’t gone into all the possible details. We pick up material for the 9 – 12 year olds where they left off in the 6 – 9 class. We have to **review and think about what they really know.** Certainly as the student progress through the elementary class they **should arrive at more and more abstract understandings**, though they enter at different levels of understanding.

The prepared environment should differ from the lower level 6-

9 classroom. It should have a few **more textbooks, more focus on community** service (going outs). The room should be **fit for larger bodies. Rid the room of some of the simpler materials**, bells, bead frames, simplest nomenclature materials (the simple picture cards.) etc. It would be good to **add a cooking area, science lab, and a more comprehensive set of literature.**

Most of the **materials are the same, but the 9 and 12 year olds can go further** with them.

Geography

In **geography**, the composition of the chemical elements in the geography experiments, they can study **crystallization** to a further degree, the history of the **water supply**, the composition and different types of **soil**, the composition of different **rock formations**, the **rivers** of the world, and their significance to the various countries and where they are. They might study the history of the location of particular **cities** - why are they there, the ages of various **mountains** and mountain ranges, how do they each form and at what period of time in our earth's history they developed. Important passes through the mountains and their connections with history are intriguing for them. The 9 – 12 year old should study **climate** in more detail.

The 9 to 12 year old might want to keep a **temperature or a rainfall graph** for the year. They can look into the naming of the **winds, and ocean currents**, droughts and other **climatic conditions** through out the world. They can look into **forest decomposition, industries, economic geography** to a deeper levels - world commerce to its furthest extent.

Biology

The upper elementary child may want to study the **developmental lines of animal species** from the beginning of the Timeline of Life to the end of it - which ones are still in existence today. They may want to find what changes have occurred in a plant or animal's existence on earth. What plant or animal species have been lost, why, how and when? What plant or animal **species are endangered** today? What is being done to prevent this?

When possible, the 9 – 12 year olds should **have a garden:** planning and maintaining it. This is part of conservation and our part in it. The **domestication of animals** can be researched, and explored - where, when, why and how. What changes did that bring to those people? How did this affect society? There should be on-going **classification** work in the 9 – 12 class. The children should work with our classification materials but also other systems used through various scientific disciplines.

History

The children in **the 9 -12 class should study human beings further**, delving into archeological studies, paleontology, the discoveries of the earliest human beings, when they were made, where they were made, by whom were they made, and what have we learned from each previous discovery. A **deeper understanding of the Stone Age cultures**, where they were found, what was significant about this, what were the differences

between them and other people, who were the earliest travelers and explorers, and what the discoveries these groups made did to the lives of society.

The upper elementary should **investigate the earliest inhabitants of a particular region**; how did they get to those places, the history of their tools. Any tool you think of has a history that the upper elementary child can investigate. In mechanics, **the lever, the wheel, the pulley - each of those has a history**. What about the plow and similar machinery? What is the history of the tractor, the combine harvester, the spinning wheel, the printing press, the assembly line? Anything imaginable among mechanical apparatuses - **what went into the development of each and how did its development change the people**. Other areas of study include electricity and magnetism, some of the organizations formed by the UN, OPEC and the EU. Why were they formed? Have they been successful in meeting their goals and have their missions changed over time?

Language

Other **language scripts** may interest the upper elementary child such as Arabic, Chinese, etc. Besides our own, **study other language families**. Besides that of Indo-European writers, the 9 -12 should study Anglo-Saxon writings; Chaucer or Shakespeare for example, **putting on a production** of either. Undertake a deeper study of **English literature** and develop a timeline for it or for the history of American literature; they should turn out a perfectly edited final copy of a piece of writing. It should be of the highest quality and free of mistakes.

Math and Geometry

The children should look into the **stories of some of the great mathematicians**. When were their contributions made? What is the **history of particular math concepts**? When did they first appear and what were some of the first uses? What is the history of the language of mathematics - the mile, the meter, the stone weights? What is the history of those and other **terms**. What about the relationship between **art and architecture**, the building of bridges and great structures.

They should be ready to investigate other types of geometry, non-Euclidian geometry. They can become familiar with the use of mathematical formulas for getting space ships into space, the computer, the binary system which is being used, with blue prints and with drawings to scale - **making use of their understanding of similarity and proportion** and knowing how to apply them.

Music

Upper elementary children may want to deepen their know of the **history of music**. They may investigate different **culture's styles** of music - their place in their own history. The child can investigate the **music's meaning and origin**, the history of national anthems, for example. The **dancing** and music of any different groups of people. Teach dance.

Art

The **history of art, pottery, sculpture, painting, working with metal, jewelry design.** Investigate the **art of different groups** from around the world. Compare and contrast what different groups have done. The 9 to 12 aged child may want to examine different **art movements in history** and what brought them on.

Summary

These are just some indications of how the work and the **lessons need to be extended for the 9 – 12 year old.** The children cannot have a beginning knowledge of all of these things, many of them will be introduced for the first time.

Always make sure we are keeping an eye on the **public school curriculum.** This work of the second plane is this plan that Maria Montessori talked about as **cosmic education;** it show that when human beings arrived on the scene that **humans possessed intelligence, reason and love.**

These traits carry a responsibility and we need to make sure we understand how we use them. The significance of this is for us is this: to be fully human we need to use them properly.

Through cosmic education, we help the elementary child **come to a recognition of his own humanity and the humanity of others.** Through the studies of human beings and their achievements over the ages, we can show the children the grandeur of the human being and the heights, to which, they can achieve and rise up to, in service to one another.

If we remember that this **second plane of development is the age of compassion,** the time when they are **sensitive to justice** and they are looking for **heroes to admire,** we can **show them these characteristics from people around the world and through the ages.** This is how we can **call the spirit of the child** and it's because of this Maria Montessori called the child the **cosmic agent of peace.** This is why she thought that, through the children, we could bring about peace on earth.

February 28, 2005

Running An Elementary Montessori Classroom

Size of the Class

Eventually, it should be a **class of 30-35 children**. Dr. Montessori said she had the best results with 40 children. The big difference is we have mixed ages so a larger class size can work. This does not mean we want 30 brand new children. One should build a class up to 30 or 35.

In starting a new class, one would **begin with approximately 10-12 children** coming from primary classes. The next year, then there are 10 or 12 more and the third year, add another 10-12; factoring in attrition, at the end of 3 years, one will have a full class.

One thing to watch out for is that the children don't move up too young. They **should not move to elementary at 5 ½** because the school has so many applications for primary and so they want to promote some children early. They need to be second plane children. When during the year, can children move from one class to another? These children can visit the elementary class in preparation for the transition if school policy allows.

Ideally, we hope the children have worked through the entire set of primary lessons and materials, so there aren't many gaps to fill in in the elementary class; this would generally take to 6 or 6 ½. There is nothing wrong with a child spending 4 years in primary class, especially if they've begun young enough. Then they enter elementary class well prepared and after second plane characteristics have begun to appear. Normally, parents are the ones who have a problem with this – but generally this is not a problem. Focus on the psychological characteristics to determine when promotion should occur.

There should be a **proper division of 6-9 and 9-12 age groups**. There can be some overlapping but it should not be 6-8 and 8-10! Working with too few children in a class can be difficult and we may encounter positions like this. This happens when starting a new elementary program – where perhaps there are 6-8 children at the start. In this situation, it is too easy to interfere since one will know what is going on all the time; this class becomes too dependent on the teacher. Avoid this by giving the lessons and help but then pulling away to let the children work on their own. Set up a material-making corner for yourself where you can watch the children but can 'work' in such a way that the children learn to let you work and they also can work on their own.

If there are **too few children, the incentive for follow-up work is not there** coming from the other children. With 30 children, they see other work being done and become inspired by each other. The **teacher will have to play the role of catalyst**; with a small class this will be the case for a longer period.

Try to develop the class with a **reasonable age spread – some balance**

where there are approximately the same number of each age child. If looking for a job, look at the age spread in the primary classes; it should be about even among ages. If primary class becomes unbalanced one year, that will work its way through the entire school.

There is a **difference between a 6-9 class plus a 9-12 class versus a 6-12 class**. Dr. Montessori herself spoke of 3-6, 6-9, and 9-12 classes. Sometimes there's a situation where a school never builds up enough children to divide the elementary class into two parts, especially in small towns. **If they only have 25 children of 6-12, then one would keep the class together into a single 6-12 class**. It is reasonable to wait until there is a group of perhaps 40 before separating them into two separate 6-9 and 9-12 classes. This is seen as a **natural growth pattern for an elementary program**. Remember we want the whole school to be a community. We have low walls between the primary and elementary classes for this reason and the 6-12 class is somewhat in this vein.

If all of the children coming into elementary class are from the primary classes, this is preferable to taking children who have not had any primary experience. Especially as new teachers, **ideally we want all of the children to have come through Montessori primary class**. After a few years of experience, where most of the second period in development as a Montessori teacher is done, one might take in one or two children who have not gone through the primary program. But we need to minimize the amount of remedial work to be done – and this is unavoidable for those not coming through the primary program. If considering taking in some who've not come through the primary class, make sure you can interview both the child and the parents. See if it is possible for the child to visit for a day to assess how that child will fit into the environment. Present a piece of material to this child and see if she catches on, how she manipulates material, assess her independence if possible. Then ask if there will be enough time to work with this child, to help her fit in – without taking too much from the rest of the class.

We may be asked to take a child who was in the primary but then left for another program along the way. After a year or two, the child has become miserable and the family wants to bring the child back. Again, one would interview the child and parents, find out what difficulties the child has been having. Have the child visit for a day or two. Be honest with the parents about how it is felt the child will adjust – if habits must be broken or newly developed – and whether the parents will be supportive. Try not to take too many odd situation placements since one also has special-need children who need extra time as well. We want to **preserve our ability to present cosmic education**.

Before committing to a position, **visit and see the space that will be your classroom**. Make sure that it will be possible to do elementary work in that room. Is there room to spread out? Is there room for timeline work? Is there a place for experiments? Does the classroom have windows? Think about what it will be like ahead of time. We spend lots of time in the classroom.

Questions to Ask Before Accepting a Job

Make sure to **speak with the primary or lower elementary class teachers about the children promoted into your class**. Even if just considering a job, ask about test scores the children have received, in a general sense. What is the most advanced work the children have reached in the classrooms? The more you can find out ahead of time, the better one can prepare for the first day of school. Talk to the administrator and find out what policies they have about: going out, special teachers for music, etc., book and supplies buying, use of chemicals, additions to the classroom. Who makes those decisions? Adding sex education materials for a 9-12 class is appropriate whereas workbooks are not.

Find out what the parents know about the elementary class; who's been doing parent education and what topics have been covered? Where do going-out companions come from? Does the school allow parents to be the companions on going-out? Do parents understand that elementary work is very different from primary work? Does the administrator understand this as well? Does the school have a policy for taking work home? **It's not a good idea to send the child home with work each and every week since this sets up the child for parental pressure**. What is the school's policy? Does the school have a policy for report cards? Is the school AMI-recognized or are they interested in becoming AMI-recognized. Find out about salary and benefits.

Find out about the school's policy for teacher's children's tuition. Do they provide a reduced tuition or tuition waiver? If there is a chance to visit the school ahead of time, try to observe the classes for several days to assess what kind of discipline the children have. Is it inner discipline or is there discipline dependent on the teacher? What kind of independence do they demonstrate? Do they choose their own work and making appropriate choices?

The more one can find out, the more one is able to plan for the start of the school year. So use every means possible to find out about the children, classes, the school as a whole, what expectations are for the teacher outside of the classroom such as committees, and similar types of involvement in the school community.

The First Day

What you do on the first day depends on whether this is a new class or whether one is taking over an established class. Find out from the children more information to add to what the school itself has told you. Assess exactly how many children need remedial work, what kind of remedial work is needed and where the children is at exactly with that earlier work. Are there any children at nearly the same level that can be grouped for remedial lessons?

Are there any subjects the children have not been exposed to at all in the primary class? Sometimes one finds they got no music, or biology. Those are then the

areas we need to take care to put into the environment early-on. Get the nomenclature work going right away.

Are there some of the children who've had a solid foundation and are ready to begin the elementary work? If taking over a class with two or three age levels in it, one needs to divide his time between the children at different levels of work. Keeping the balance between different levels of work, getting new children working and setting up remedial lessons – all must be planned and coordinated.

Think about **how you want to present**. Sometimes this is a presentation table and this is fine as long as **you know you're not glued to that table. Nor does one want to move around every 5 minutes**. We set the model and should be settled into various work as they should be. But **we do need to get around, passing by to see that there is purposeful work going on**. If you don't want a presentation, it is possible to move around the room to give presentations. Either system works fine. Sometimes, once the lesson is complete, you want the children to keep working but cannot do so if materials have been laid out at the teacher's presentation table. **Make sure, when presenting a lessons, that you are facing the rest of the class**.

Remember most presenting is done to small groups. We give rare lessons to entire groups – such as Great Lessons – and occasionally individual lessons such as remedial work or with older children who've moved into more individual work – or if there is only one ready or interested in a certain subject – such as cubed roots. Sometimes we give individual lessons because there is only a single student ready.

In planning for the beginning of the school year, the most straightforward planning is starting a new class. If starting a new elementary class, one builds up the lessons from Great Lessons, through Key Lessons and so on. If possible, on the first day of school, give the very first Great Lesson, to build up that anticipation and excitement from the first day.

The difference between what we did involves math and language. We cannot wait six weeks to get math and language started. They have been introduced in primary so we are simply picking up where the child was last working.

There are **five things to keep in mind and work towards** in the classroom. The first is **freedom of choice**. This depends on how much freedom of choice they've developed

The **second thing is the public school curriculum**. This must be presented to the children. Show them the chart of booklet we've made and what it means. How is it to be used?

The **third thing is the child's independent work**. The **fourth is the children working in groups** – cooperative group work. The **fifth is the individual work the child has to do for himself** to perfect a skill – something he still needs to master.

The second plane child is strong and tough. This means we can speak to

them about these things. We don't have to treat them in the same soft gentle way a primary child is dealt with – though we remain respectful. **Elementary children need to understand that they don't have a choice about working.** This is what they come to school for – and this is a class that works. This expectation must be set from the first day of school.

This is the plane of development when intellect is at its greatest capacity and we don't want to waste these years. This is where our responsibility lies and must be part of what we explain to the children. **We are facilitators but they must do the exploration** and investigations and related work. We explain that we can speak directly to the children, we will be direct about how they treat each other and the environment. We don't speak in anger but as a matter of fact. We cannot wait until we are angry because we will not deal well with issues. **Set the expectations early** so we can remind the children as the year progresses.

Remember that now that we're **dealing with the reasoning mind, change involves an act of the will. We need their agreement** and will to behave according to expectations – and we should explain the reasons why we have these expectations. We have to seduce their cooperation since we cannot force it. One cannot make a child work; we can **entice them and explain why they should** but they have to make the decision to work.

The work of an elementary teacher is different from the primary teacher's role. **We have to expect the children to be involved in the class, to work, to keep up on what is required by the public curriculum.** If they refuse to do any of those things, we have a problem and can be held accountable years down the road for what the child failed to learn. This is a time to bring the child into a conference, with parents and others who may help in the school. **Use reason them with and help them reason toward working.** When a child is not working, call him over. **“Do you want to make your choice of what to do or do you want my choice of work?”** Then make sure the actually begin working. If they cannot decide what to do, offer them two choices. Then they have to choose one of their own. Make them do one of those until suggestions from the child are forthcoming in the future.

From day-one, keep the language and math working going. If we extend the work in language, it should mean we are also extending the work in history, biology and geography – since language is connected with all of those. Keep the work in math and geometry since they help develop the reasoning mind; we don't do math and geometry just for those subject areas! Math and geometry help develop logical thinking and deductive reasoning. We do this to help the children learn to come to their own conclusions after having been given stated facts. Much of the math work helps us move from a concrete sensorial representation of it to an abstract understanding of it. This is one of the greatest gifts we can give these children.

Language and math are critical areas – the tools that allow us to explore cosmic education but we have to **be careful not to over-emphasize them** in the classroom. This is easy - to fall into making math and language the two subjects they work on every day, allowing them to work on other things afterward. It is a **much better idea to begin each day with a story** presentation. If we can **excite the children with**

stories, keeping the **theme of cosmic education** going, they will be using language and math to follow-up the stories. If their skills are not where they should be, we can use other work in the form of stories, to stimulate their reading and logical analysis. Think about how each day is structured and the difference it can make in a classroom. Don't be too accommodating of late arrivals.

In an elementary class, **expect to teach or provide help for nearly the entire day**. Teach from a plan. You cannot walk in each day and do whatever strikes one and expect to be successful. **Be flexible in a plan, but create a plan and follow it.**

At the end of each week, plan for the next week. Decide which presentations will carry-forward that were not done this week – or which one's will be dropped due to a change in interest or perceived readiness. **Plans and records have to be written.** What happens if something happens to you? The children's class will be disorganized and progress will be interrupted in that event. Sometimes the children will jump ahead of where we thought they were and we must be ready to jump ahead as well, including recording the change.

At the end of each day, consider what was done compared with that day's plan. Did you do the lessons that were planned? Was it more or less? Why did we only do 3 lessons, for example? Sometimes, our **lessons are just running too long or we try to cover too much in one lesson**. Be sure to break down the lessons to one thing at a time, keeping them short and sweet. Note what kind of follow-up happens if we overload them versus when we have given them a taste of a lesson.

Remember certain lessons require more planning ahead than others, sometimes **announcing the lesson ahead of time. We hope to get them thinking, exploring or reading ahead of time.** We can invite them to bring artifacts involved to that lesson.

Following the Child's Interest

In India in 1936, speaking of the elementary class and the work there, Dr. Montessori said, **“There is no need for subjects to be marshaled collaterally. Each learner should be free to go ahead in each subject when he is ripe for it, exceptions being made for those subjects which are linked together by their very nature.”** If we as teachers fall into marshaling the subjects collaterally, we hold the children back. We've got to be ready to follow a child's interest, to let them follow their own interest. If they begin to explore something on their own and find they don't understand it get stuck, we **have to trust that they are going to come and ask for help.**

The **children at the second plane become very adept at assessing what they can do themselves and where they need help.** This is exactly what we want. They need to know themselves – what they're good at and what they need more work in, what they can help others with and what they can get help from other in. If we tell them, “no you cannot start that, you're not ready”, this is **likely to discourage them**. Sometimes we know they need other presentations and in that case, we can offer to give the next

presentation that leads up to the one they want.

Dr. Montessori in London in 1937 said, “In many schools, the question of respect for the child has been interpreted by the teacher to mean that she must stand aside entirely, and just give the apparatus. This is not so. The **teacher should be the animating spirit of the class**, and should awaken interest in the class with her teaching, and give it direction. And then comes free-choice and repetition of the exercise and so on. **The secret is to always present these things in the most interesting way possible and then allow free-choice when he has understood.**” This means we have to be teaching, otherwise the children have nothing to choose from. They can't choose something if they know nothing about it. They will not choose something if they've not understood the presentation. So we must make sure they've understood. Then, it's their repetition that will lead to further knowledge.

A commonly made mistake in the elementary is to not teach enough, to look around and think, “they're all working so I don't need to give presentations.” Are they working out of interest and excitement or are they doing busy-work. This is where observation skills come in. They should be working from interest and excitement; if not, you need to keep presenting until this has been generated in the children – especially in the beginning of the school year – except when you take several minutes to observe, you should be presenting.

In 1947 in Adyar India, talking about the elementary class and children and comparing them to first plane children, Montessori said, “In the first period, the ideal was that the child was so active that the teacher could disappear. There was a sensorial world in which the child took through the senses and not through teaching. But now the teacher cannot disappear. Here, the child absorbs, not with the senses, but with the intelligence. And the teacher must be there to give the food. As before, the teacher was the director to activity. Now, instead, she must be the guide into this super-world – the world of culture created by the intelligence of man. So here, the figure of the teacher changes but we must not forget that the child has not ceased to develop. There are two parts, the continuation of help to the development of the individual and the giving of culture. Culture must be given to aid the forming of the individual. The technique should be as follows. The items of knowledge we wish to impart must be given so as to impress the mind, to strike the imagination which so arouses the intelligence. Once the intelligence has been aroused, clear vision has been given and the rest must be left to the work of the sub-conscious. The teacher gives inspiration. The work is done by the individual who is inspired. Here, we cannot say, like we say in the first period, that it is best to know how to withdraw. Here, it is **necessary that the teacher know how to inspire.**”

So Montessori gives a good comparison between the roles of teacher at the first and second plane. In the first plane, the **urge of the hormone forces the child to become a being adapted** to her time, place and culture – adapted to her environment which consists of sensorial qualities. As long as the child is given freedom in that environment, she can't help but explore that environment and culture. This is the sensitive period for culture and why Montessori spoke of liberty in the environment. The role of the adult is to link the child with that environment – and then give the freedom to explore it.

In the **second plane, the child is an explorer through intelligence** and this is how her exploration need to be carried out; this is why we have to teach. The teacher must provide the food that the intelligence needs. The **child now has to have materials that allow the reason to operate**. This is why we assess lessons carefully. We want to make sure we teach enough but also **make sure we don't exhaust the subject in the lesson** as well. In either case, we can end up with bored children. If they are bored, it will likely be one of these reasons. We must give our lessons in a way that **appeals to the psychological characteristics** or they won't be of interest to the children. We must **establish good work habits right from the beginning** so the children know what our expectations are.

When you assess each day, ask yourself **how much interest was generated today** – not just how many lessons were given. The interest is a better tool for evaluating oneself. How much interest did I generate today? When the **children come back the next day, how many of them show interest in follow-up?**

Environment

The **environment needs to be completely furnished with both the materials we can purchase from the manufacturer and the hand-made materials**. **How you set up that environment will reflect one's personality**. There are two AMI-approved manufacturers: Nienhuis in Holland and Ganzagarreti in Italy. What we should be aware of is that we **cannot mix and match the materials between manufacturers**. Don't order something new that will not fit! It is often easy to stick with one or the other manufacturer.

Apart from the two approved manufacturers, there are numerous others. Be a little leery of these. Not only are the materials not always of the best material but sometimes are not made to exact specifications. **Be careful of these materials**.

We have all our materials, plants and animals in the environment. Remember limitation too. We need materials for lessons but don't add things unless it is well-justified. **Make sure it is for a key that isn't already provided. Reason through everything added to the environment**. The classroom is not a super-market, per Ms. Stephenson. Take a shopping bag and go around – taking things off the shelf that don't belong.

The **environment needs to be clean, orderly and attractive every day**. The teacher is responsible for establishing that order. Then the **teacher must teach the children how to maintain that order**. It's not enough to do clean-up on Friday – unless it is a major cleaning. Each day, we must restore the environment to its beauty and order. **Order and cleanliness must be maintained on a daily basis**. We have seen classrooms where this is not done – and suddenly, even a Friday cleanup is not enough.

Make sure the **environment is furnished with all of the art supplies that the children need, paper supplies needed, all types of pencils – everything they need to be independent**. Remember we want the whole appearance of the room to be one that

prompts, “It's beautiful”, not, “It's cluttered”. On the walls, there might be a calendar, some artwork – but not the children's work. **If you hang the children's work, then everyone's work must be hung.**

Make sure **items of interest are brought in and that those things move around to stimulate discussion** and interest, by moving into a different setting in the room. Bring in something new and remove something that has been there for a long time, to keep the environment fresh and stimulating. **You, your assistant and the children are responsible for maintaining the environment.** The assistant can wash finger-prints off the door-jamb or cleaning something too high for the children's reach – but not doing what should be and could be done by the children. **If a child or group has made a mess, they clean it up; this is the only way of developing responsibility in them.** We will of course have higher expectations for a 9-12 class than for the 6-9 class. It is especially in the 9-12 class that one should be able to get away from having assigned jobs – to the third level of work – where people see and recognize what needs to be done – and take care of it before being asked.

The Assistant

If you have an assistant in the classroom, it should be your assistance, not an assistant for the children. This would be someone who could help with cleaning, with administrative tasks such as notes or letters that must go home. The assistant can help with organization such as filing, with safety in the classroom such as when children are working with a hot-plate – which requires adult supervision in monitoring this activity.

The assistant can be another pair of eyes in the classroom, watching, letting you know who is working well and who isn't – passing that information to you. The assistant can be instrumental in making going-out possible. He or she can be a person readily available for those spontaneous trips out. **The assistant should never present materials nor help the children follow-up on presentations.** The more adult help the children receive, the less independent they will be. It is for the sake of the child that we let him alone to do for himself.

Children who need remedial work need us more than other children, so again, assistants never do this. **Assistants are never used to maintain discipline.** If you can't develop the ability to enforce discipline in the classroom on your own within a few years, consider changing professions.

Discipline

Remember Dr. Montessori spoke of **three levels of discipline.**

Is that child **capable at all of self-discipline? This is our first level to establish** this. This should have been approached in the first plane work.

The **second level is that the child is capable sometimes but not consistent.**

The **third level is the child who has developed inner-discipline and is able to make a choice based not only on self-interest but also based on the good of the society** – the class itself and its community.

The **level they are at should determine where they work in the classroom.** The children who are at the first level need to be kept close to the teacher. Children at the second level can be further away from the teacher but still within contact range. Children at the third level can work anywhere in the class they like.

From day-one, we must begin setting expectations in the class. **Talk with the children about the society in the classroom and discuss the role of laws in society.** We word these laws in a positive way and the children must see the need for the rules, verbally explained if not demonstrated. Have them do a skit to demonstrate the need. What happens if everyone runs across the classroom. Develop the list of positively expressed rules for the class society. What happens when the rules are broken?

There **must be logical consequences for broken rules and this must be enforced consistently.** If someone runs, have them go back and walk – the reaction should be related to the infraction. If they play while working, then they work instead of playing later. With rude language or violent behavior, it cannot be tolerated and must be isolated. Oftentimes, the children can be asked what they think the logical consequence should be. **Most important is to be consistent.** If you are strict one day and easy-going the next day, they will try things to test and will be confused in general.

Whatever rules one expects the children to abide by, the **teacher must also abide by.** If there is no sitting on tables, no chewing gum, no drinks or candy – then the teacher must do the same. We have to be fair across the board and this is part of consistency and respect. Dr. Montessori said in “The Child Society and the World”, we need to invest time early-on in the classroom to get these rules established – and then the children will respond.

Practical Factors

If you have new children in the class, remember that those children need an introduction to this elementary environment. Furthermore, when one takes over a class, one will probably make changes to make it yours. The children then need an introduction to how the room is set up – where things have been moved to. They need to be shown where new things added are and how to use them. They need to be oriented about rules but also about the physical environment.

Children from the Primary Class

Remember that for children coming into the 6-9 class from primary class, there is a very different setup from the primary class. We don't have each presentation set up and ready to go; we have to gather things up beforehand. So we must include the children in that, showing them where things are at. Care of the environment is different as well. We need to show the children where cleaning supplies are kept and how they are used. **Introduce the children to their part in caring for the environment**, probably on the first day of school.

The children will need to know what working in this classroom will be like since it also differs from the primary class. Make sure to explain what the expectations are. **Just as the children need to know about the environment, they also want to know about you.** Tell them a little bit about yourself.

The First Steps

Before the end of the first morning, the children should be introduced to the work journals. It could be that only new children need to be introduced but it could be that everyone is being introduced for the first time. In either case, remember that **the children need some small group presentations on how to use the work journals.** Make sure that you remind the children to write in their work journals. **Lots of positive encouragement in the beginning will help establish their use.**

You cannot wait several days before raising the issue of journals again, always explaining why this record keeping is necessary. It is a **tool to help them so they can choose whatever they want to work on** but at the same time determining if they are covering what needs to be covered according to the **public curriculum.** **Explain to the children that another purpose of journals is** so the two of you can meet together once per week; this **allows progression toward points of interest** for the child while assuring required work is covered. Without these meetings and journals, either the teacher is doing all this work or else the children end up with some real gaps in their knowledge, since we've not likely **tracked exactly what has been presented and understood.**

Another thing to carefully think through and **explain to the children is what we want them to do with completed work and incomplete work.** Depending upon the environment, we need to have something in place for both kinds of work. Often the children have some of their own personal effects and supplies in the room; where will they keep their work journal, for example? **Lockers, bins, cubby-holes and drawers** are some of the possibilities available. Paper files can work well to sort out complete from incomplete work. Sometimes we can find portable carts with files or drawers, each with two sections – one for finished and the other for unfinished. **It is important to keep complete work separate from incomplete work.** Some children are good at starting work but never complete it. Those **children need a reminder** when they arrive in the morning – to consider what they might complete today – in addition to doing new work.

Remember that **certain children will hide behind busy-work.** It's not enough to look around and see they are busy. What are they working at? Is it their 20th

booklet of multiplication tables? Time to move on if so. This helps them develop a higher and higher degree of responsibility toward educating themselves. The **children who show from the beginning that they can work and make reasonable choices can be given more freedom of choice**. Other need to be helped to develop that ability. Offer them choices just as one would do in primary.

One of the **psychological characteristics is the need for the group**. They will learn effectively if they can work in a group. But they're coming in from a primary environment that has been mainly individual work. **They have to be taught how to work together in a group, how to share activities and resources**. Very often children want to work with others but they don't know how to get along – so we have to help them negotiate this.

We want an atmosphere of variety in the classroom. We want to **get the children involved as quickly as possible with all of the different areas of work** in the classroom. So, we have to be conscious of presenting a variety as soon as possible. On the first day, we might present the Creation of the Cosmos to all of the children. But after this, one small group could be start on the nature of the elements while another is presented Composition of the Earth. Another small group may be directed to some of the follow-ups to the Creation of the Cosmos.

One of the early language sections is Word Study. We **could present suffixes to one group and present Compound Words to another group**. So, we must **keep careful records of what was presented and to whom**. And then we must note where that information will spread to; we **don't have to present that lesson to those who've gotten the information from classmates**. Always ask yourself, “What area is no one working in”? This helps determine what needs to be presented to reinforce variety and get through all the lessons. Even if a child needs remedial math, perhaps that child could get into fractions, wooden hierarchical materials or other materials and Great Lessons from math. If they have a concept of multiplication, we could bring in Multiples, Commutative and Distributive Laws, the Checkerboard and other tools that don't necessarily require them to know all facts yet. **Make selected variety available to them in every case – including remedial-level students**.

Another thing which should happen from the first day is **reading aloud to the children**. This could be poetry, short stories, historical tales of adventure, fiction and fantasy or non-fiction. Sometimes the children want to make their own anthology of their favorite poems.

As soon as the children can sit and listen, you can introduce hand-work and designing. Take the time to give them a careful exact presentation about designing. We are still modeling things for the children; elementary teachers often short-change skills like designing since they seem simple; designing is creating doodle-like patterns that decorate a page, as opposed to illustrating, which involves drawing images of reality such as faces and other things in the environment. But children need a full presentation of these kinds of things. A compass and ruler fall into the same category; take time to show them how to draw a straight line and how to use a compass. At the back of “Master Math Geometry” (“Master Math Geometry”, by Debra Anne Ross, published 2005, ISBN 1564146677), there are construction instructions that are very useful in the classroom.

All of the different areas of our classrooms should be introduced and in-progress as soon as possible. Even before we come to “The Coming of Human Beings”, we can begin speaking about needs of human beings and reading about human beings. Written compositions will begin with the interest generated from such early work. Even just looking at pictures can be research – as long as they don't simply copy what they've seen – and can lead to writing by the children.

Spelling rules can be introduced very early, picking up on the phonograms that we assume have been introduced in primary. Many spelling rules can be connected to that phonogram work. If the children have not been introduced to decorating their work, make sure this is introduced as well. Introduce written forms that writing can take – such as posters, poems, timelines, booklet, skits, scroll, definitions – but we have to introduce and show these to the children. Introduce other types of work such as diorama – where a shoe box is used to create a scene such as the solar system of cultural setting.

Remember to plan on having different sizes and kinds of paper. The children should be able to take what they need for their project but also be conscious of being wasteful. Introduce the idea of reusing paper and conservation in general. This means you have also set up the bins for recycling that are distinct from garbage.

We need to prepare the children for going out. This can be discussed on the very first day; let them know it's coming and what the expectations about it are. This can also build excitement in them. Review the guidelines used while out of the classroom. Let them know that behavior in the classroom determines whether they are deemed ready to go out into public. Make sure they realize that going out is prepared by them and is logistically handled by them. Let them know that going-out is an extension of work in the classroom and that they must bring something from the outside back into the classroom through either taking notes or sketching what is seen.

Factors in Class Organization

The children have to work and you have to get them working. This is basic. The **teacher must give the suggestions and ideas the children can do** – until they come up with their own follow-up. Give them pictures to look at, places to go, things to read – as well as **describe the forms their work can take: reports, pictures, speeches, movies, plays or skits, music lyrics** and so on. All of this initially comes from the teacher and must be planned ahead of time – including **how the initiatives wind down and turn into fruitful endeavor by the children directly**.

The **Great Lessons are to open doors to their investigations and to all areas** of work in Cosmic Education. After the Great Lessons, the **children should have the chance to think about them and ponder** – the rest that Dr. Montessori talked about – before beginning to explore. The **public school curriculum**, either posted on the wall or otherwise visible. If we don't have these things visible, we end up carrying the responsibility for them rather than helping them take charge of moving through this material. If we don't make sure they don't keep up with that standard, we can become

uneasy about how things are going in our work.

Make sure that we are doing parent education and that we **help parents realize that Montessori is not a preparatory school for the gifted** and talented or to get into the best university down the road. Everything else being equal, we can assure them they will be up to standard by 9 or 12 years old. We cannot simply guarantee excellence across the board as **we are concerned about all aspects of the child – not just academic.**

As far as making sure the child is up to the public standard by age 9 or 12, we need to **make sure the children are prepared for what they'll encounter in a traditional classroom**, since this is where they will end up when they leave us. How much we worry about the public curriculum depends on where one teaches and where the children will go after their Montessori years. If heading to a Montessori adolescent program, this is rarely a problem. For others, make sure they know **how to use textbooks and work books** – standard tools in traditional classrooms. Make sure they know **how to take a spelling test**. Let them know they are not to talk unless called on in a traditional setting or that one cannot simply go to the restroom when nature calls, for two examples. Also, the authoritarian attitude one encounters is a surprise for them. This can all be done in a game-like fashion – perhaps one day per week for the last six weeks of school.

With respect to **homework**, some elementary classes think they should begin sending homework home to get the child ready for a traditional setting. This is best done in these last weeks before the end of the year and the transition is upon them. Montessori children are independent and responsible; they may not like the amount of homework they begin to encounter but they will do it if they are assigned it. Don't stress this too much.

Dr. Montessori spoke of **freedom of the individual intellect**. Only if the intellect is free, can the imagination work. Our job is to **make sure we don't tie down that imagination and limit it by our expectations. Don't give the children assignments or contracts** – except for those that cannot choose work any other way. **Don't give a special time for when language and math** need to be done. We have to trust in the spontaneous organizing power of the intellect; language and math are the two areas that often show up on tests so one does have to make sure that the children are up to public standard – but do it in other ways that DOL (daily oral language) and other types of tools that bring monotony.

Remember the **Montessori formula includes limitation of material**. Remember this **material should be limited to keys**. If you consider adding something, **make sure you can justify what key it includes that cannot be found elsewhere** in the class. This means when you go to workshops and continuing education, be careful about what ideas are adopted. We are there for the children, not the other way around. We are to follow the intellects of the children, guarding against hampering their free operation by our own conceptions of how education should work.

Consider human tendencies in relation to the work in the elementary class. These must be kept in mind and observed in the classroom. The children must be **given the possibility of activity**. If we think of exploration in the elementary class, it should be

carried out through the material as key, and it should be carried out outside the classroom as well. How far can we take this piece of work? What can this idea lead to? Is there an end to it or can it go on and on? How was this interest generated in the first place? The **tendency for orientation** means to put oneself in relation with one's surroundings. Orientation for these children is to the universe in Cosmic Education, as it was prepared for life and humanity over billions of years, and is to the society in which the child lives. The **tendency for order** is toward the laws and patterns of the universe and the laws and patterns that underlie the development of human beings, including its society. The **tendency for communication**, since the child now works through reason, uses language that is fundamentally significant to exploring the cosmos. **Imagination must be stimulated** in the child to see what he does not perceive sensorially, to understand what cannot be comprehended sensorially. The significance of imagination must be understood as the basis of humanity's achievements from their beginning of their time on earth. Repetition, precision, control of error – all of these are interesting to the child's development – though they take different forms – immense and varied form – in the elementary class. This is why we let the children find their own examples – to determine their own challenges and boundaries. The tendency of precision is what helps the child come to a thorough understanding of any given subject; **he will not be satisfied unless he has really grasped an idea and understood it well and to get to this stage he must do his own work, driven by his own sensibilities and curiosities.** Control of error comes about in discussion of the reasons for arriving at conclusion of work. The child himself determines when work is done, after making his own **check for error against a standard he also must assess and control.** The child's work in elementary leads him to the possibility of arriving at the **capacity for abstraction.**

The **AMI standard is a 2-3 hour uninterrupted period in the morning and afternoon**, one in each, protecting as many long periods of exploration as possible. We need to do our utmost to protect this working time. Special classes present logical impediments to investing one's effort and time with abandonment. This must be managed so that both special classes and the Great Work periods do not detract from one another.

March 4, 2005

Planning the First Three Days

Scenario One

There are 12 children, ages 6-7 years old, in a brand new class of 6-9 year olds. Among them, 2 need remedial language and 3 need remedial math.

Day One

8-830 Full group in circle, discusses safety and emergency procedures, daily procedures, introduce teacher and announce Creation of the Cosmos story in the afternoon. Discuss names, the date and day of the week. Record birthdays.

8:30-9:00 While class sits, walk around the room once, pausing to touch each of the materials, explaining care for them and their purpose, briefly. Invite children to use materials they already know how to use. Announce there will be lessons on how to use them.

9:00 Ask for 3 volunteers to serve snacks. Ask them to choose the next three for tomorrow, after they are through today. Assistant supervises.

9:00-10:00 Small groups: Format for Journals, find out who reads and understands time, 4 lessons.

10-11 Small groups: Wooden Hierarchical Materials Quantity, 3 lessons

11:00-12:30 Clean up, lunch, recess.

12:30-1 Discuss jobs. Ask for small groups to determine jobs. List for later.

1-1:40 Creation of the Cosmos

1:45-2:15 Bathroom break and quiet time. Introduction to Nouns with small group.

2:15-2:45 1 remedial language and 1 remedial math lesson, during quiet time.

2:40-3:00 Clean-up and dismissal

Day Two

8-8:30 Full group in circle: Where do we keep work? Where do art materials come from? Discuss what the bell means (practice using the bell several times during the day to condition the whole class' reaction). Establish mutual respect. Speak about community.

8:30 Snack prepared and on-going during morning

8:40-9 Full group in circle: Geometry Story, draw pictures or write about how we measure things

9-10 Small groups: Congruency, Similarity and Equivalence

10-11 Small groups doing first journal entry, Creation of the Cosmos, and activity for yesterday. Discuss when entries should be done. Dispel intimidation.

11:00-12:30 Clean up, lunch, recess.

12:30-1 Read aloud

1-1:40 Small groups: 1 remedial language and 1 remedial math lesson. Follow-up is on geometry, journals, creation of the cosmos demos and charts that are left displayed.

1:45-2:30 Small groups: Wooden Hierarchical Material, comparison of dimensions

2:30-2:40 Circle on carpet: sharing counting the number of syllables in our names.

2:40-3:00 Clean up and dismissal

Day Three

8-8:30 Full group: Chit chat, complete rules in the classroom. Reminder on journals.

8:30 Snack prepared and on-going during morning w/ assistant's watchful eye.

8:30-9:20 Small groups: Suffixes, 2 lessons and Compound Words, 1 lesson

9:30-11 Small groups: Fundamental Needs of Plants, 3 lessons

11:00-12:30 Clean up, lunch, recess.

12:30-1 Read aloud. Remind to update journals.

1-2:00 Assistant takes children for nature walk in three turns– one specimen per person as subject of follow-up.

1-1:45 Small groups: 2 Bells (up-down, high-low, whole and half steps), 2 Movement for Control

1:45-2:40 Small groups: introduce materials and invite tempera works of nature scenes, 2 lessons

2:40-3:00 Clean up and dismissal

March 4, 2005

Scenario Two

There are 21 children, 9 6-7 year olds and 12 7-8 year olds. Among them, 3 need remedial math and 3 need remedial language.

Day One

8-830 Full group in circle, discusses safety and emergency procedures, daily procedures, introduce teacher and announce Creation of the Cosmos story in the afternoon. Discuss names, the date and day of the week. Record birthdays.

8:30-9:00 While class sits, walk around the room once, pausing to touch each of the materials, explaining care for them and their purpose, briefly. Invite children to use materials they already know how to use. Announce there will be lessons on how to use them.

9:00 Ask for 3 volunteers to serve snacks. Ask them to choose the next three for tomorrow, after they are through today. Assistant supervises.

9:00-10:00 Small groups: present format for journals, find out who reads and understands time, 4 lessons.

10-11 Small groups: Wooden Hierarchical Materials Quantity, 3 lessons

11:00-12:30 Clean up, lunch, recess.

12:30-1 Discuss jobs. Ask for small groups to determine jobs. List for later.

1-1:40 Creation of the Cosmos

1:45-2:15 Bathroom break and quiet time. Explore the room quietly.

2:15-2:45 1 remedial language and 1 remedial math lesson, during quiet time.

2:40-3:00 Clean-up and dismissal

Day Two

8-8:30 Full group in circle: Where do we keep work? Where do art materials come from? Discuss what the bell means. Establish mutual respect. Speak about community.

8:30 Snack prepared and on-going during morning

8:40-9 Full group in circle: Geometry Story, draw pictures or write about how we measure things

9-10 Small groups: Congruency, Similarity and Equivalence

10-11 Small groups doing first journal entry, Creation of the Cosmos, and activity for yesterday. Discuss when entries should be done. Dispel intimidation.

11:00-12:30 Clean up, lunch, recess.

12:30-1 Read aloud

1-1:40 Small groups: 1 remedial language and 1 remedial math lesson. Follow-up is on geometry, journals, creation of the cosmos demos and charts that are left displayed.

1:45-2:30 Small groups: Wooden Hierarchical Material, comparison of dimensions

2:30-2:40 Circle on carpet: sharing counting the number of syllables in our names.

2:40-3:00 Clean up and dismissal

Day Three

8-8:30 Full group: Chit chat, complete rules in the classroom. Reminder on journals.

8:30 Snack prepared and on-going during morning w/ assistant's watchful eye.

8:30-9:20 Small groups: Commutative and Distributive Law, 2 lessons

9:30-10:30 Small groups: Fundamental Needs of Plants, 3 lessons

10:30-11 Small groups: remedial language (1) and remedial math (1)

11:00-12:30 Clean up, lunch, recess.

12:30-1 Read aloud. Remind to update journals.

1-2:00 Assistant takes children for nature walk in three turns— one specimen per person as subject of follow-up.

1-1:45 Small groups: 2 Bells (up-down, high-low, whole and half steps), 2 Movement for Control

1:45-2:40 Small groups: introduce materials and invite tempera works of nature scenes, 2 lessons

2:40-3:00 Clean up and dismissal

March 4, 2005

Scenario Three

There are 32 children, ages 6 to 9 years old. Among them, 4 need remedial language and 2 need remedial math. There are 11 6-7 year olds, 9 7-8 year olds, and 12 8-9 year olds.

Day One

8-830 Full group in circle, discusses safety and emergency procedures, daily procedures, introduce teacher and announce Creation of the Cosmos story in the afternoon. Discuss names, the date and day of the week. Record birthdays.

8:30-9:30 While class sits, walk around the room once, pausing to touch each of the materials, explaining care for them and their purpose, briefly. Invite children to use materials they already know how to use. Announce there will be lessons on how to use them.

9:30 Ask for 3 volunteers to serve snacks. Ask them to choose the next three for tomorrow, after they are through today.

9:40-11:00 Small groups: Format for Journals, find out who reads and understands time.

11:00-12:30 Clean up, lunch, recess.

12:30-1 Discuss jobs. Ask for small groups to determine jobs. List for later.

1-1:40 Creation of the Cosmos

1:45-2:15 Bathroom break and quiet time.

2:15-2:45 1 remedial language and 1 remedial math lesson, during quiet time.

2:40-3:00 Clean-up and dismissal

Day Two

8-8:30 Full group in circle: Where do we keep work? Where do art materials come from? Discuss what the bell means. Establish mutual respect. Speak about community.

8:30 Snack prepared and on-going during morning

8:40-9 Geometry Story: draw pictures or write about how we measure things

9-10 Small groups: Congruency, Similarity and Equivalence

10-11 Small groups doing first journal entry, Creation of the Cosmos, and activity for yesterday. Discuss when entries should be done. Dispel intimidation.

11:00-12:30 Clean up, lunch, recess.

12:30-1 Read aloud

- 1-1:40 Small groups: 1 remedial language and 1 remedial math lesson. Follow-up is on geometry, journals, creation of the cosmos demos and charts that are left displayed.
- 1:45-2:30 Small groups: Logical Analysis, three lessons
- 2:30-2:40 Circle on carpet sharing something we know about the room or the school that we want to share with others.
- 2:40-3:00 Clean up and dismissal

Day Three

- 8-8:30 Full group: Complete rules in the classroom. Remind to complete journals.
- 8:30 Snack prepared and on-going during morning
- 8:30-9:20 Small groups: Suffixes, 3 lessons
- 9:30-11 Small groups: Fundamental Needs of Plants, 3 lessons
- 11:00-12:30 Clean up, lunch, recess.
- 12:30-1 Read aloud. Remind to update journals.
- 1-2:30 Assistant take older children for nature walk – one specimen per person as subject of follow-up.
- 1-1:45 Small groups: 2 Bells (up-down, high-low, whole and half steps), 2 Movement for Control
- 1:45-2:40 Small groups: 1 Flat Bead Frame lesson and 2 Bank Game lessons
- 2:40-3:00 Clean up and dismissal